

Università degli Studi di Salerno



Dipartimento di Informatica

Penetration Testing & Ethical Hacking

Information Gathering

Parte 3

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Utilizzo di Crawler

Surface Mapping ed Asset Discovery – Maltego

- Sviluppato originariamente dall'azienda Sudafricana Paterva
 - Successivamente è diventata Maltego Technologies GmbH, un'azienda con sede a Monaco (Germania)

- Permette di estrarre, raccogliere e rappresentare informazioni in modo significativo
 - Si basa fortemente sul concetto di *Open Source Intelligence (OSINT)*

- Consente di
 - Identificare le relazioni chiave tra le informazioni raccolte
 - Visualizzare graficamente le relazioni tra i dati
 - Così che sia più facile individuare aspetti comuni tra le informazioni



Utilizzo di Crawler

Maltego – Caratteristiche Principali

- Strumento interattivo per il Data Mining
 - Utilizzato nelle investigazioni online per trovare relazioni tra informazioni provenienti da varie fonti di dati sulla rete Internet

- Si basa sul concetto di **Trasformata** (*transform*)

- **Trasformata**: serie di operazioni che permettono di
 - Automatizzare il processo di ricerca e raccolta informazioni su diverse fonti di dati
 - Mostrare i risultati della raccolta in maniera grafica
 - Così da evidenziare le relazioni semantiche tra i dati



MALTEGO

Utilizzo di Crawler

Maltego – Caratteristiche Principali

- Consente di raccogliere molte informazioni su un determinato asset
 - DNS
 - Whois
 - Blocchi di Rete
 - Indirizzi IP
 - Sottodomini
 - Etc



MALTEGO

Utilizzo di Crawler

Maltego – Caratteristiche Principali

- Consente anche di raccogliere informazioni sulla «componente umana» appartenente ad un determinato asset
 - Aziende, enti ed organizzazioni
 - Indirizzi fisici
 - Indirizzi e-mail
 - Siti Web
 - Social Network
 - Numeri di telefono
 - Etc



MALTEGO

Utilizzo di Crawler

Maltego – Caratteristiche Principali

- Consente anche di raccogliere informazioni sulla «componente umana» appartenente ad un determinato asset
 - Aziende, enti ed organizzazioni
 - Indirizzi fisici
 - Indirizzi e-mail
 - Siti Web
 - Social Network
 - Numeri di telefono
 - Etc

E molto altro ancora...



MALTEGO

Utilizzo di Crawler

Maltego – Caratteristiche Principali

- Esistono tre versioni di Maltego
 - **Maltego Community Edition (CE)** [Che utilizzeremo per il corso]
 - Costo: gratis
 - Professional
 - Costo: a partire da 6000 euro all'anno
 - Organization
 - Costo: i dettagli sui costi per questo piano non sono disponibili pubblicamente e possono variare in base alle dimensioni dell'organizzazione, ai requisiti e alle funzionalità desiderate
- Tutte le versioni di Maltego hanno accesso ad una libreria di trasformate standard
 - Per la raccolta di informazioni da una vasta gamma di fonti pubbliche, comunemente utilizzate durante le indagini online e nella Digital Forensics



Utilizzo di Crawler

Maltego – Community Edition

- Utilizzeremo la «Community Edition» di Maltego

- Fornisce un sottoinsieme limitato delle funzionalità di Maltego
 - Supporta al massimo 200 interrogazioni (*Credits*)
 - Numero ridotto di trasformate rispetto a tutte quelle disponibili
 - Non può essere usata per fini commerciali
 - È eseguita su Server condivisi tra tutti gli utenti della «Community»
 - Talvolta, potrebbe essere non molto efficiente
 - Fornisce supporto limitato all'utente

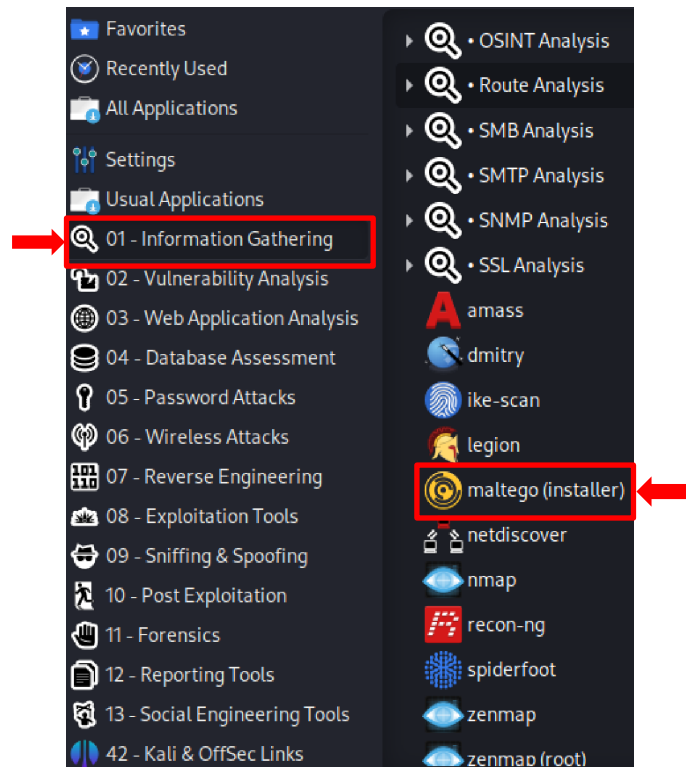


MALTEGO

Utilizzo di Crawler

Maltego – Primo Avvio

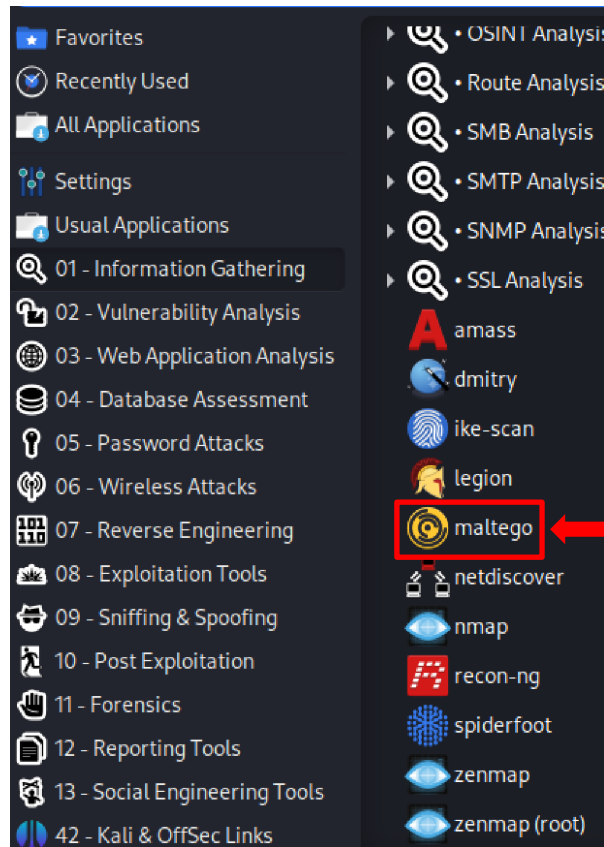
- Maltego va installato in Kali Linux, dove è presente già un installer
- Selezionare **maltego (installer)** dal menu «01 – Information Gathering»



Utilizzo di Crawler

Maltego – Primo Avvio

- Dopo l'installazione, può essere avviato tramite lo stesso menù



Utilizzo di Crawler

Maltego – Primo Avvio

- Successivamente, è necessario effettuare la registrazione del prodotto
- <https://www.maltego.com/pricing/>

Community	Professional	Organization
Free Community Edition (CE) for Maltego explorers	Ideal for individuals investigators or small teams—up to 5 users (billed per seat)	Best for larger teams and government organizations (for teams of 5 or more users)
USE FOR FREE	GET STARTED	CONTACT US
The CE plan includes:	Everything in CE, plus:	Everything in Professional, plus:
● Link analysis software (limited) ● Out-of-the box access to commercial data (limited) ● 200 Maltego Credits ● Connectors to external data (limited) ● Maltego Academy with on-demand and live training	● Investigation tool for quick OSINT searches on the go ● Unmatched access to commercial data ● 20,000 Maltego Credits / month ● User management system (soon) ● Encrypted investigation storage (soon)	● Real-time social media monitoring ● Deep and anonymous social network analysis ● 250,000+ Maltego Credits / month ● Easily integrate your own data (external and local) ● Custom services ● Priority support
See all features	See all features	See all features



MALTEGO

Utilizzo di Crawler

Maltego – Primo Avvio

- Successivamente, è necessario effettuare la registrazione del prodotto
- <https://www.maltego.com/pricing/>

Complete Your Profile

First Name *

Last Name *

Are you a student? ☒ No ☐ Yes

Country *

Italy ▼

 Due to export laws, Maltego products are unavailable in certain countries.

Area code *

IT +39 ▼

Phone number *

☐ I would like to receive updates about Maltego product updates, news, events and offers.

CONTINUE →



MALTEGO

Utilizzo di Crawler

Maltego – Primo Avvio

- Successivamente, è necessario effettuare la registrazione del prodotto
- <https://www.maltego.com/pricing/>

Complete Your Profile

Organization Name *	Private
Organization Size *	1-10 ▼
Business Role *	Trainer ▼
Your Role *	Professor
Organization website e.g. https://www.company.com	
Main Use Cases *	Penetration Testing × ▼
How did you first learn about Maltego?	▼
Focus of Business Unit *	Trust & Safety ▼

CONTINUE →

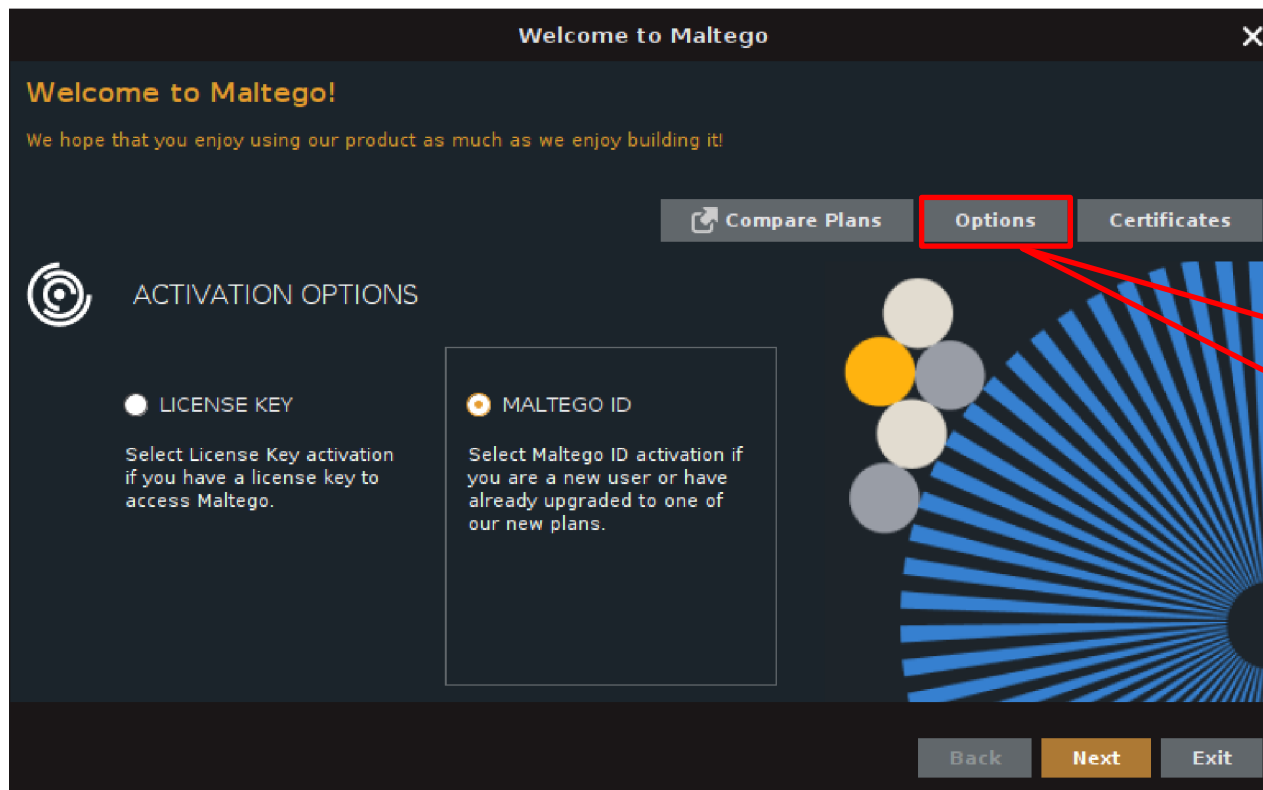


MALTEGO

Utilizzo di Crawler

Maltego – Primo Avvio

- Una volta terminata la registrazione del prodotto, è possibile tornare alla schermata di Maltego, così da poterlo attivare



**Assicurarsi
che non sia
impostato un
proxy**



Utilizzo di Crawler

Maltego – Primo Avvio

- Una volta terminata la registrazione del prodotto, è possibile tornare alla schermata di Maltego, così da poterlo attivare



Assicurarsi che non sia impostato un proxy



MALTEGO

Utilizzo di Crawler

Maltego – Primo Avvio

- È necessario attivare il prodotto, utilizzando le credenziali inserite in fase di registrazione



Utilizzo di Crawler

Maltego – Primo Avvio

- È necessario attivare il prodotto, utilizzando le credenziali inserite in fase di registrazione

STEPS

1. **License Agreement**
2. Login Link Options
3. Login
4. Maltego ID Activation
5. Select Data Sources
6. Download Data Sources
7. Data Sources T&Cs
8. Install Data Sources
9. Help Improve Maltego
10. Web Browser Options
11. Privacy Mode Options
12. Ready

LICENSE AGREEMENT: Please read and accept the following License Agreement.

General Terms and Conditions for Software Licenses and Accompanying Services

Effective November 2022

These General Terms and Conditions for Software Licenses and Accompanying Services ("General Terms and Conditions") apply to software distributed and accompanying services provided by Maltego Technologies GmbH, a company registered in the district court Munich, Germany under no. HRB 236523 (hereinafter referred to as "Licensor") to its customers (hereinafter referred to as "Licensee").

Unless agreed otherwise Licensor distributes software licenses by way of Subscription Plans. By subscribing to a Subscription Plan Licensee acquires temporary rights to use software and will be given access to optional Accompanying Services. The specific scope of each Subscription Plan is specified on Licensor's website. In addition, these General Terms and Condition apply. (Licensor and Licensee also referred to as "Party" and collectively the "Parties").

I. Software Licenses

1. Scope of Software Licenses

1.1. By subscribing to a Subscription Plan Licensee will be granted a worldwide, non-exclusive, non-transferrable and non-sublicensable right to temporarily use the respective Software for Licensee's

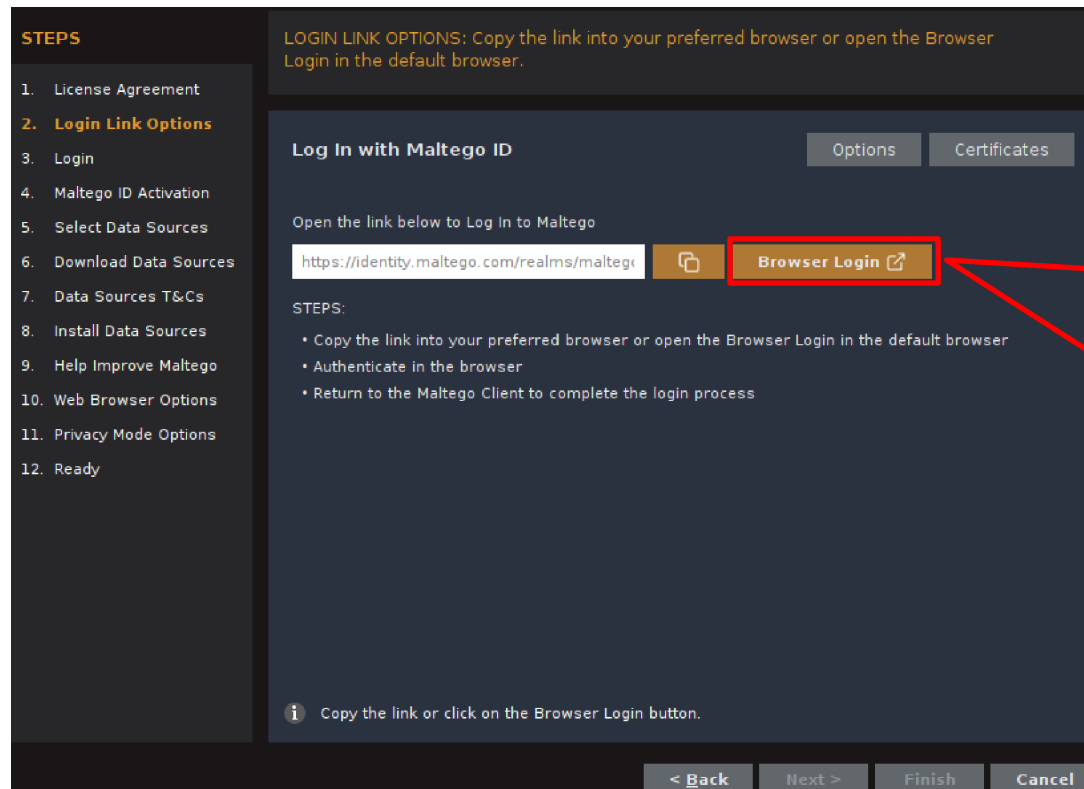
☒ Accept

< Back **Next >** Finish Cancel

Utilizzo di Crawler

Maltego – Primo Avvio

- È necessario attivare il prodotto, utilizzando le credenziali inserite in fase di registrazione



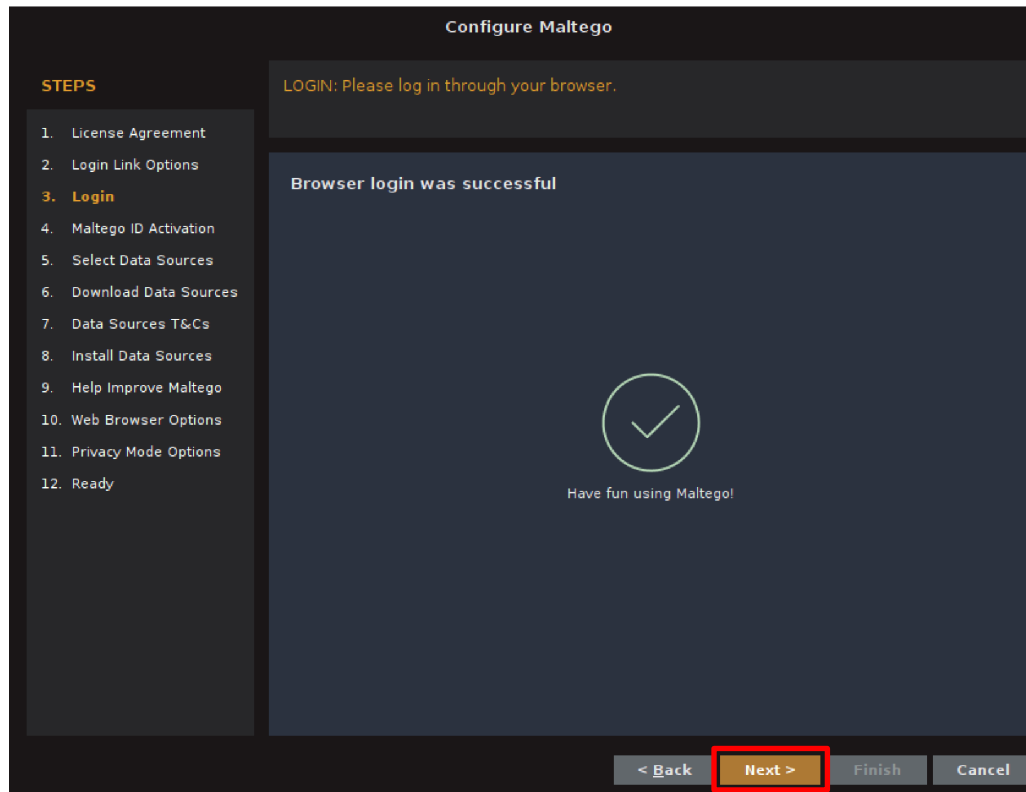
**Effettuare il login
tramite browser,
utilizzando le
credenziali
inserite durante
la registrazione**



Utilizzo di Crawler

Maltego – Primo Avvio

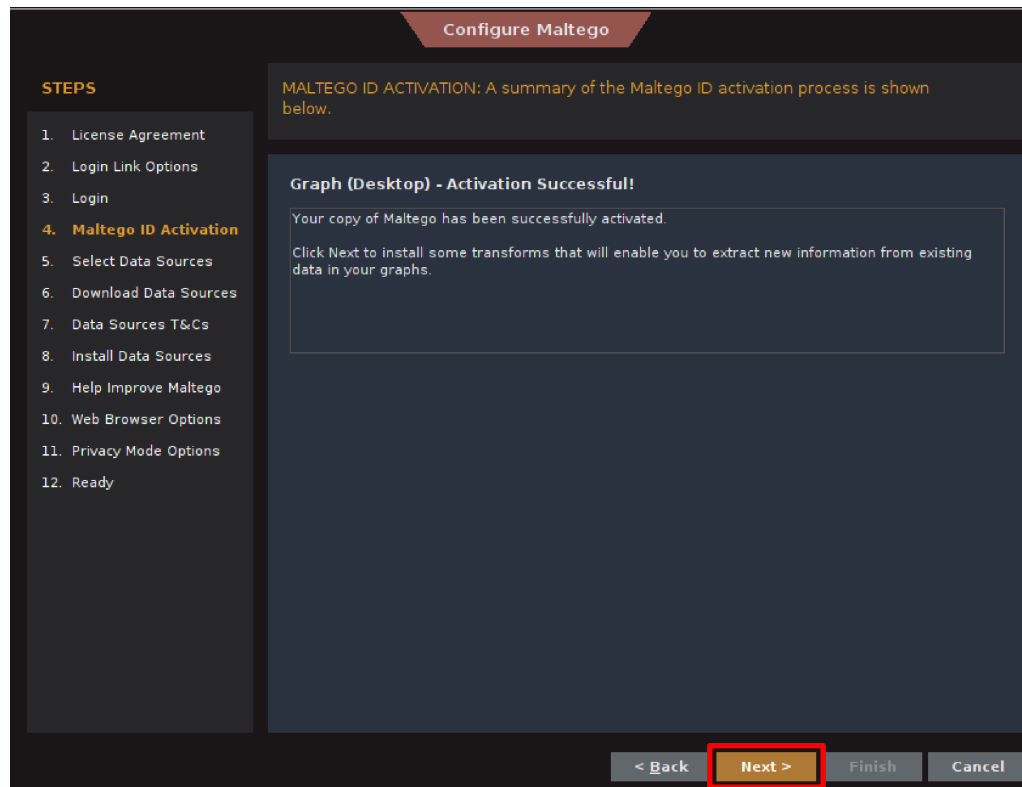
- È necessario attivare il prodotto, utilizzando le credenziali inserite in fase di registrazione



Utilizzo di Crawler

Maltego – Primo Avvio

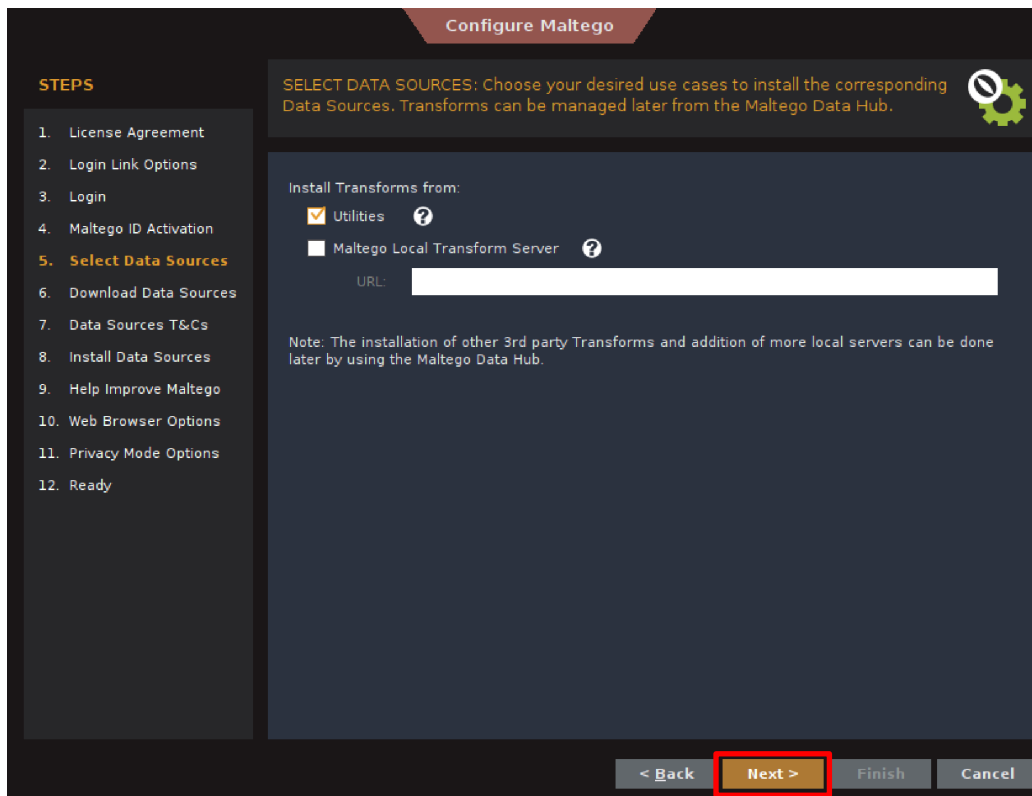
- È necessario attivare il prodotto, utilizzando le credenziali inserite in fase di registrazione



Utilizzo di Crawler

Maltego – Primo Avvio

- È necessario attivare il prodotto, utilizzando le credenziali inserite in fase di registrazione

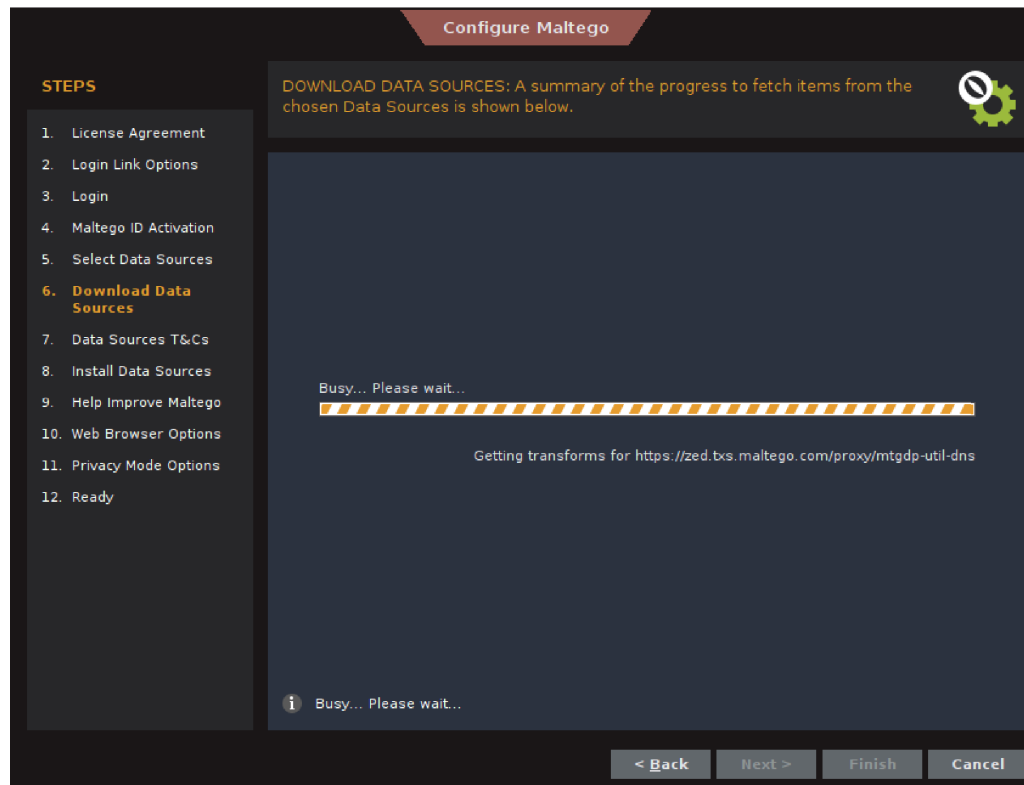


The screenshot shows the 'Configure Maltego' window. On the left, a 'STEPS' list includes: 1. License Agreement, 2. Login Link Options, 3. Login, 4. Maltego ID Activation, 5. **Select Data Sources**, 6. Download Data Sources, 7. Data Sources T&Cs, 8. Install Data Sources, 9. Help Improve Maltego, 10. Web Browser Options, 11. Privacy Mode Options, and 12. Ready. The main area is titled 'SELECT DATA SOURCES: Choose your desired use cases to install the corresponding Data Sources. Transforms can be managed later from the Maltego Data Hub.' It contains the section 'Install Transforms from:' with two options: 'Utilities' (checked with a checkbox and a help icon) and 'Maltego Local Transform Server' (unchecked with a checkbox and a help icon). Below these is a 'URL:' text input field. A note at the bottom states: 'Note: The installation of other 3rd party Transforms and addition of more local servers can be done later by using the Maltego Data Hub.' At the bottom right, there are four buttons: '< Back', 'Next >' (highlighted with a red box), 'Finish', and 'Cancel'.

Utilizzo di Crawler

Maltego – Primo Avvio

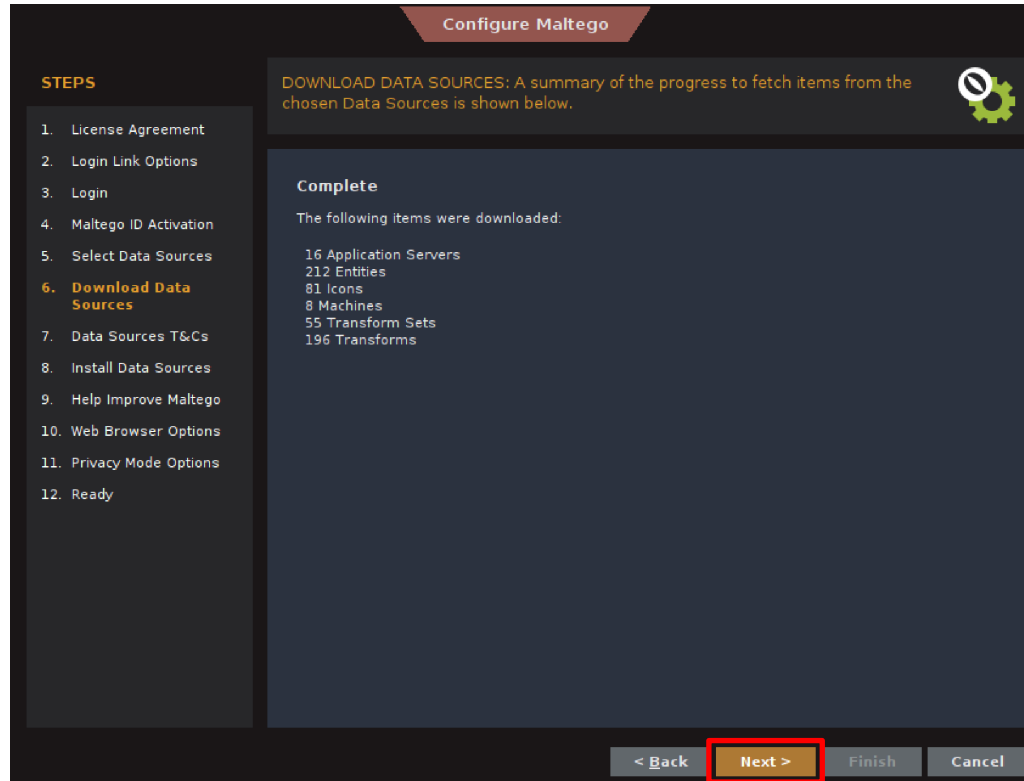
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Utilizzo di Crawler

Maltego – Primo Avvio

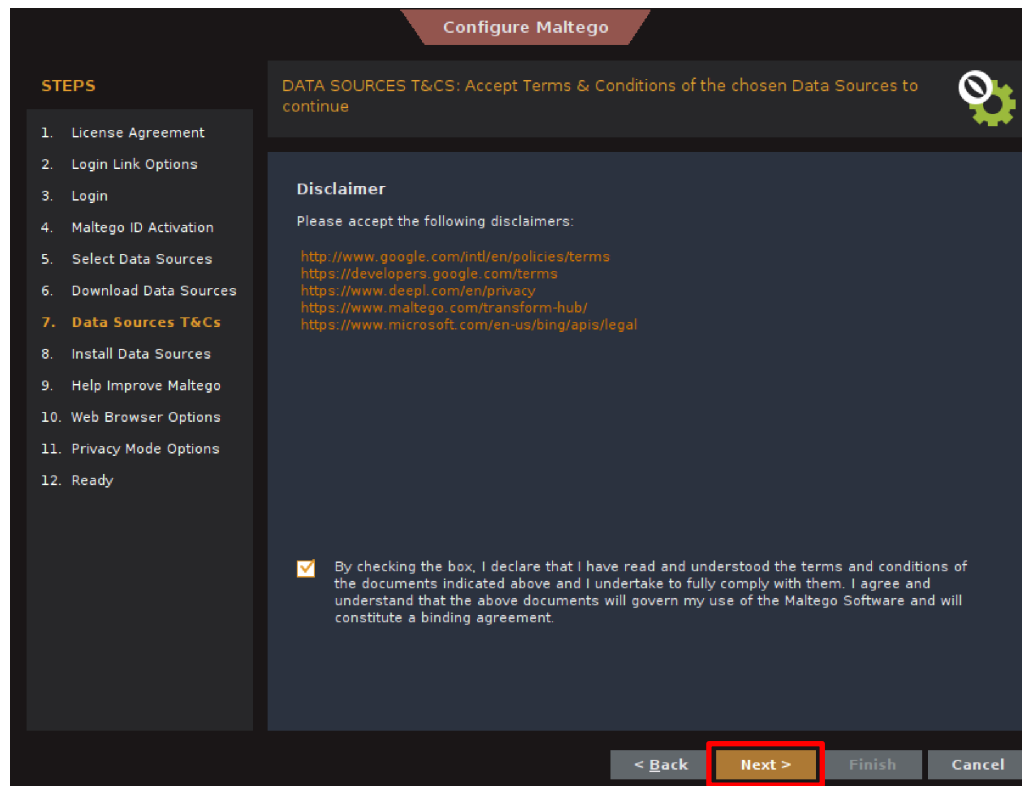
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Utilizzo di Crawler

Maltego – Primo Avvio

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Utilizzo di Crawler

Maltego – Primo Avvio

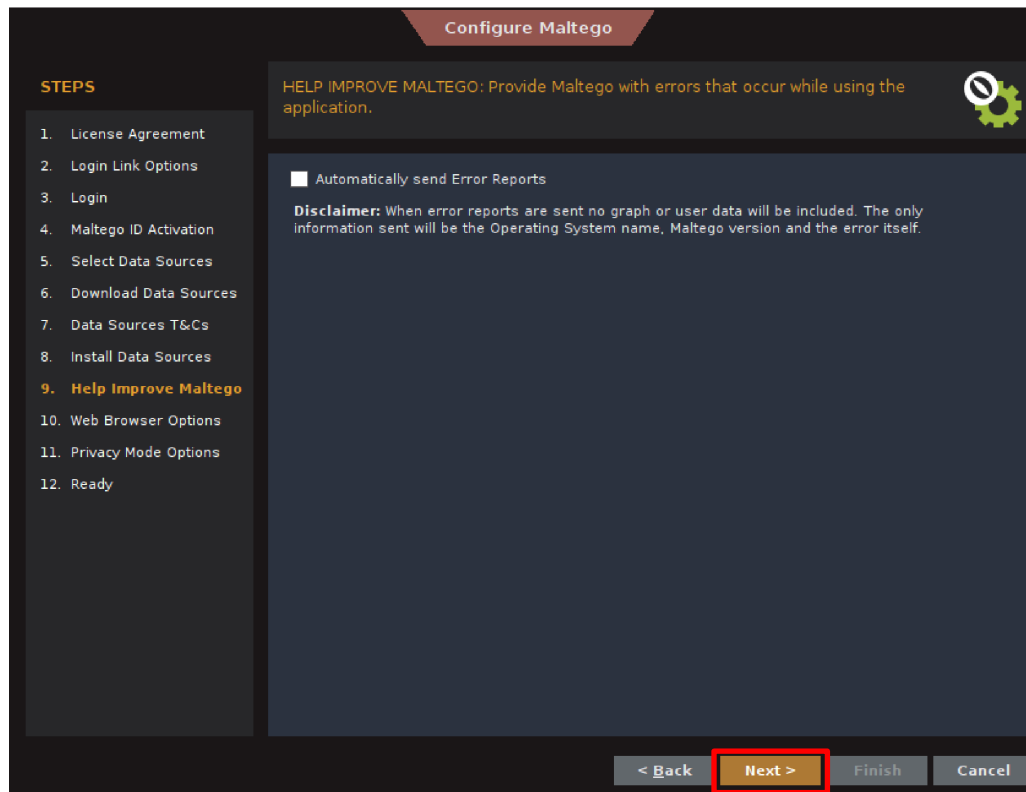
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Utilizzo di Crawler

Maltego – Primo Avvio

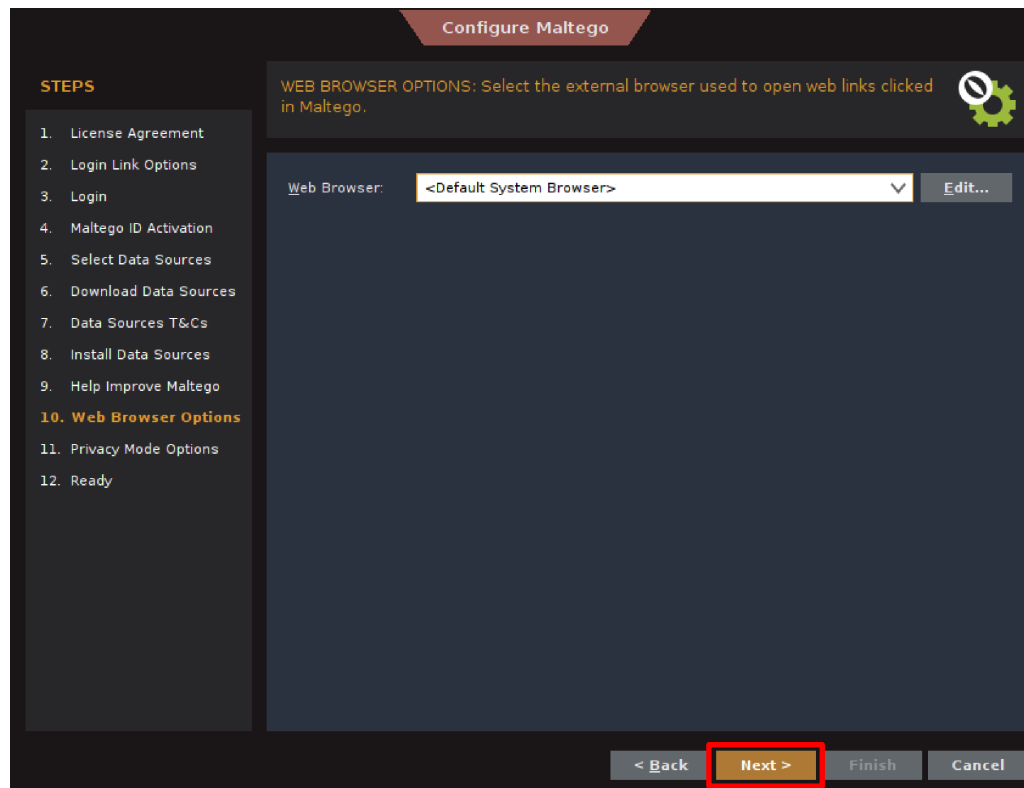
- È necessario attivare il prodotto, utilizzando le credenziali inserite in fase di registrazione



Utilizzo di Crawler

Maltego – Primo Avvio

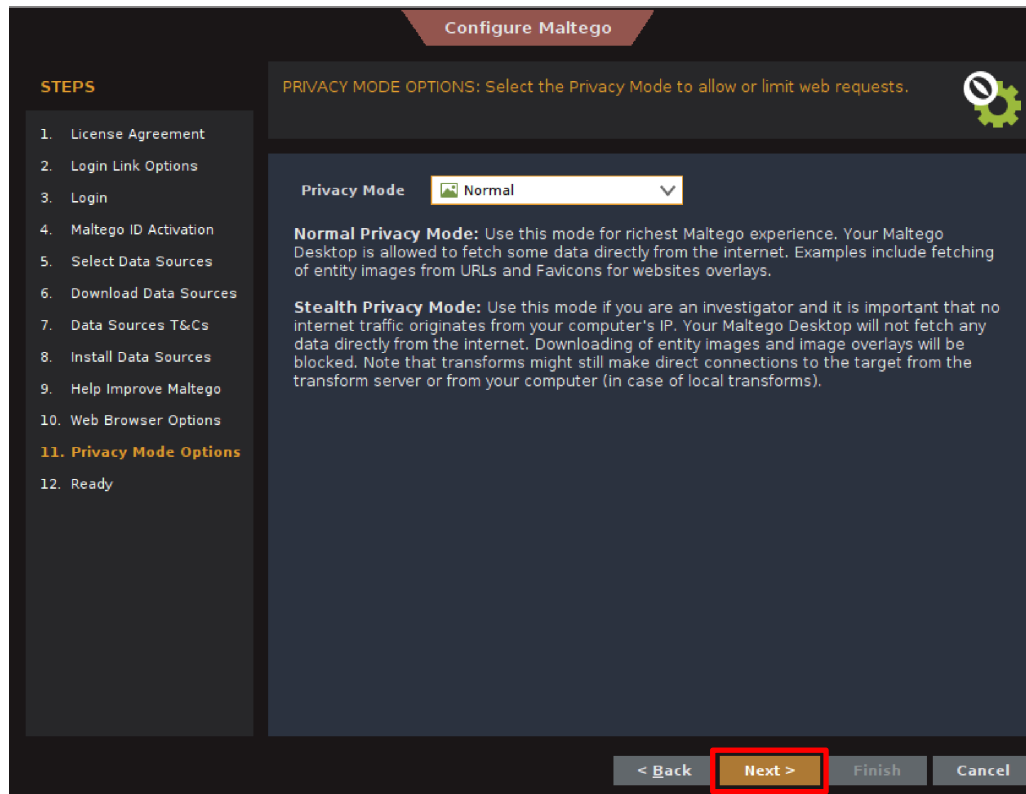
- È necessario attivare il prodotto, utilizzando le credenziali inserite in fase di registrazione



Utilizzo di Crawler

Maltego – Primo Avvio

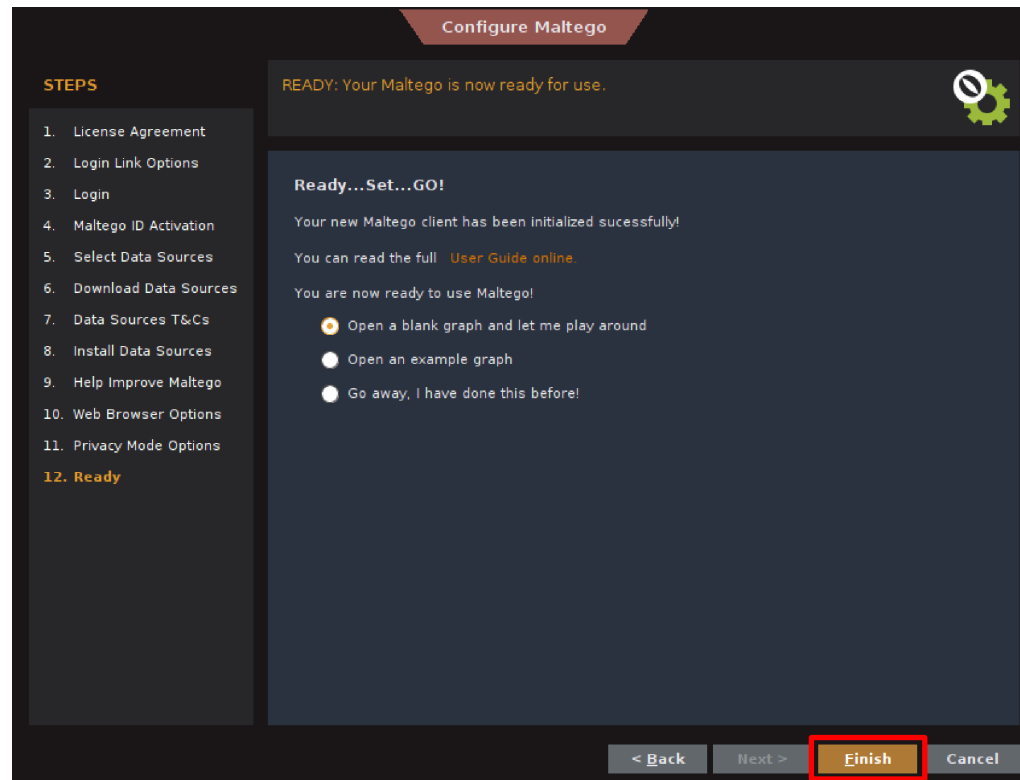
- È necessario attivare il prodotto, utilizzando le credenziali inserite in fase di registrazione



Utilizzo di Crawler

Maltego – Primo Avvio

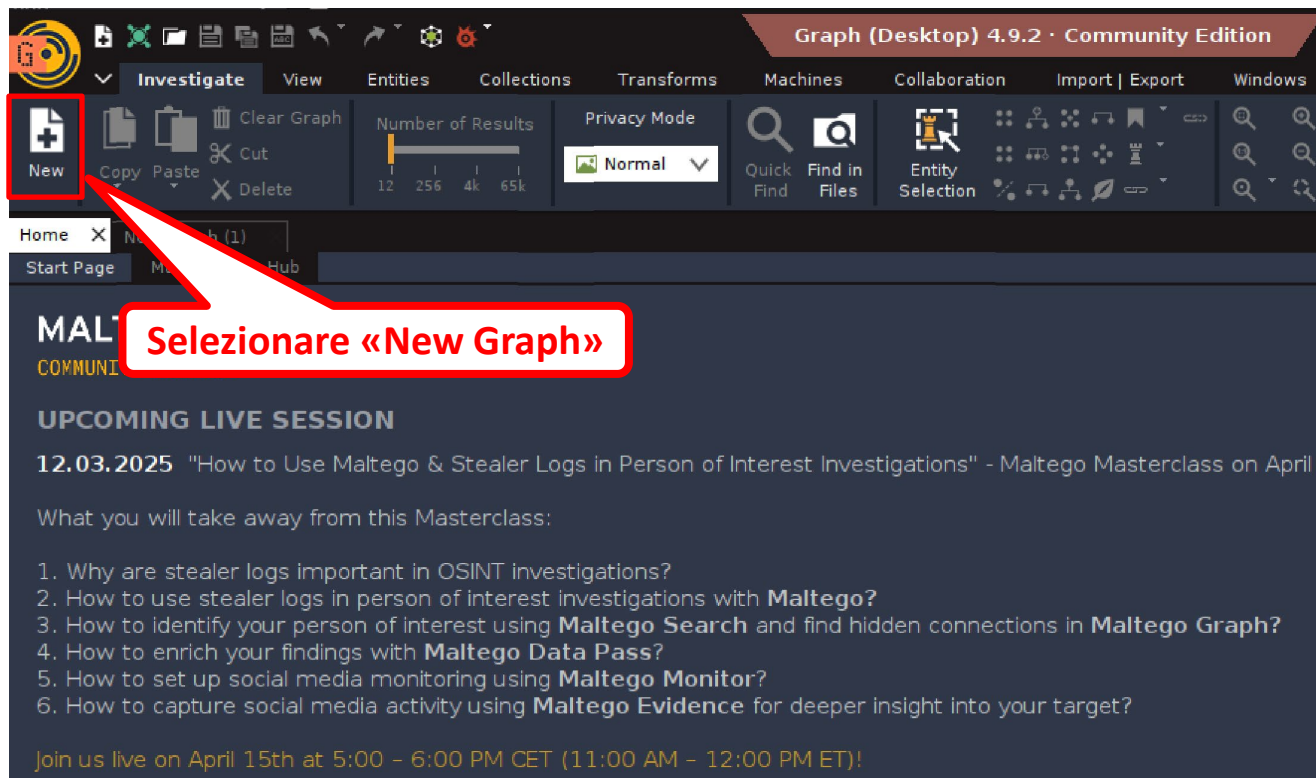
- È necessario attivare il prodotto, utilizzando le credenziali inserite in fase di registrazione



Utilizzo di Crawler

Maltego – Primo Avvio

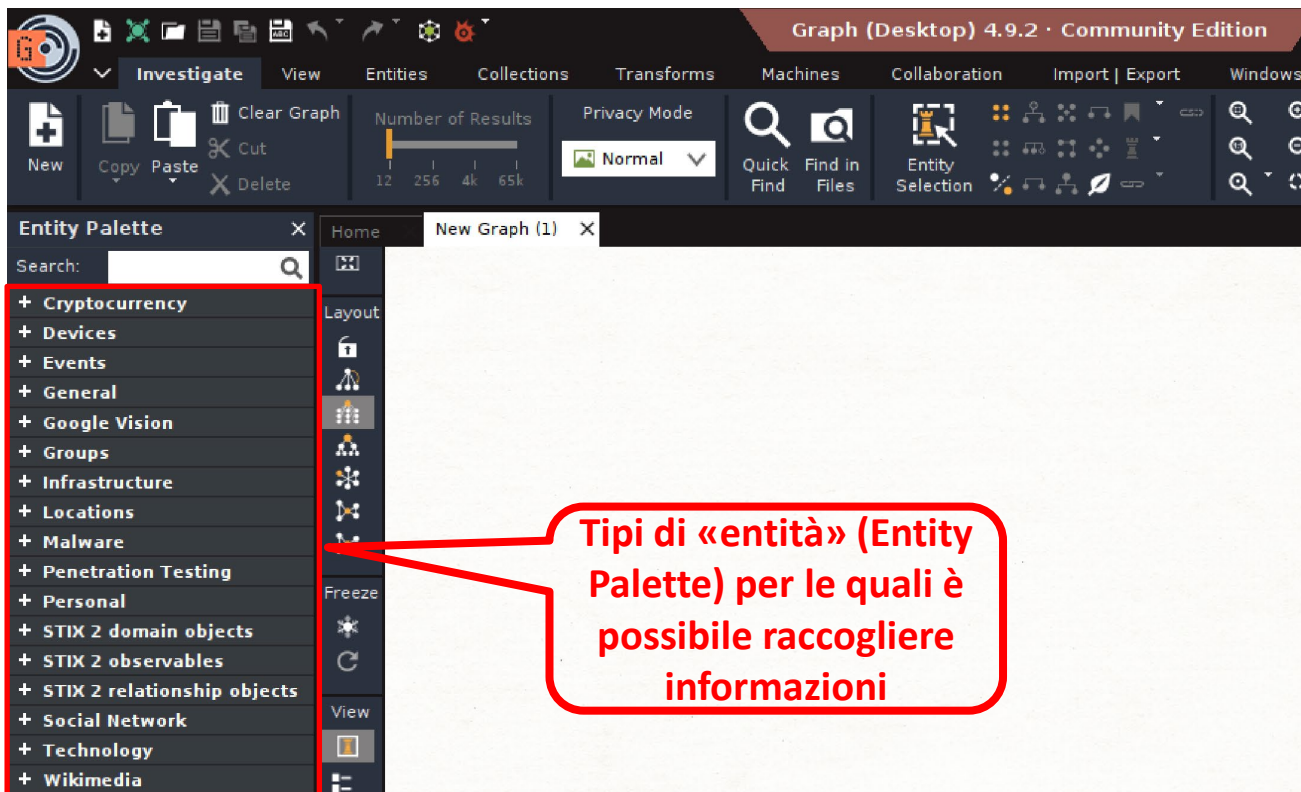
- Una volta terminata la registrazione del prodotto, è possibile tornare alla schermata di Maltego per poterlo utilizzare



Utilizzo di Crawler

Maltego – Primo Avvio

- Una volta terminata la registrazione del prodotto, è possibile tornare alla schermata di Maltego per poterlo utilizzare



Entity Palette

Utilizzo di Crawler

Maltego – Entity Palette

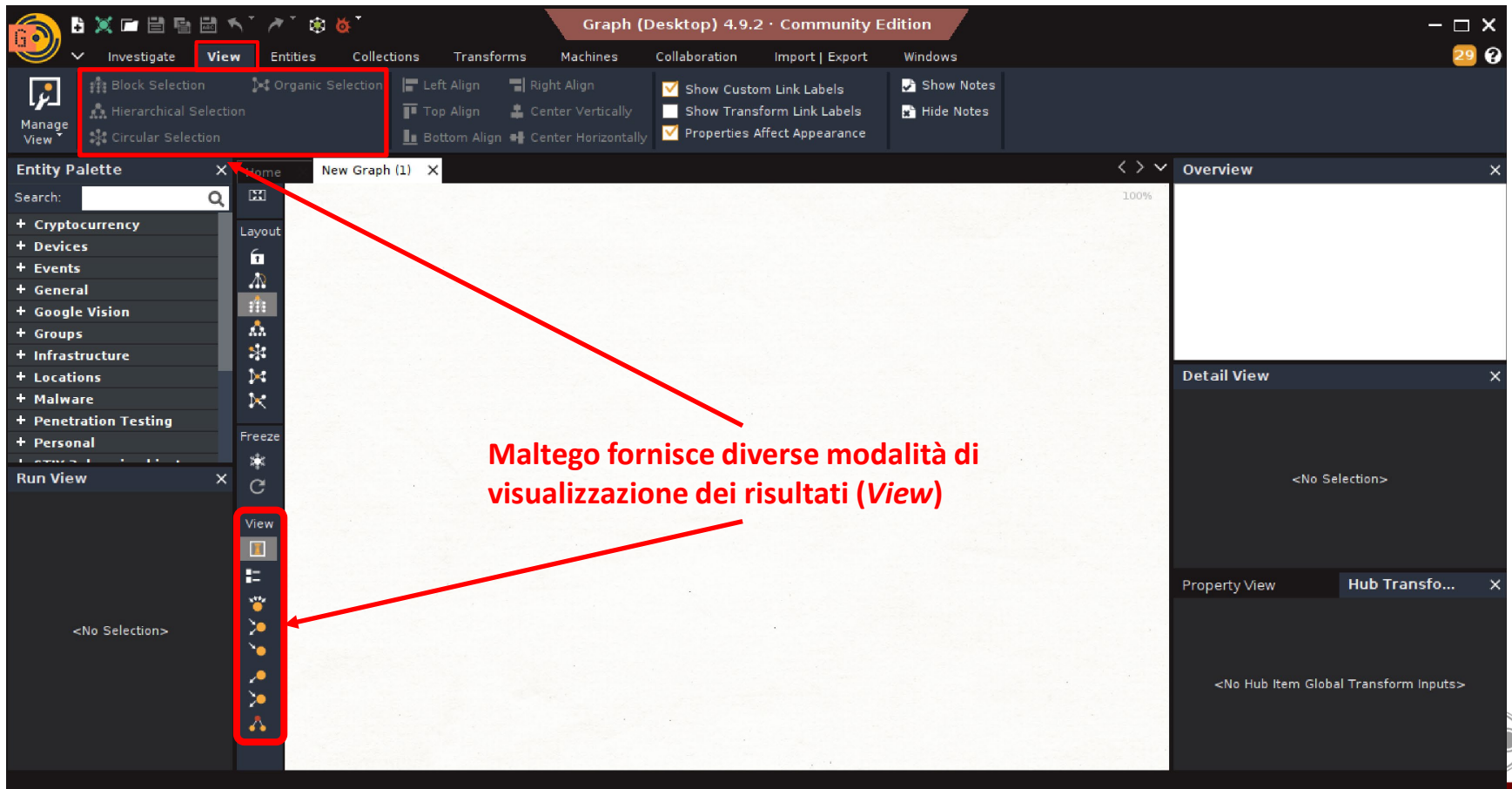
- Le entità (*Entity Palette*) sono suddivise in 17 gruppi
 - Cryptocurrency
 - Devices
 - Events
 - Groups
 - General
 - Google Vision
 - Infrastructure
 - Locations
 - Malware
 - Penetration Testing
 - Personal
 - Social Network
 - Technology
 - Wikimedia
 - STIX 2 domain objects
 - STIX 2 observables
 - STIX 2 relationship objects



MALTEGO

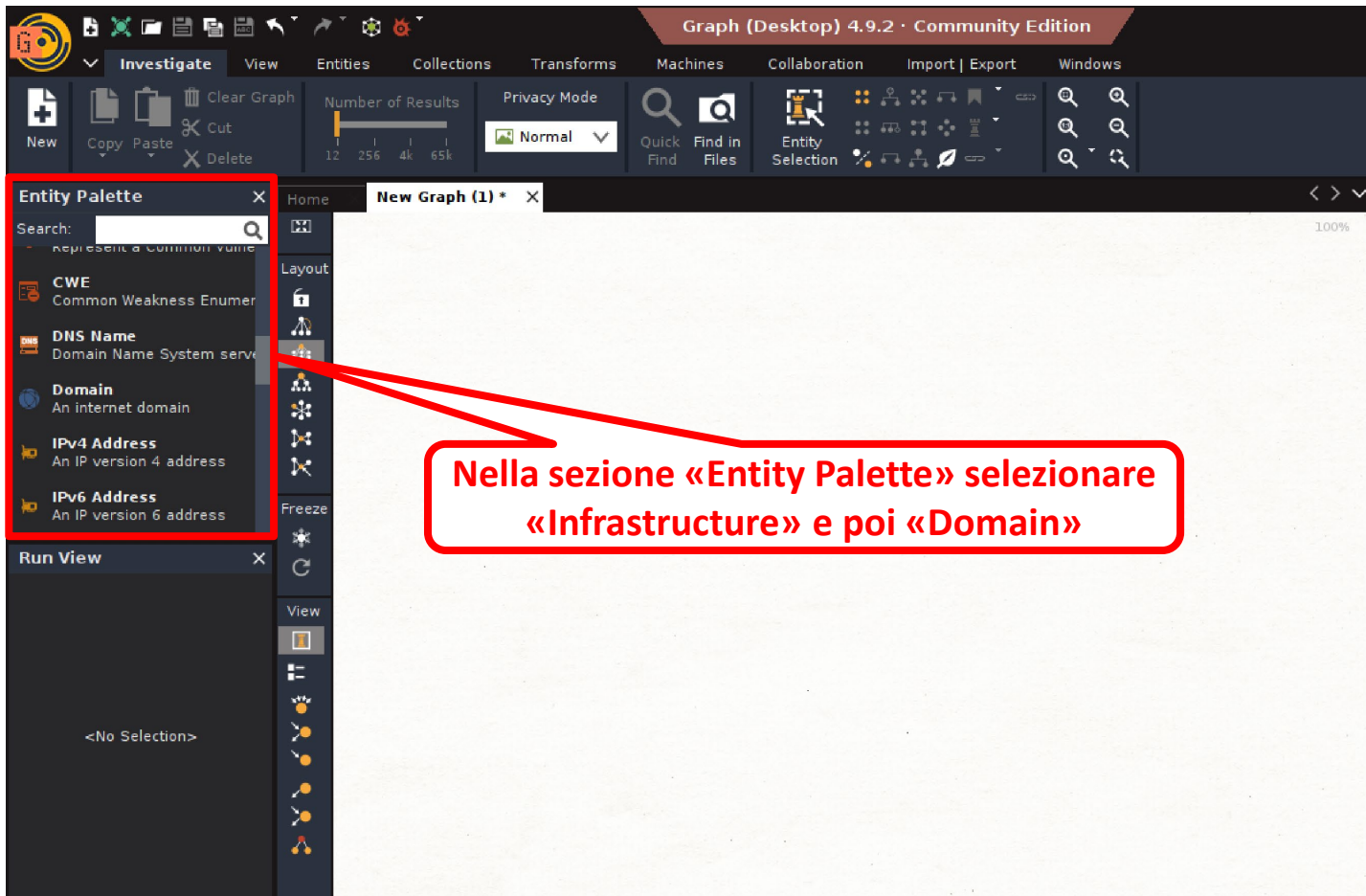
Utilizzo di Crawler

Maltego – View



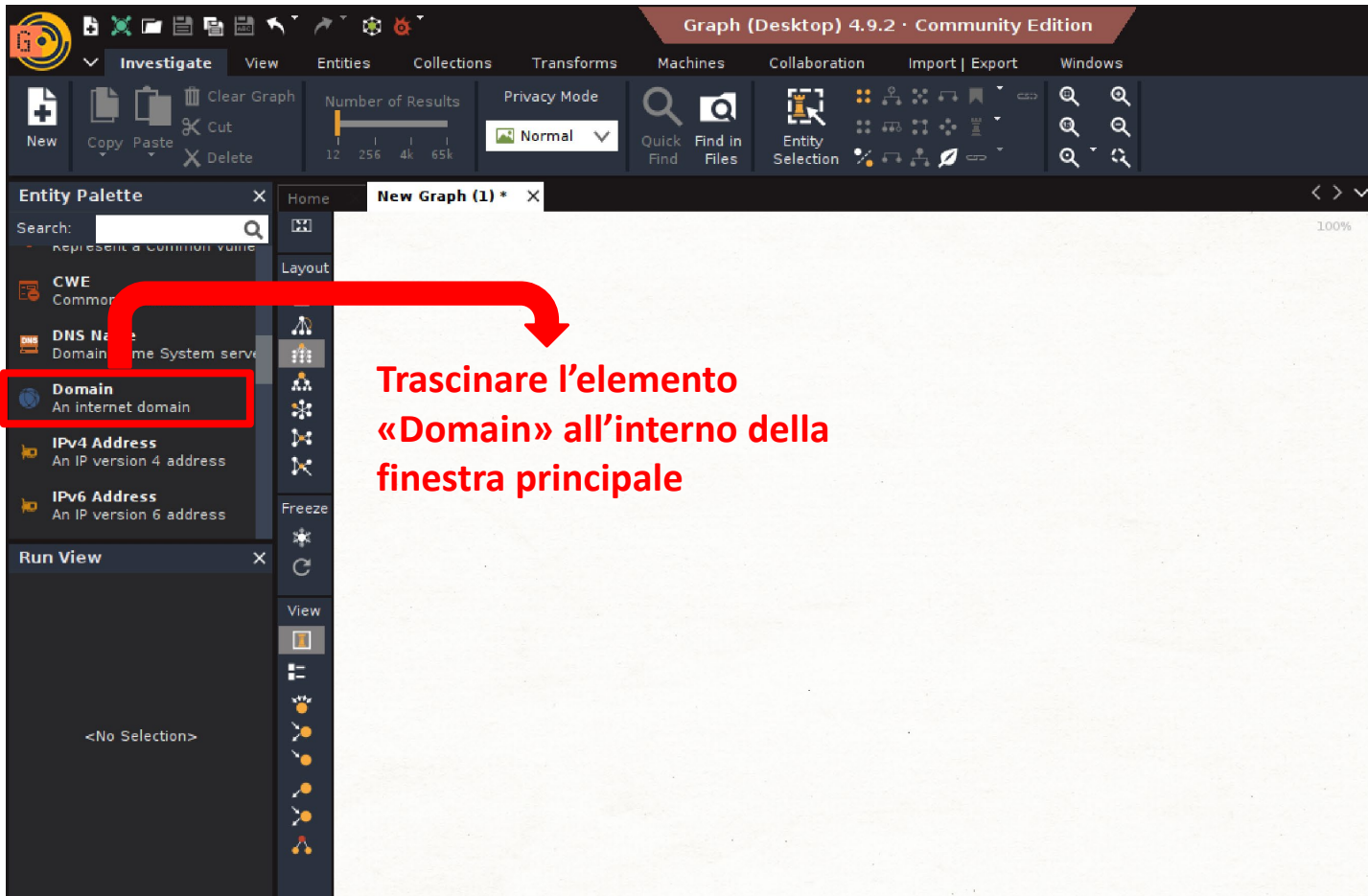
Utilizzo di Crawler

Maltego – Esempio 1



Utilizzo di Crawler

Maltego – Esempio 1



Utilizzo di Crawler

Maltego – Esempio 1

Graph (Desktop) 4.9.2 · Community Edition

Investigate View Entities Collections Transforms Machines Collaboration Import | Export Windows

Entity Palette

Search: []

+ Penetration Testing

+ Personal

+ STIX 2 domain objects

+ STIX 2 observables

+ STIX 2 relationship obje...

+ Social Network

+ Technology

+ Tracking

- Wikimedia

Page

Wikimedia Page

Run View

+ Transforms

- Machines

Company Stalker

This machine wil...

Find Wikipedia...

This machine tak...

Footprint L1

This performs a ...

Footprint L2

This performs a ...

Footprint L3

This performs a ...

Footprint XXL

This machine is ...

Layout

Freeze

View

Overview

Detail View

Domain

maltego.Domain

maltego.com

Property View

Hub Transform Inputs

Properties

Type

Domain

Domain Name

maltego.com

WHOIS Info

Graph info

Weight

0

Incoming

0

Outgoing

0

Bookmark

maltego.com

Nella finestra principale comparirà un dominio, chiamato `maltego.com`



MALTEGO

Utilizzo di Crawler

Maltego – Esempio 1

Graph (Desktop) 4.9.2 · Community Edition

Entity Palette

Search

+ Penetration Testing

+ Personal

+ STIX 2 domain objects

+ STIX 2 observables

+ STIX 2 relationship obje...

+ Social Network

+ Technology

+ Tracking

- Wikimedia

Page

Wikimedia Page

Run View

+ Transforms

- Machines

Company Stalker

This machine wil...

Find Wikipedia...

This machine tak...

Footprint L1

This performs a ...

Footprint L2

This performs a ...

Footprint L3

This performs a ...

Footprint XXL

This machine is ...

Facendo doppio click su **maltego.com** è possibile cambiare il nome del dominio

➤ Useremo **unisa.it**

Detail View

Domain

maltego Domain

maltego.com

Property View

Hub Transform Inputs

Properties

Type

Domain

Domain Name

maltego.com

WHOIS Info

Graph info

Weight

0

Incoming

0

Outgoing

0

Bookmark

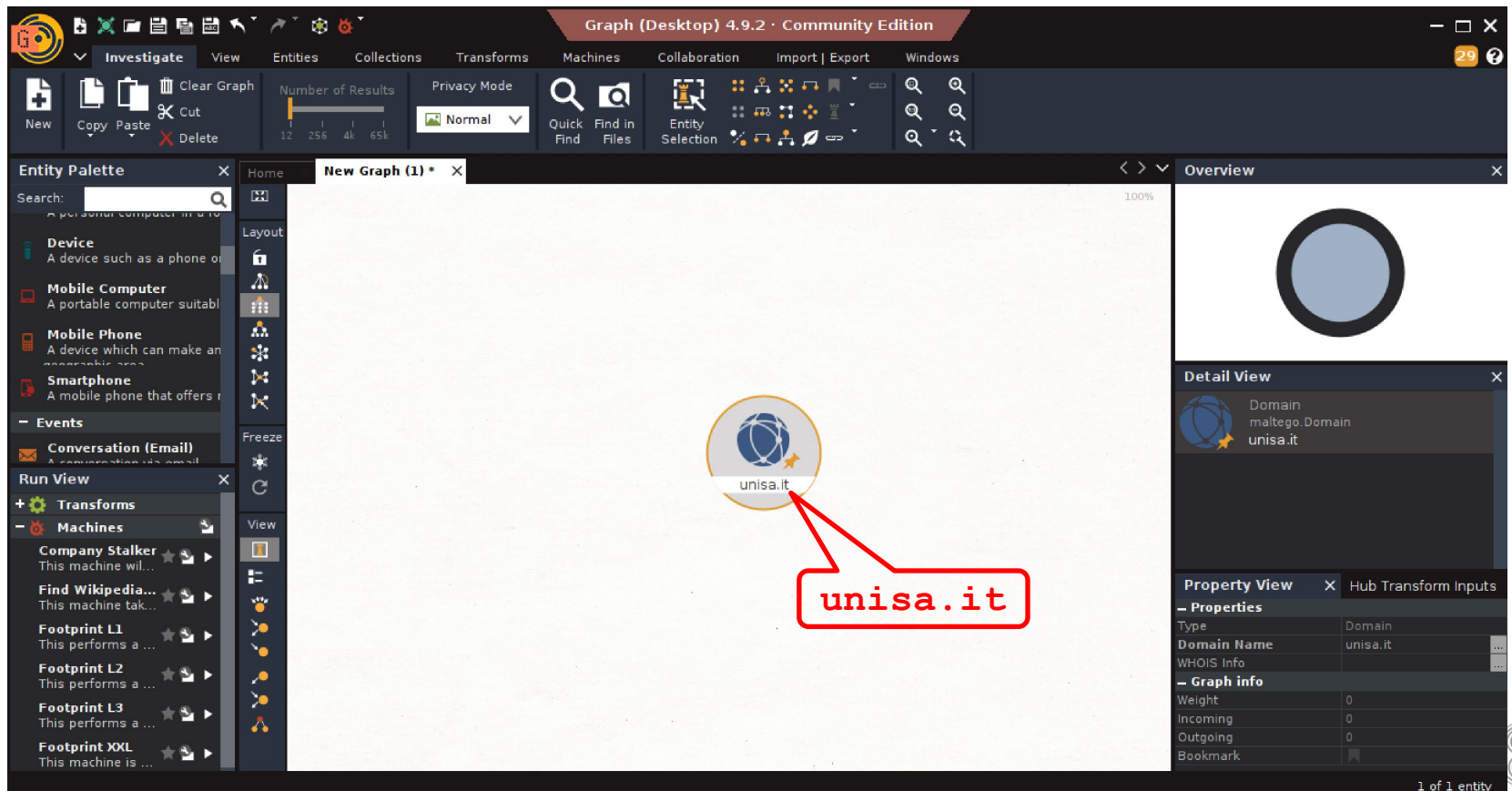
0



MALTEGO

Utilizzo di Crawler

Maltego – Esempio 1

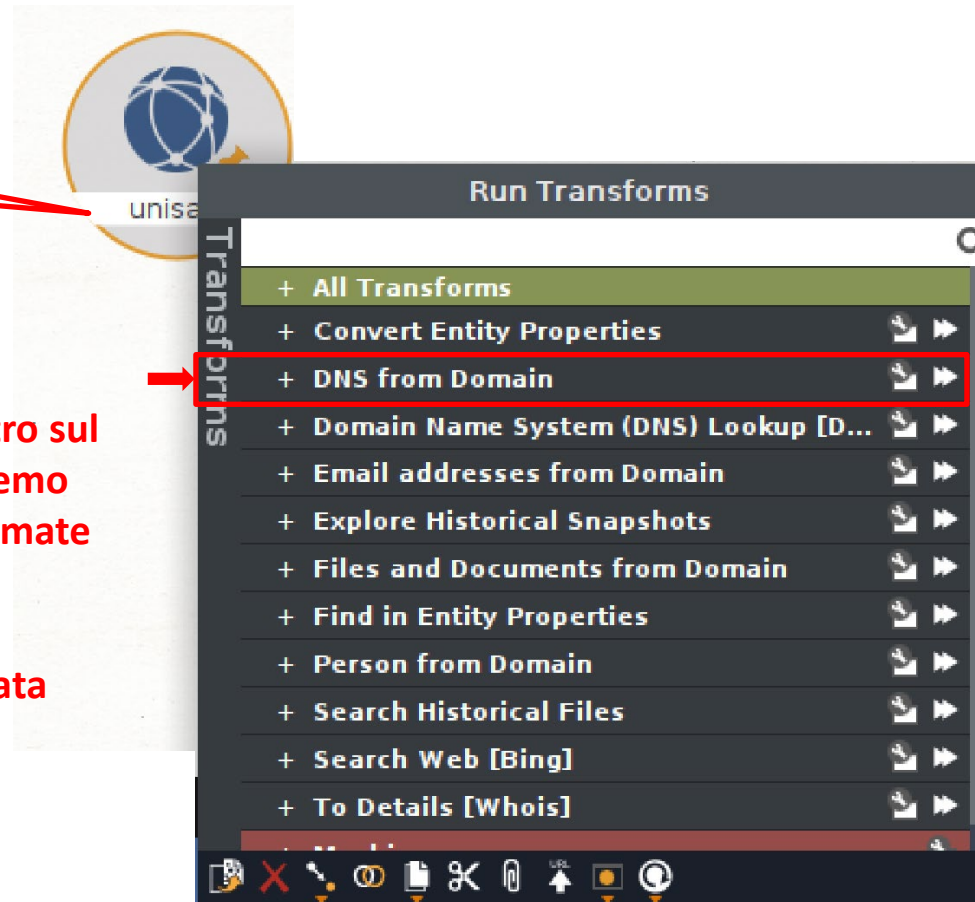


MALTEGO

Utilizzo di Crawler

Maltego – Esempio 1

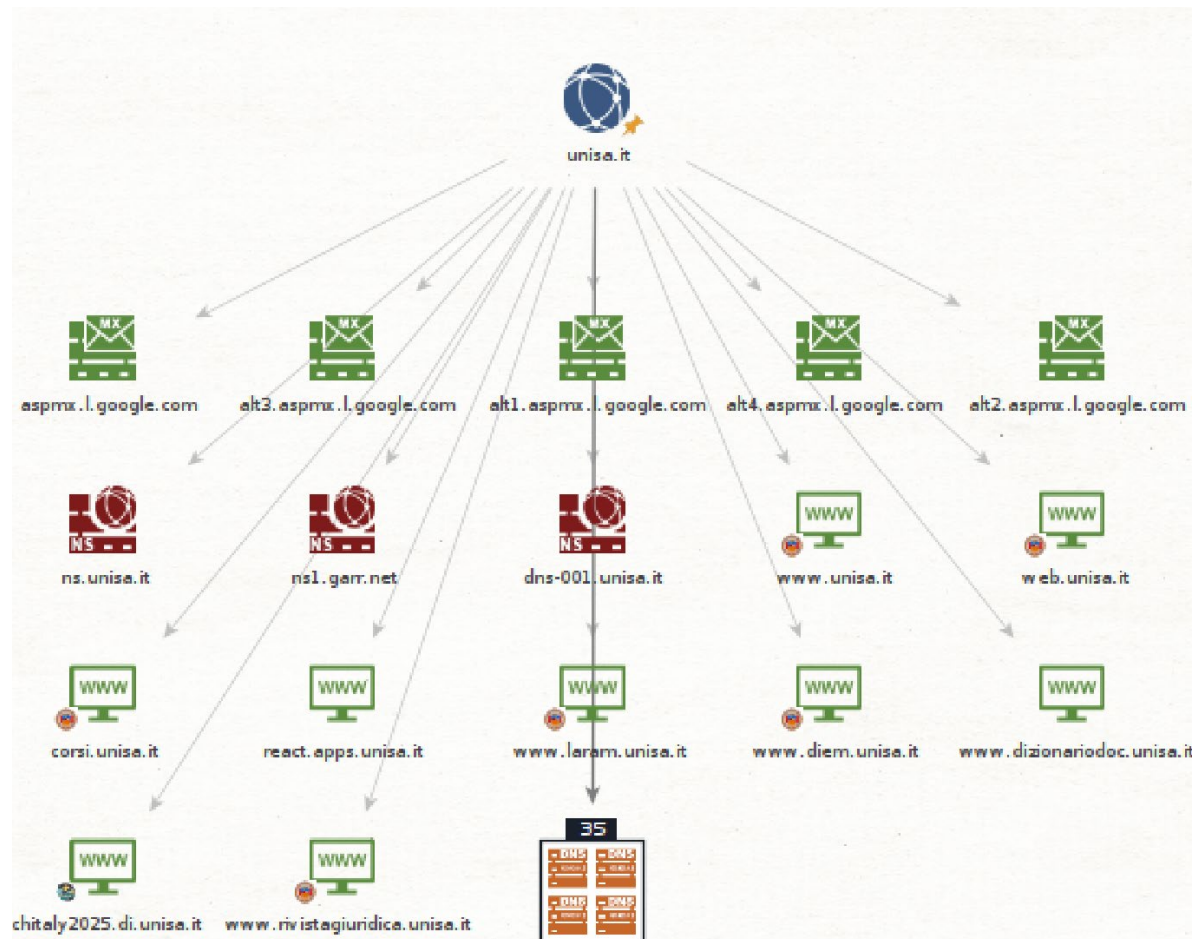
unisa.it



- Cliccando con il tasto destro sul dominio di interesse potremo accedere a tutte le trasformate applicabili su di esso
- In questo esempio applicheremo la trasformatata «DNS from Domain»

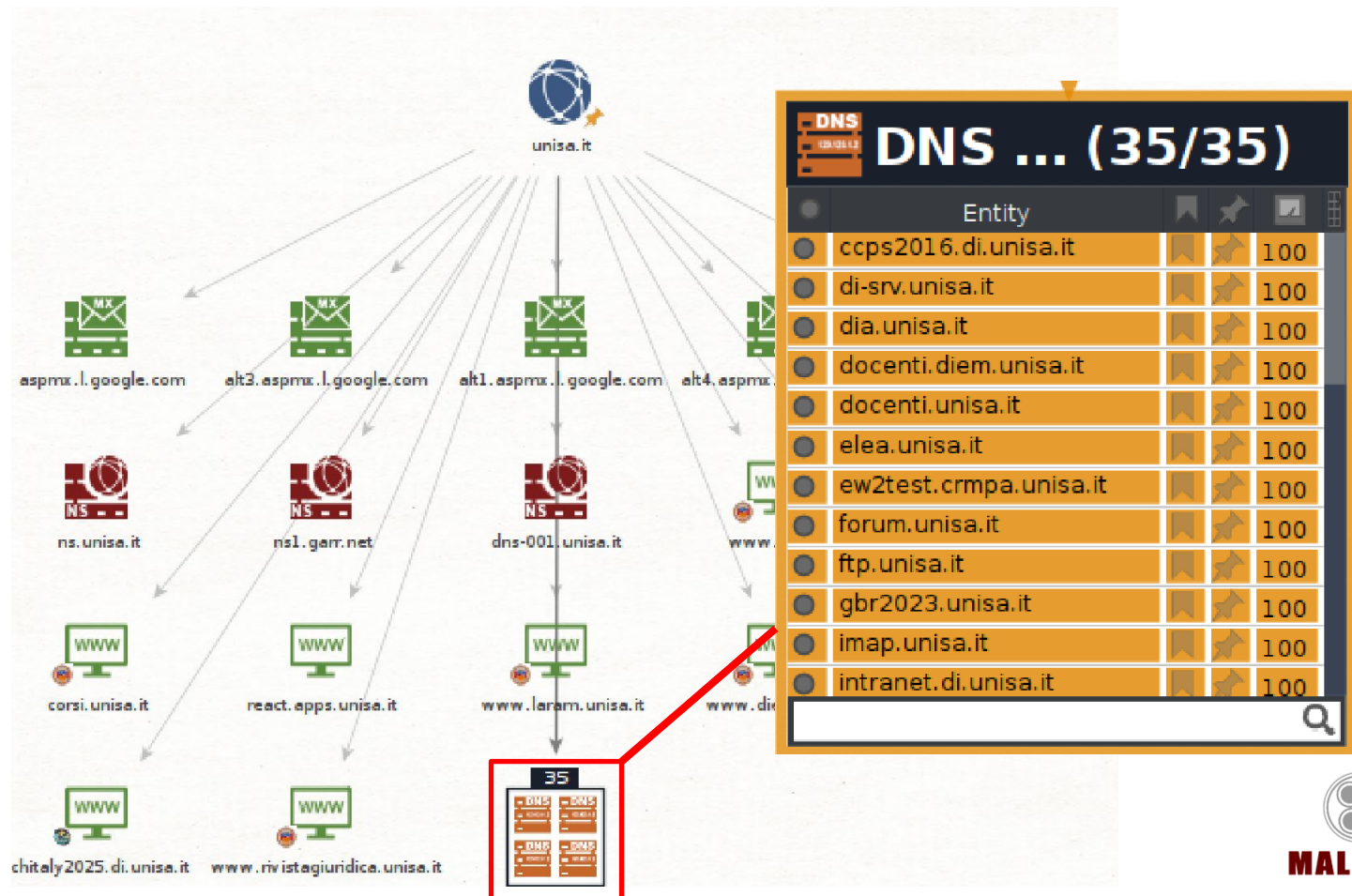
Utilizzo di Crawler

Maltego – Esempio 1



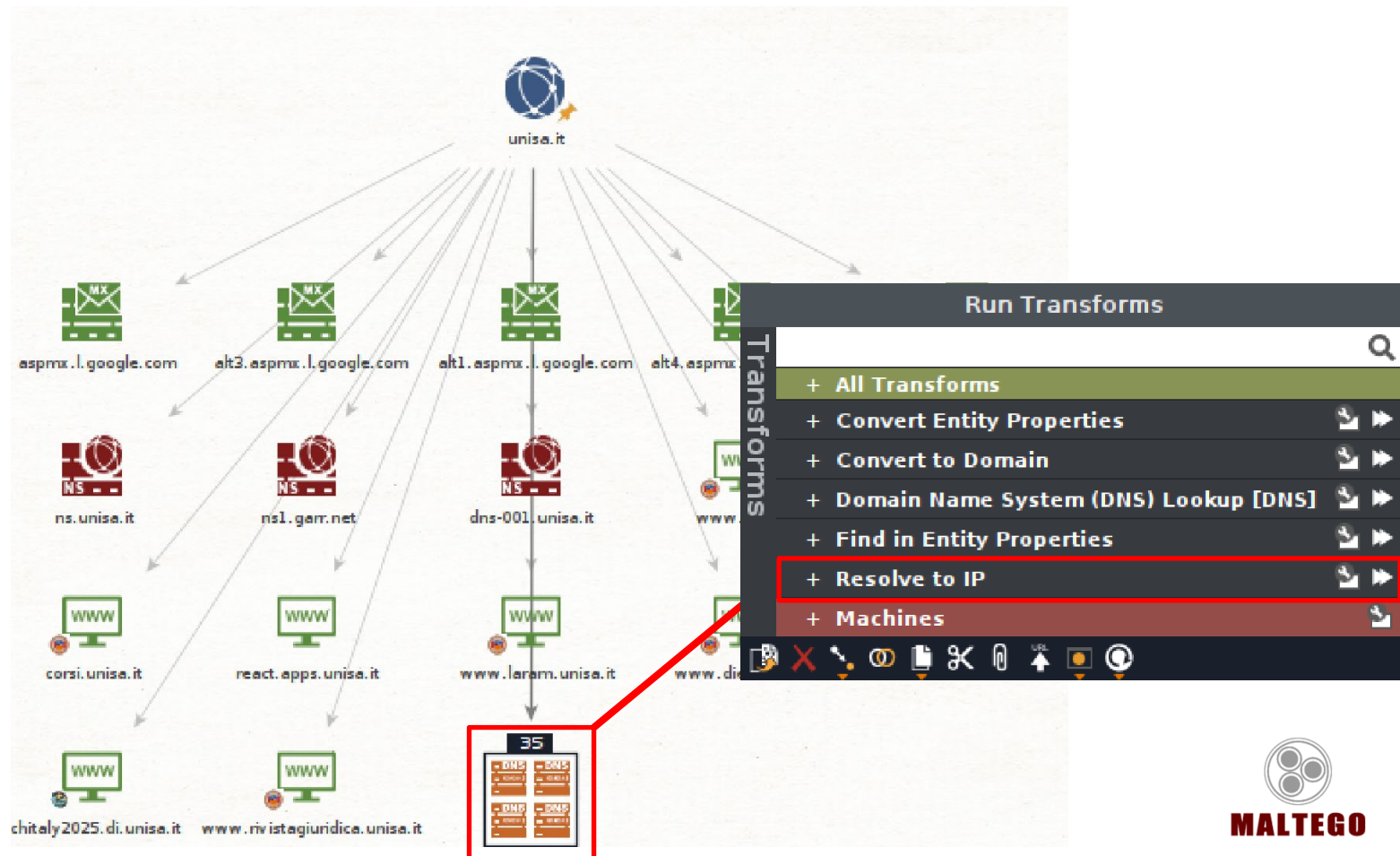
Utilizzo di Crawler

Maltego – Esempio 1



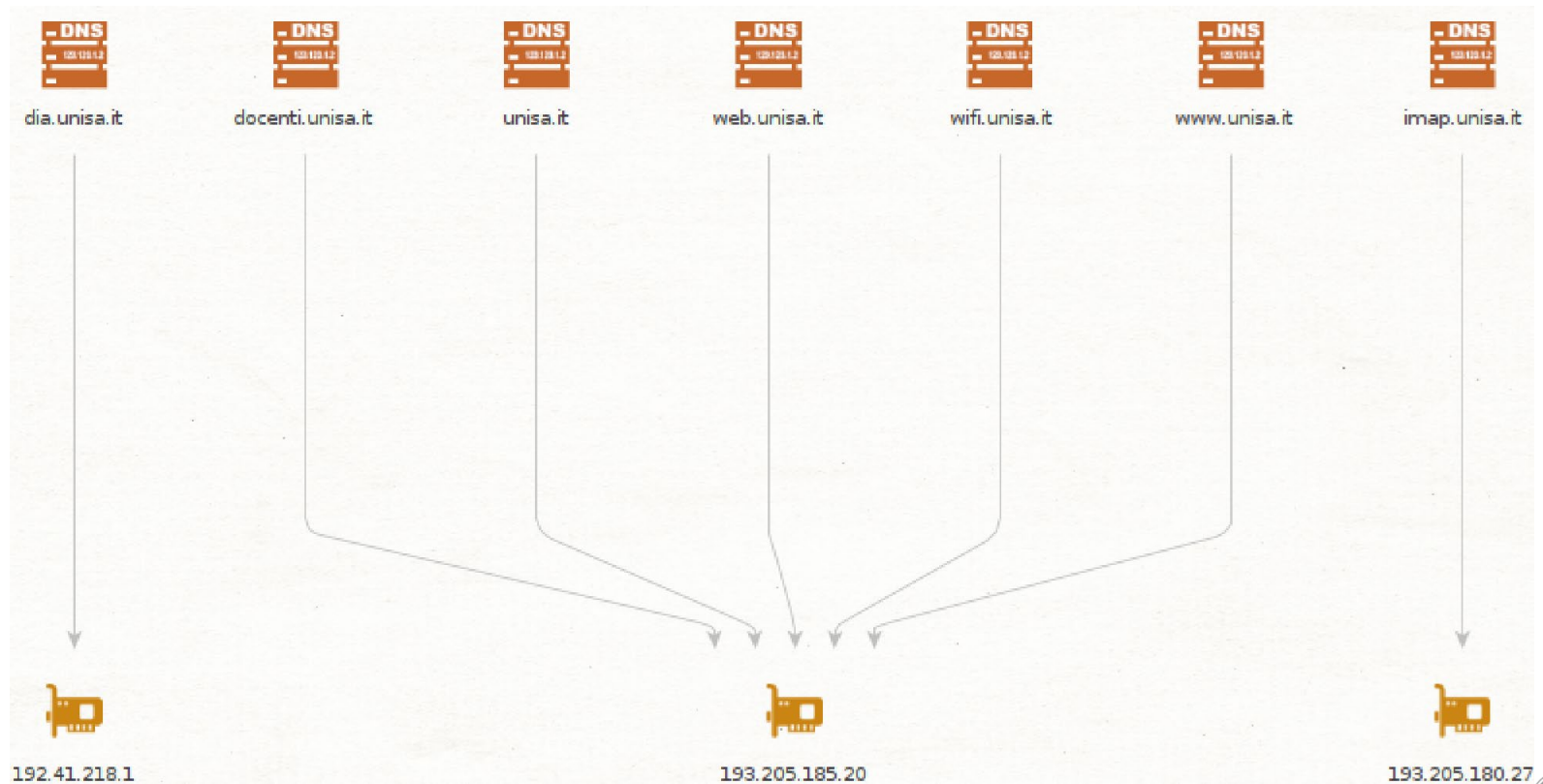
Utilizzo di Crawler

Maltego – Esempio 2



Utilizzo di Crawler

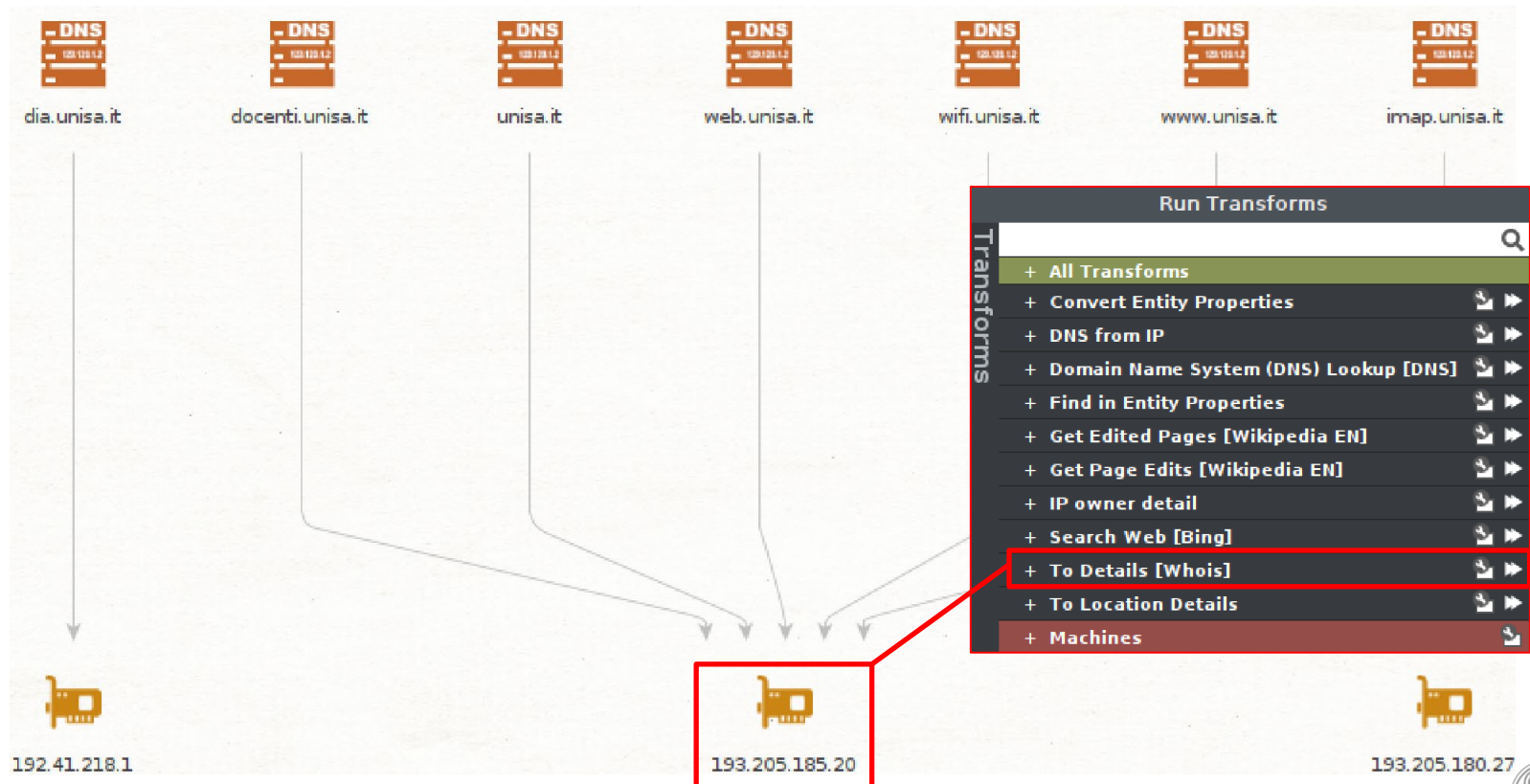
Maltego – Esempio 2



MALTEGO

Utilizzo di Crawler

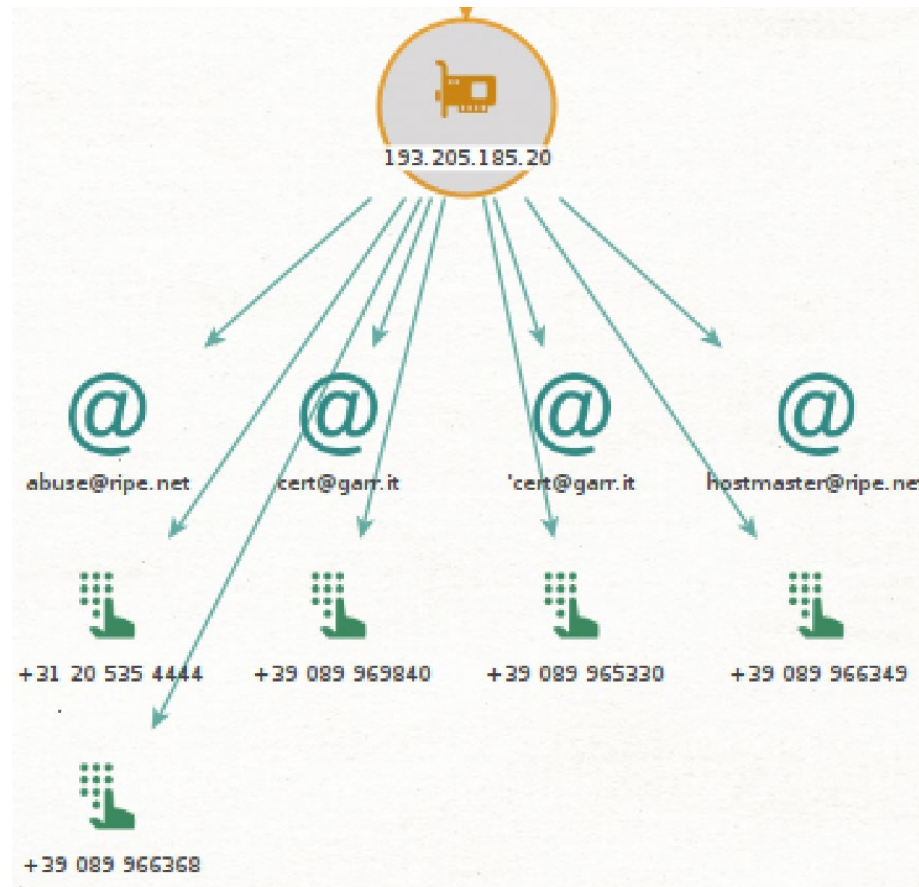
Maltego – Esempio 3



MALTEGO

Utilizzo di Crawler

Maltego – Esempio 3



Utilizzo di Crawler

Maltego – Esempio 4

The screenshot displays the Maltego Desktop 4.9.2 Community Edition interface. The main graph area shows a single entity, 'owasp.org', represented by a blue globe icon. A red speech bubble points to this entity with the text 'Dominio owasp.org'. On the left, the 'Entity Palette' lists various entity types such as Domain, Page, Website, Netblock, DNS Name, Cryptocurrency, and Bitcoin Cash Address. The top menu bar includes options like Investigate, View, Entities, Collections, Transforms, Machines, Collaboration, Import, Export, and Windows. The 'Run Transforms' dialog box is open on the right, showing a list of transforms. The transform 'Email addresses from Domain' is highlighted with a red rectangle. The bottom panel shows the 'Output - Transform Output' log with various system messages and warnings.

Graph (Desktop) 4.9.2 · Community Edition

Investigate View Entities Collections Transforms Machines Collaboration Import Export Windows

Entity Palette

Search: [Search Bar]

Recently Used *

- Domain: An internet domain
- Page: Wikimedia Page
- Website: An internet website
- Netblock: A range of IP version 4 address
- DNS Name: Domain Name System server name
- Cryptocurrency
- Bitcoin Cash Address

Run View

- Transforms
- Machines

Layout

Freeze

View

Transforms

- + All Transforms
- + Convert Entity Properties
- + DNS from Domain
- + Domain Name System (DNS) Lookup [D...]
- + Email addresses from Domain
- + Explore Historical Snapshots
- + Files and Documents from Domain
- + Find in Entity Properties
- + Person from Domain
- + Search Historical Files
- + Search Web [Bing]
- + To Details [Whois]

Hub Transform Inputs

Domain: owasp.org

Output - Transform Output

```
[4/7/25, 10:34 AM] INFO Running transform [Utilities] To Email Addresses [P...]  
[4/7/25, 10:34 AM] INFO ---[ Unified Credits ]--- : 5 of 200 credits remain  
[4/7/25, 10:34 AM] WARNING HTTP response status code: 500 | Internal Server  
[4/7/25, 10:34 AM] INFO Transform [Utilities] To Email Addresses [PGP] comp  
[4/7/25, 10:36 AM] INFO Running transform [Utilities] To Email Addresses [P...]  
[4/7/25, 10:36 AM] INFO ---[ Unified Credits ]--- : 4 of 200 credits remain  
[4/7/25, 10:36 AM] WARNING timeout (from entity "nsa.gov")  
[4/7/25, 10:36 AM] INFO Transform [Utilities] To Email Addresses [PGP] comp
```



MALTEGO

Utilizzo di Crawler

Maltego – Esempio 4



MALTEGO

Utilizzo di Crawler

Surface Mapping ed Asset Discovery – SpiderFoot

- Strumento di *ricognizione*, basato su OSINT, per la raccolta e l'analisi automatica di dati
 - Interroga oltre 100 fonti di dati per raccogliere informazioni su indirizzi IP, nomi di dominio, indirizzi e-mail e molto altro ancora
- Si integra con numerose fonti di informazioni ed utilizza una vasta gamma di metodi per l'analisi dei dati, rendendoli di facile consultazione
- Grazie ad un Web server integrato può essere utilizzato mediante un'interfaccia Web-based semplice e intuitiva
 - Ma può anche essere anche usato interamente da riga di comando
- È scritto in Python e rilasciato sotto licenza MIT

Utilizzo di Crawler

SpiderFoot

- Per utilizzare lo strumento SpiderFoot è sufficiente specificare l'asset di interesse e scegliere i moduli da abilitare per tale strumento
 - SpiderFoot si occupa automaticamente di raccogliere i dati, per comprendere le caratteristiche dell'asset e come esse sono in relazione
- I risultati restituiti da una scansione tramite SpiderFoot possono rivelare molte informazioni sull'asset
 - Anche eventuali data leak, vulnerabilità o altre informazioni sensibili che potrebbero essere sfruttate durante un processo di penetration testing

Utilizzo di Crawler

SpiderFoot – Esempio

- Spiderfoot può essere avviato da terminale nel modo seguente
 - `sudo spiderfoot -l 127.0.0.1:5001`

```
(kali㉿kali)-[~]
└─$ sudo spiderfoot -l 127.0.0.1:5001
[sudo] password for kali:

*****
Use SpiderFoot by starting your web browser of choice and
browse to http://127.0.0.1:5001/
*****

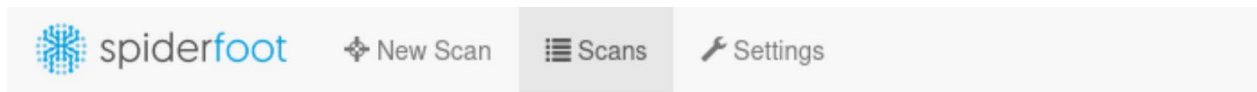
2024-04-08 12:19:15,712 [INFO] sf : Starting web server at 127.0.0.1:5001 .
..
2024-04-08 12:19:15,717 [WARNING] sf :
*****
Warning: passwd file contains no passwords. Authentication disabled.
Please consider adding authentication to protect this instance!
Refer to https://www.spiderfoot.net/documentation/#security.
*****
```

Utilizzo di Crawler

SpiderFoot – Esempio

➤ Spiderfoot può essere utilizzato tramite Web browser, accedendo all'URL seguente

➤ `http://127.0.0.1:5001/`



Scans

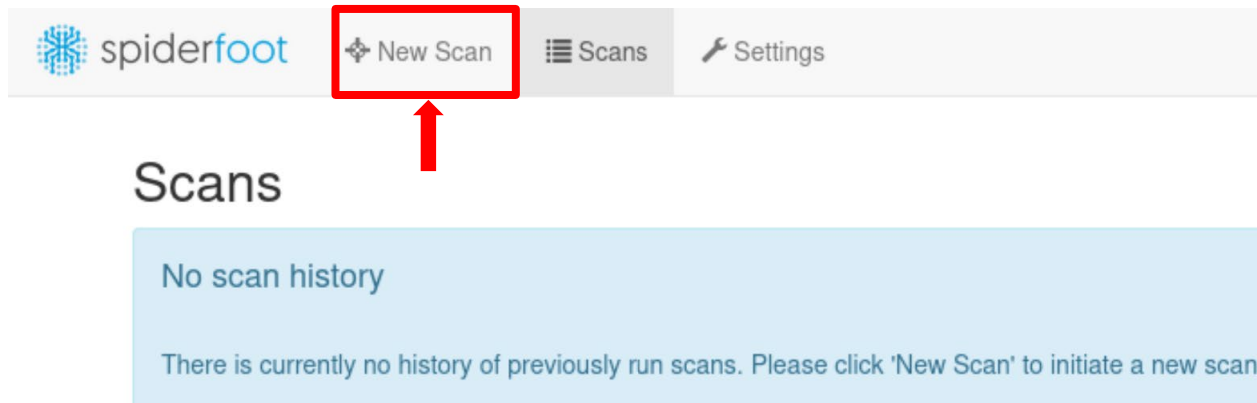
No scan history

There is currently no history of previously run scans. Please click 'New Scan' to initiate a new scan.

Utilizzo di Crawler

SpiderFoot – Esempio

- Configurazione di una nuova scansione



Utilizzo di Crawler

SpiderFoot – Esempio

➤ Scelta del tipo di analisi da effettuare («By Use Case»)

New Scan

Scan Name

Scan Target

? Your scan target may be one of the following. SpiderFoot will automatically detect the target type based on the format of your input:

Domain Name: e.g. <code>example.com</code>	E-mail address: e.g. <code>bob@example.com</code>
IPv4 Address: e.g. <code>1.2.3.4</code>	Phone Number: e.g. <code>+12345678901</code> (E.164 format)
IPv6 Address: e.g. <code>2606:4700:4700::1111</code>	Human Name: e.g. <code>"John Smith"</code> (must be in quotes)
Hostname/Sub-domain: e.g. <code>abc.example.com</code>	Username: e.g. <code>"jsmith2000"</code> (must be in quotes)
Subnet: e.g. <code>1.2.3.0/24</code>	Network ASN: e.g. <code>1234</code>
Bitcoin Address: e.g. <code>1HesYJSP1QqcyPEjnQ9vzBL1wujruNGe7R</code>	

➔ **By Use Case**

By Required Data

By Module

N.B. Ciascuna analisi utilizzerà un determinato insieme di moduli

☒ All
Get anything and everything about the target.

All SpiderFoot modules will be enabled (slow) but every possible piece of information about the target will be obtained and analysed.

☐ Footprint
Understand what information this target exposes to the Internet.

Gain an understanding about the target's network perimeter, associated identities and other information that is obtained through a lot of web crawling and search engine use.

☐ Investigate
Best for when you suspect the target to be malicious but need more information.

Some basic footprinting will be performed in addition to querying of blacklists and other sources that may have information about your target's maliciousness.

☐ Passive
When you don't want the target to even suspect they are being investigated.

As much information will be gathered without touching the target or their affiliates, therefore only modules that do not touch the target will be enabled.

Run Scan Now



Information Gathering

Utilizzo di Crawler

SpiderFoot – Esempio

➤ Scelta del tipo di analisi da effettuare («Footprint»)

New Scan

Scan Name

The name of this scan.

Scan Target

The target of your scan.

📌 Your scan target may be one of the following. SpiderFoot will automatically detect the target type based on the format of your input:

Domain Name: e.g. `example.com`

IPv4 Address: e.g. `1.2.3.4`

IPv6 Address: e.g. `2606:4700:4700::1111`

Hostname/Sub-domain: e.g. `abc.example.com`

Subnet: e.g. `1.2.3.0/24`

Bitcoin Address: e.g. `1HesYJSP1QqcyPEjnQ9vzBL1wujuNGe7R`

E-mail address: e.g. `bob@example.com`

Phone Number: e.g. `+12345678901` (E.164 format)

Human Name: e.g. `"John Smith"` (must be in quotes)

Username: e.g. `"jsmith2000"` (must be in quotes)

Network ASN: e.g. `1234`

By Use Case

By Required Data

By Module

☐ All

Get anything and everything about the target.

All SpiderFoot modules will be enabled (slow) but every possible piece of information about the target will be obtained and analysed.

☒ Footprint

Understand what information this target exposes to the Internet.

Gain an understanding about the target's network perimeter, associated identities and other information that is obtained through a lot of web crawling and search engine use.

☐ Investigate

Best for when you suspect the target to be malicious but need more information.

Some basic footprinting will be performed in addition to querying of blacklists and other sources that may have information about your target's maliciousness.

☐ Passive

When you don't want the target to even suspect they are being investigated.

As much information will be gathered without touching the target or their affiliates, therefore only modules that do not touch the target will be enabled.

Run Scan Now

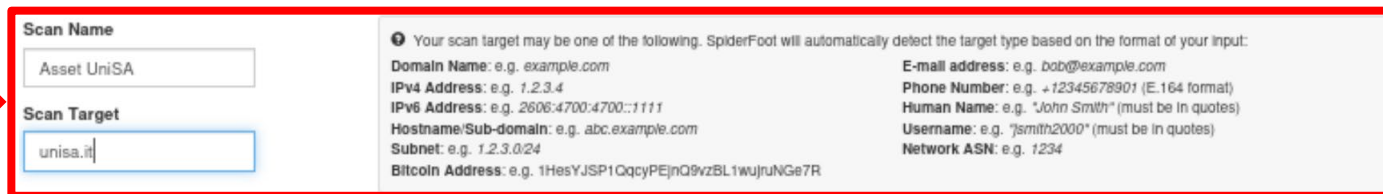


Utilizzo di Crawler

SpiderFoot – Esempio

➤ Definizione dell'asset («unisa.it»)

New Scan

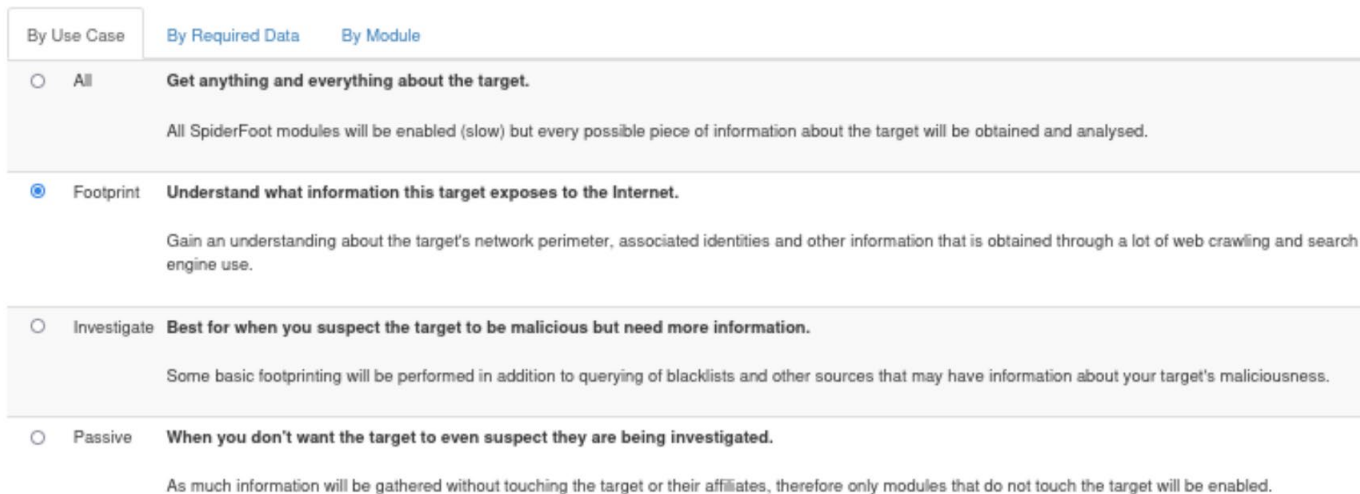


Scan Name
Asset UniSA

Scan Target
unisa.it

❗ Your scan target may be one of the following. SpiderFoot will automatically detect the target type based on the format of your input:

- Domain Name: e.g. example.com
- IPv4 Address: e.g. 1.2.3.4
- IPv6 Address: e.g. 2606:4700:4700::1111
- Hostname/Sub-domain: e.g. abc.example.com
- Subnet: e.g. 1.2.3.0/24
- Bitcoin Address: e.g. 1HesYJSP1QocyPEjnQ9vzBL1wujruNGe7R
- E-mail address: e.g. bob@example.com
- Phone Number: e.g. +12345678901 (E.164 format)
- Human Name: e.g. "John Smith" (must be in quotes)
- Username: e.g. "jsmith2000" (must be in quotes)
- Network ASN: e.g. 1234



By Use Case | By Required Data | By Module

☐ All **Get anything and everything about the target.**
All SpiderFoot modules will be enabled (slow) but every possible piece of information about the target will be obtained and analysed.

☒ Footprint **Understand what information this target exposes to the Internet.**
Gain an understanding about the target's network perimeter, associated identities and other information that is obtained through a lot of web crawling and search engine use.

☐ Investigate **Best for when you suspect the target to be malicious but need more information.**
Some basic footprinting will be performed in addition to querying of blacklists and other sources that may have information about your target's maliciousness.

☐ Passive **When you don't want the target to even suspect they are being investigated.**
As much information will be gathered without touching the target or their affiliates, therefore only modules that do not touch the target will be enabled.

Run Scan Now

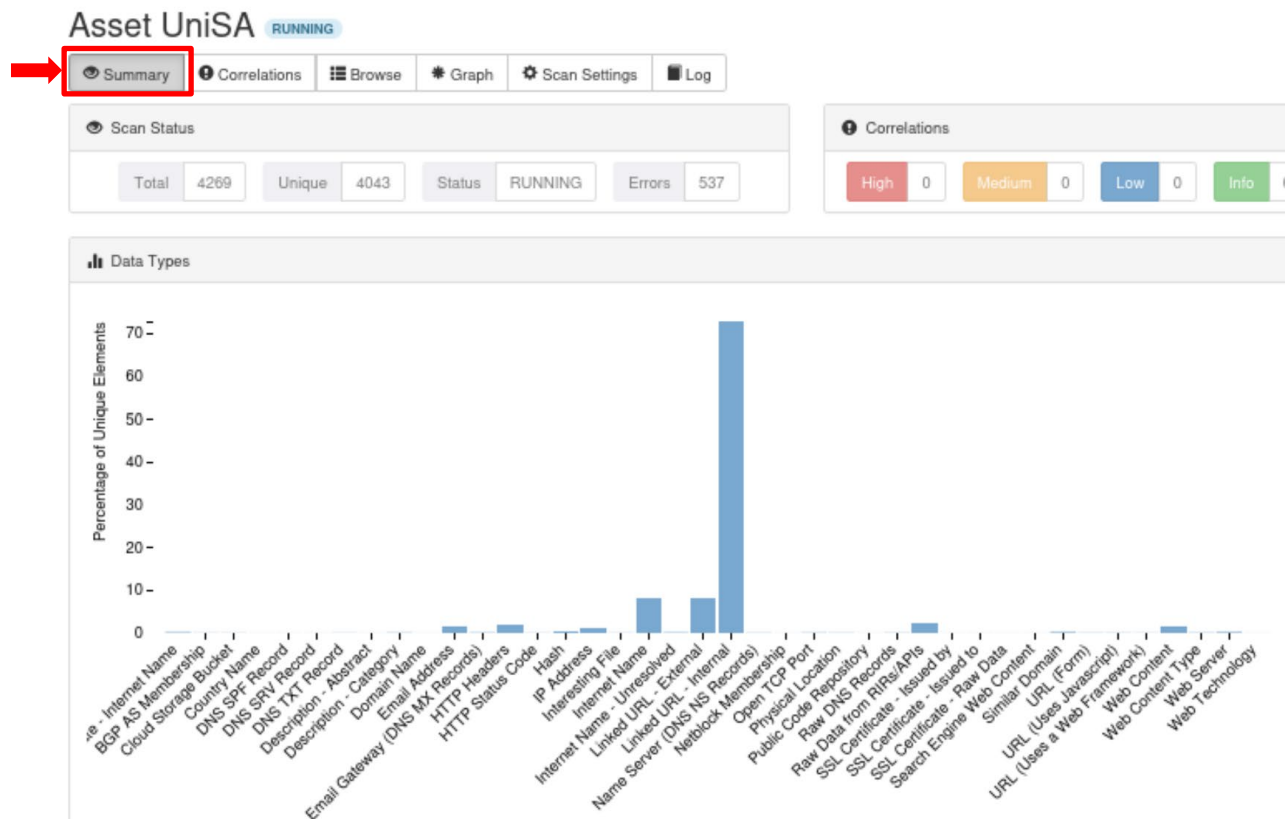
Avviato il processo di scansione, potranno subito essere visualizzati i relativi risultati intermedi



Utilizzo di Crawler

SpiderFoot – Esempio

➤ Riepilogo dei risultati ottenuti dalla scansione



Utilizzo di Crawler

SpiderFoot – Esempio

➤ Risultati ottenuti dalla scansione (*Indirizzi IP*)

Asset UniSA RUNNING

Summary Correlations Browse Graph Scan Settings Log

Browse / IP Address

☐	Data Element	Source Data Element	Source Module	Identified
☐	193.205.160.9	esa-mx-1.unisa.it	sfp_dnsresolve	2024-03-23 08:04:17
☐	193.205.163.20	isem20.unisa.it	sfp_dnsresolve	2024-03-23 07:56:48
☐	193.205.165.100	193.205.165.101	sfp_dnsneighbor	2024-03-23 08:00:27
☐	193.205.165.101	mail-out-101.unisa.it	sfp_dnsresolve	2024-03-23 07:19:05
☐	193.205.165.102	193.205.165.101	sfp_dnsneighbor	2024-03-23 08:00:27
☐	193.205.165.103	193.205.165.101	sfp_dnsneighbor	2024-03-23 08:00:27
☐	193.205.165.104	193.205.165.101	sfp_dnsneighbor	2024-03-23 08:00:27
☐	193.205.165.105	193.205.165.101	sfp_dnsneighbor	2024-03-23 08:00:27
☐	193.205.165.106	193.205.165.101	sfp_dnsneighbor	2024-03-23 08:00:27

Utilizzo di Crawler

SpiderFoot – Esempio

➤ Risultati ottenuti dalla scansione (*Sottodomini*)

Asset UniSA RUNNING

Summary Correlations Browse Graph Scan Settings Log

Browse / Internet Name

☐	Data Element	Source Data Element	Source Module	Identified
☐	accessocampus.unisa.it	unisa.it	sfp_sslcert	2024-03-23 07:23:12
☐	alternanza.unisa.it	unisa.it	sfp_sslcert	2024-03-23 07:22:47
☐	ambiente.unisa.it	unisa.it	sfp_sslcert	2024-03-23 07:23:04
☐	analytics.unisa.it	unisa.it	sfp_dnsbrute	2024-03-23 07:18:55
☐	auth.dev.unisa.it	unisa.it	sfp_urlscan	2024-03-23 07:23:18
☐	auth.syntonia.unisa.it	unisa.it	sfp_urlscan	2024-03-23 07:23:17
☐	auth.unisa.it	unisa.it	sfp_dnsbrute	2024-03-23 07:18:56
☐	auth.unisa.it	unisa.it	sfp_urlscan	2024-03-23 07:23:19
☐	auth1.dev.unisa.it	unisa.it	sfp_urlscan	2024-03-23 07:23:20

Utilizzo di Crawler

SpiderFoot – Esempio

➤ Risultati ottenuti dalla scansione (*Indirizzi e-mail*)

Asset UniSA RUNNING

Summary Correlations Browse Graph Scan Settings Log

Browse / Email Address

☐	Data Element	Source Data Element	Source Module	Identified
☐	20centroictstaff@unisa.it	unisa.it	sfp_emailformat	2024-03-23 07:19:07
☐	20ritcas@unisa.it	unisa.it	sfp_emailformat	2024-03-23 07:19:07
☐	20rpelella@unisa.it	unisa.it	sfp_emailformat	2024-03-23 07:19:07
☐	G.MADONNA3@studenti.unisa.it	unisa.it	sfp_grep_app	2024-03-23 07:18:24
☐	GMadonna@studenti.unisa.it	unisa.it	sfp_grep_app	2024-03-23 07:18:28
☐	Gianni@studenti.unisa.it	unisa.it	sfp_grep_app	2024-03-23 07:18:34
☐	GianniMorandiManiGrandi@studenti.unisa.it	unisa.it	sfp_grep_app	2024-03-23 07:18:26
☐	ProvaUser@studenti.unisa.it	unisa.it	sfp_grep_app	2024-03-23 07:18:32
☐	a.prova@studenti.unisa.it	unisa.it	sfp_grep_app	2024-03-23 07:18:24
☐	a.raiola14@studenti.unisa.it	unisa.it	sfp_grep_app	2024-03-23 07:18:30

Utilizzo di Crawler

Surface Mapping ed Asset Discovery – OWASP Amass

- Framework che consente di effettuare operazioni di Surface Mapping ed Asset Discovery mediante
 - Information gathering passiva, basata su OSINT
 - Information gathering attiva, basata su *bruteforcing*
- Fornisce
 - Vari strumenti per l'asset discovery
 - Un database integrato per la memorizzazione dei risultati
 - Modelli per la formalizzazione dell'asset

Utilizzo di Crawler

OWASP Amass – Caratteristiche Principali

- **Raccolta delle informazioni:** automatizza la raccolta di informazioni sull'asset
 - Nomi di dominio, sottodomini, indirizzi IP, certificati SSL/TLS, server e-mail, etc
- **Integrazione delle informazioni:** integra dati provenienti da varie fonti di informazioni
 - Motori di ricerca, database di record DNS, servizi di cloud hosting, social network, etc
- **Analisi della superficie di attacco:** consente di identificare potenziali punti di ingresso e vulnerabilità
 - Rilevando host «non autorizzati» o servizi esposti che potrebbero essere «presi di mira da attacchi»
- **Esportazione dei dati:** permette di esportare in diversi formati i risultati della raccolta di informazioni
 - Facilitandone l'analisi e l'utilizzo da parte di altri strumenti e processi di sicurezza

OWASP Amass – Comandi

- Composto da due comandi
 - **amass intel** – Si occupa di raccogliere informazioni di varia tipologia riguardanti l'asset
 - **amass enum** – Si occupa dell'enumerazione, attiva e passiva, dei sottodomini relati all'asset

[illegible]

Utilizzo di Crawler

OWASP Amass – Comando intel

➤ **amass intel -h**

```
Usage: amass intel [options] [-whois -d DOMAIN] [-addr ADDR -asn ASN -cidr CIDR]

-active
    Attempt certificate name grabs
-addr value
    IPs and ranges (192.168.1.1-254) separated by commas
-asn value
    ASNs separated by commas (can be used multiple times)
-cidr value
    CIDRs separated by commas (can be used multiple times)
-config string
    Path to the YAML configuration file. Additional details below
-d value
    Domain names separated by commas (can be used multiple times)
-demo
    Censor output to make it suitable for demonstrations
-df value
    Path to a file providing root domain names
-dir string
    Path to the directory containing the output files
```

Output parziale

Utilizzo di Crawler

OWASP Amass – Comando enum

➤ **amass enum -h**

```
Usage: amass enum [options] -d DOMAIN

-active
    Attempt zone transfers and certificate name grabs
-addr value
    IPs and ranges (192.168.1.1-254) separated by commas
-alts
    Enable generation of altered names
-asn value
    ASNs separated by commas (can be used multiple times)
-aw value
    Path to a different wordlist file for alterations
-awm value
    "hashcat-style" wordlist masks for name alterations
-bl value
    Blacklist of subdomain names that will not be investigated
-blf string
    Path to a file providing blacklisted subdomains
-brute
    Execute brute forcing after searches
-cidr value
    CIDRs separated by commas (can be used multiple times)
```

Output parziale

Utilizzo di Crawler

OWASP Amass – enum – Esempio

➤ `amass enum -d unisa.it`

```
(kali㉿kali)-[~]  
$ amass enum -d unisa.it  
unisa.it (FQDN) → ns_record → ns1.garr.net (FQDN)  
unisa.it (FQDN) → ns_record → dns-001.unisa.it (FQDN)  
unisa.it (FQDN) → ns_record → ns.unisa.it (FQDN)  
unisa.it (FQDN) → mx_record → alt3.aspmx.l.google.com (FQDN)  
unisa.it (FQDN) → mx_record → aspmx.l.google.com (FQDN)  
unisa.it (FQDN) → mx_record → alt4.aspmx.l.google.com (FQDN)  
unisa.it (FQDN) → mx_record → alt1.aspmx.l.google.com (FQDN)  
unisa.it (FQDN) → mx_record → alt2.aspmx.l.google.com (FQDN)  
corsi.unisa.it (FQDN) → cname_record → www.unisa.it (FQDN)  
biblioteche.unisa.it (FQDN) → cname_record → www4.unisa.it (FQDN)  
jobincampus.unisa.it (FQDN) → cname_record → www.unisa.it (FQDN)  
tfia.diem.unisa.it (FQDN) → cname_record → docenti.diem.unisa.it (FQDN)  
h40.cla.unisa.it (FQDN) → a_record → 193.205.173.90 (IPAddress)  
digitalstories.ictateneo.unisa.it (FQDN) → cname_record → ict219.ictateneo.unisa.it (FQDN)  
linuxas.seda.unisa.it (FQDN) → a_record → 193.205.167.86 (IPAddress)  
placement.unisa.it (FQDN) → cname_record → webgroup01-debzgtdscyfr9ff.z01.azurefd.net (FQDN)  
6000r.unisa.it (FQDN) → a_record → 193.205.160.190 (IPAddress)  
archeo5.dbc.unisa.it (FQDN) → a_record → 193.205.170.90 (IPAddress)  
allievo32.cla.unisa.it (FQDN) → a_record → 193.205.173.232 (IPAddress)  
provola.cbs.unisa.it (FQDN) → a_record → 193.205.187.129 (IPAddress)
```

Utilizzo di Crawler

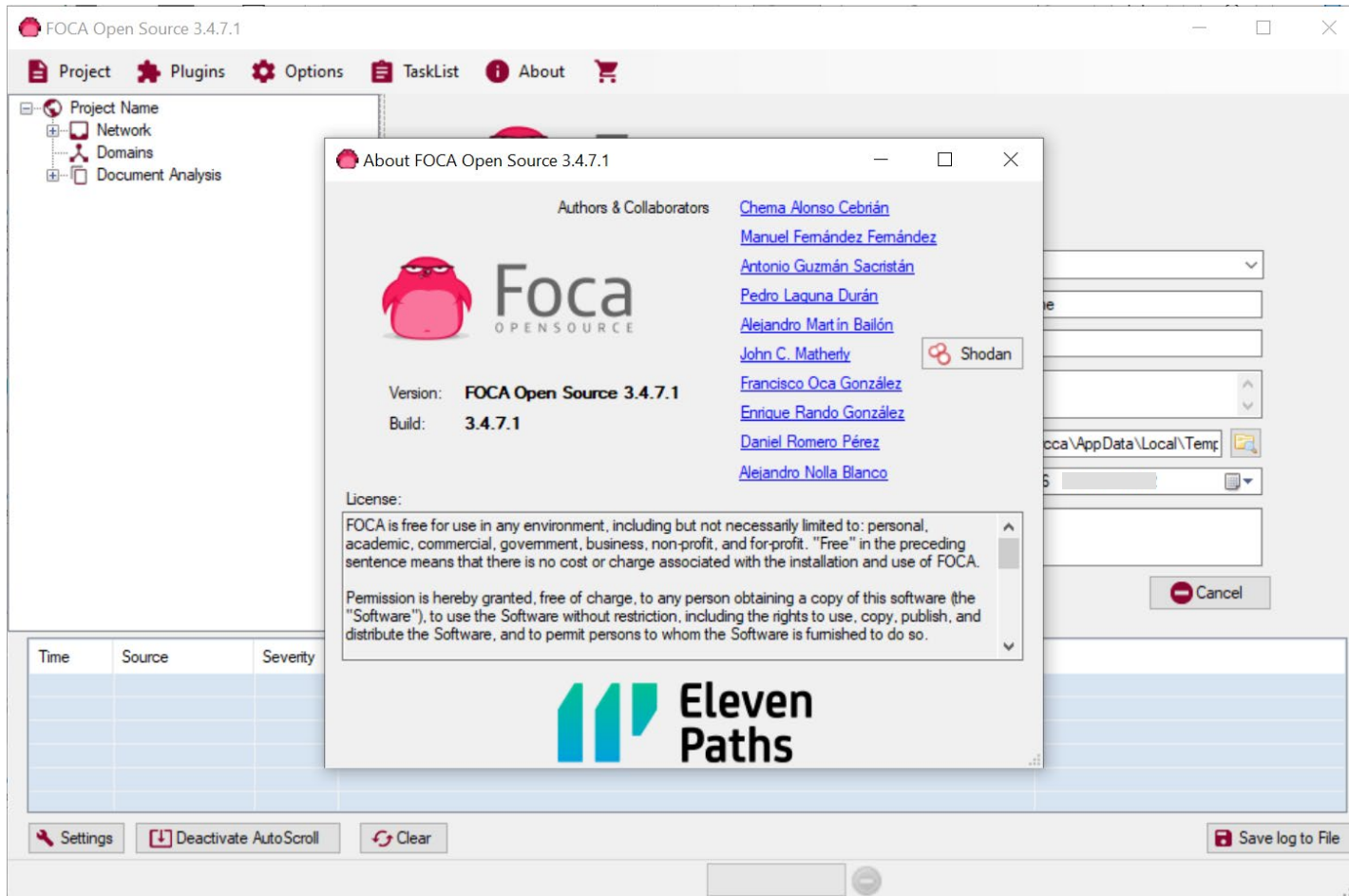
Ricerca di Metadati – FOCA

- **FOCA** (***F**ingerprinting **O**rganizations with **C**ollected **A**rchives*)
 - Permette di trovare metadati ed informazioni nascoste nei file
 - I file vengono scaricati ed analizzati automaticamente
 - Scaricabile gratuitamente
 - <https://github.com/ElevenPaths/FOCA/releases>
 - Richiede l'installazione di un database SQL per poter funzionare
 - <https://github.com/ElevenPaths/FOCA/wiki/How-to-set-up-a-SQL-database-connection>



Utilizzo di Crawler

Ricerca di Metadati – FOCA



Ricerca di Metadati – FOCA

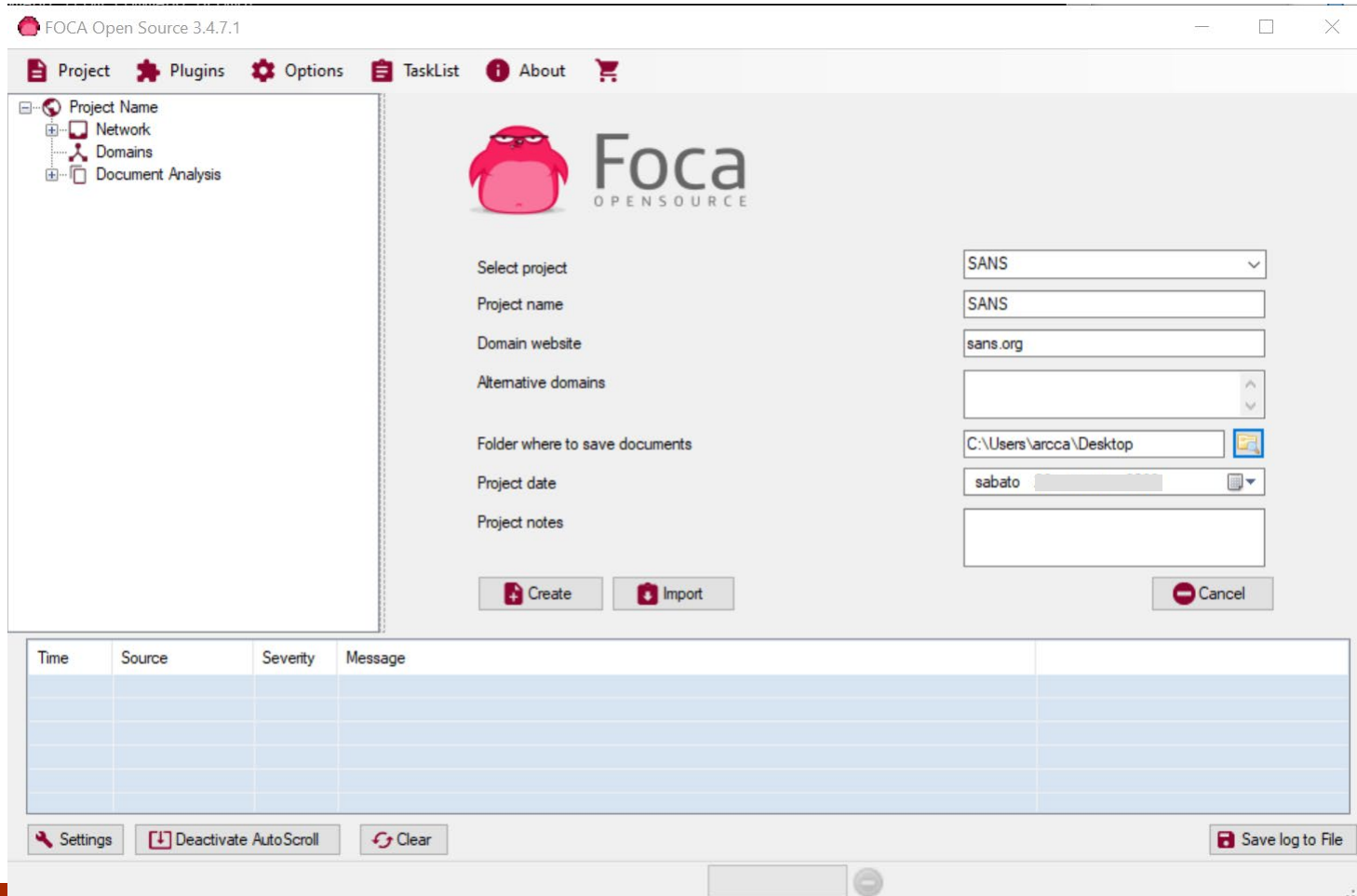


Ricerca di Metadati – FOCA



Utilizzo di Crawler

Ricerca di Metadati – FOCA – Esempio



Ricerca di Metadati – FOCA – Esempio

[illegible]

Ricerca di Metadati – FOCA – Esempio

[illegible]

Ricerca di Metadati – FOCA – Esempio

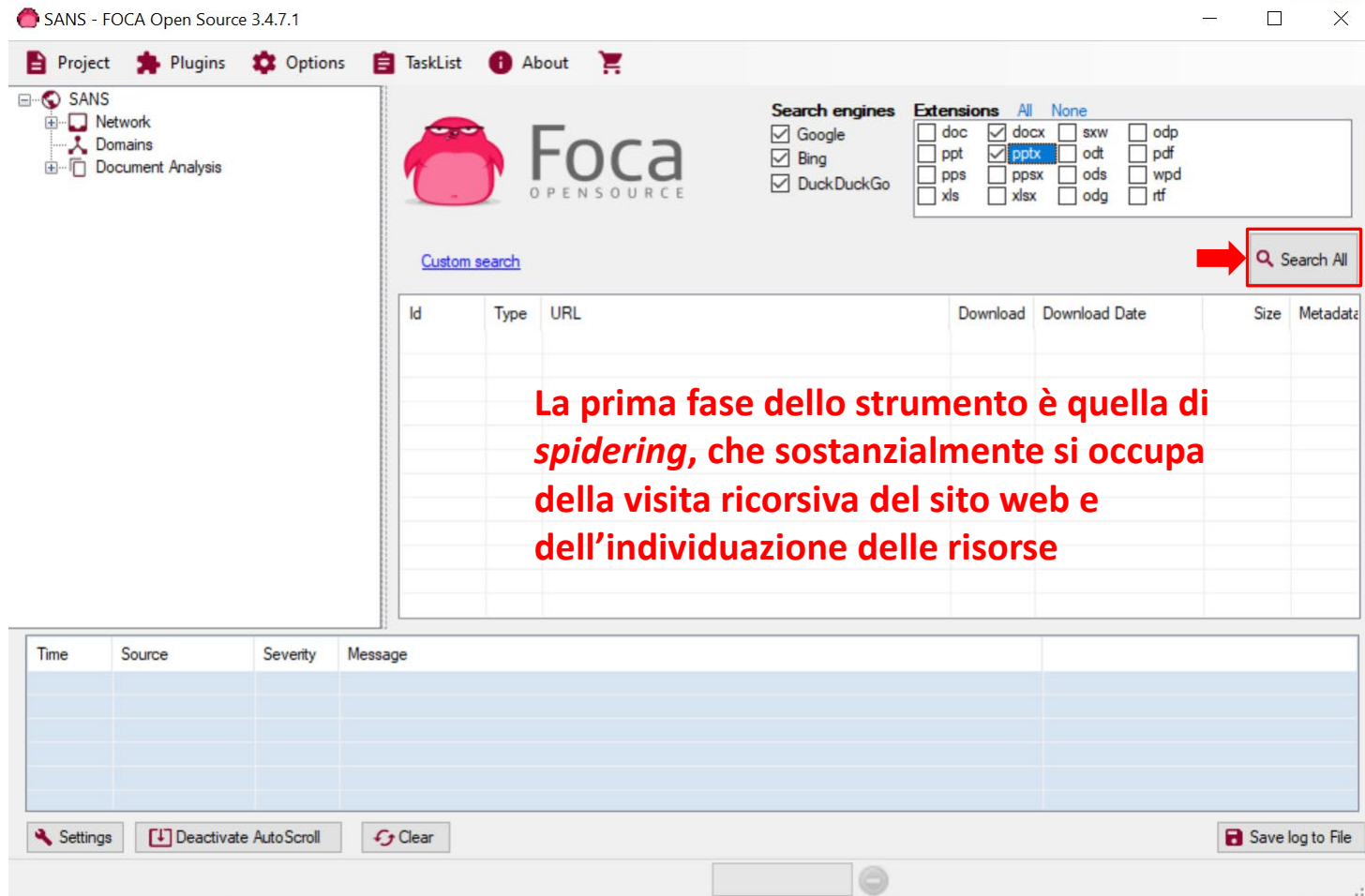


Ricerca di Metadati – FOCA – Esempio



Utilizzo di Crawler

Ricerca di Metadati – FOCA – Esempio



Utilizzo di Crawler

Ricerca di Metadati – FOCA – Esempio

SANS - FOCA Open Source 3.4.7.1

Project Plugins Options TaskList About

SANS
Network
Domains
Document Analysis

File docx e pptx individuati sull'asset

Search engines
☒ Google
☒ Bing
☒ DuckDuckGo

Extensions All None
☐ doc ☒ docx ☐ sxw ☐ odp
☐ ppt ☒ pptx ☐ odt ☐ pdf
☐ pps ☐ ppsx ☐ ods ☐ wpd
☐ xls ☐ xlsx ☐ odg ☐ rtf

Custom search Search All

Id	Type	URL	Download	Download Date	Size	Meta
0	docx	https://www.sans.org/event-downloads/35800/agenda....	×	-	19,49 KB	×
1		https://www.sans.org/blog/office-2007-metadata/	×	-	150,48 KB	×
2	docx	https://www.sans.org/media/score/checklists/ios-platfor...	×	-	138,04 KB	×
3		https://www.sans.org/blog/finding-unknown-malware-wit...	×	-	160,97 KB	×
4		http://www.sans.org/security-resources/policies/general...	×	-	-	×
5		https://www.sans.org/blog/making-reviewing-files-from-d...	×	-	157,4 KB	×
6	docx	https://www.sans.org/media/justify-training/Justify-Your...	×	-	26,92 KB	×
7		https://www.sans.org/blog/extracting-vb-macro-code-fro...	×	-	161,25 KB	×
8	docx	https://www.sans.org/media/justify-training/Justify-Your...	×	-	24,17 KB	×
9	docx	https://www.sans.org/media/justify-training/Justify-Your...	×	-	23,91 KB	×
10	docx	https://www.sans.org/media/justify-training/Justify-Your...	×	-	26,79 KB	×

Time	Source	Severity	Message
16:31:39	MetadataSearch	error	An error has occurred on DuckDuckGoWeb: Errore del server remoto: (403) Non consentito..
16:31:46	MetadataSearch	medium	BingWeb search finished successfully!! Total found result count: 96
16:31:50	MetadataSearch	medium	GoogleWeb search finished successfully!! Total found result count: 17

Settings Deactivate AutoScroll Clear Save log to File

All searchers have finished

Utilizzo di Crawler

Ricerca di Metadati – FOCA – Esempio

Successivamente è possibile scaricare una o più risorse

The screenshot displays the FOCA Open Source 3.4.7.1 application window. The interface includes a sidebar with navigation options (Project, Plugins, Options, TaskList, About), a search engines section (Google, Bing, DuckDuckGo), and an extensions section (doc, docx, ppt, pps, xls, etc.). A search results table is visible, listing various documents found. A context menu is open over the table, highlighting the 'Download' option for a specific document.

Id	Type	URL	Download	Download Date	Size	Meta
0	docx	https://www.sans.org/event-downloads/35800/agenda...	x	-	19,49 KB	x
1		https://www.sans.org/blog/office-2007-metadata/	x	-	150,48 KB	x
2	docx	https://www.sans.org/media/ecsp/charlie/insider/	x	-	138,04 KB	x
3		https://www.sans.org/	x	-	160,97 KB	x
4		http://www.sans.org/	x	-	-	x
5		https://www.sans.org/	x	-	157,4 KB	x
6	docx	https://www.sans.org/	x	-	26,92 KB	x
7		https://www.sans.org/	x	-	161,25 KB	x
8	docx	https://www.sans.org/	x	-	24,17 KB	x
9	docx	https://www.sans.org/	x	-	23,91 KB	x
10	docx	https://www.sans.org/	x	-	26,79 KB	x

The context menu for the selected document (Id 2) includes the following options:

- Download
- Extract Metadata
- Analyze Malware
- Delete
- Download All
- Extract All Metadata
- Analyze All Metadata
- Analyze All Malware
- Delete All
- Add file
- Add folder
- Add URLs from file
- Link(s)

At the bottom of the window, a log shows the status of the searchers:

Time	Source	Severity	Message
16:31:39	MetadataSearch	error	An error has occurred on DuckDuckGoWeb: Error
16:31:46	MetadataSearch	medium	BingWeb search finished successfully!! Total for
16:31:50	MetadataSearch	medium	GoogleWeb search finished successfully!! Total for

Buttons at the bottom include Settings, Deactivate AutoScroll, Clear, and Save log to File.

Utilizzo di Crawler

Ricerca di Metadati – FOCA – Esempio

SANS - FOCA Open Source 3.4.7.1

Project Plugins Options TaskList About

SANS

- Network
- Domains
- Document Analysis
 - Files (1/47)
 - docx (1)
 - Metadata Summary
 - Malware Summary (DIARIO)

Foca
OPEN SOURCE

Search engines
☒ Google
☒ Bing
☒ DuckDuckGo

Extensions All None
☐ doc ☒ docx ☐ sxw ☐ odp
☐ ppt ☒ pptx ☐ odt ☐ pdf
☐ pps ☐ ppsx ☐ ods ☐ wpd
☐ xls ☐ xlsx ☐ odg ☐ rtf

Custom search Search All

Id	Type	URL	Download	Download Date	Size	Meta
0	docx	https://www.sans.org/event-downloads/35800/agenda....	✗	-	19,49 KB	✗
1		https://www.sans.org/blog/office-2007-metadata/	✗	-	150,48 KB	✗
2	docx	https://www.sans.org/media/score/checklists/ios-platfor...	•	03/26/2022 16:38:00	138,04 KB	✗
3		https://www.sans.org/blog/finding-unknown-malware-wit...	✗	-	160,97 KB	✗
4		http://www.sans.org/security-resources/policies/general...	✗	-	-	✗
5		https://www.sans.org/blog/making-reviewing-files-from-d...	✗	-	157,4 KB	✗
6	docx	https://www.sans.org/media/justify-training/Justify-Your...	✗	-	26,92 KB	✗
7		https://www.sans.org/blog/extracting-vb-macro-code-fro...	✗	-	161,25 KB	✗
8	docx	https://www.sans.org/media/justify-training/Justify-Your...	✗	-	24,17 KB	✗
9	docx	https://www.sans.org/media/justify-training/Justify-Your...	✗	-	23,91 KB	✗
10	docx	https://www.sans.org/media/justify-training/Justify-Your...	✗	-	26,79 KB	✗

Time	Source	Severity	Message
16:31:39	MetadataSearch	error	An error has occurred on DuckDuckGoWeb: Errore del server remoto: (403) Non consentito..
16:31:46	MetadataSearch	medium	BingWeb search finished successfully!! Total found result count: 96
16:31:50	MetadataSearch	medium	GoogleWeb search finished successfully!! Total found result count: 17

Settings Deactivate AutoScroll Clear Save log to File

All documents have been downloaded

Utilizzo di Crawler

Ricerca di Metadati – FOCA – Esempio

È possibile estrarre i metadati relativi alla risorsa scaricata

The screenshot displays the Foca Open Source 3.4.7.1 application window. The interface includes a sidebar with a tree view showing the project structure: SANS, Network, Domains, Document Analysis, Files (1/47), Metadata Summary, and Malware Summary (DIARIO). The main area features the Foca logo and a search results table. A context menu is open over the table, showing options like Download, Extract Metadata, Analyze Malware, Delete, Download All, Extract All Metadata, Analyze All Metadata, Analyze All Malware, Delete All, Add file, Add folder, Add URLs from file, and Link(s). The table has columns for Id, Type, URL, Download, Download Date, Size, and Meta. The bottom status bar indicates 'All documents have been downloaded'.

Id	Type	URL	Download	Download Date	Size	Meta
0	docx	https://www.sans.org/event-downloads/35800/agenda...	x	-	19,49 KB	x
1		https://www.sans.org/blog/office-2007-metadata/	x	-	150,48 KB	x
2	docx	https://www.sans.org/media/...		03/26/2022 16:38:00	138,04 KB	x
3		https://www.sans.org/blog/...		-	160,97 KB	x
4		http://www.sans.org/secure...		-	-	x
5		https://www.sans.org/blog/...		-	157,4 KB	x
6	docx	https://www.sans.org/media/...		-	26,92 KB	x
7		https://www.sans.org/blog/...		-	161,25 KB	x
8	docx	https://www.sans.org/media/...		-	24,17 KB	x
9	docx	https://www.sans.org/media/...		-	23,91 KB	x
10	docx	https://www.sans.org/media/...		-	26,79 KB	x

Time	Source	Severity	Message
16:31:39	MetadataSearch	error	An error has occurred on DuckDuckGoWeb: Errore del serv
16:31:46	MetadataSearch	medium	BingWeb search finished successfully!! Total found result c
16:31:50	MetadataSearch	medium	GoogleWeb search finished successfully!! Total found resu

Settings Deactivate AutoScroll Clear

All documents have been downloaded

Utilizzo di Crawler

Ricerca di Metadati – FOCA – Esempio

SANS - FOCA Open Source 3.4.7.1

Project Plugins Options TaskList About

SANS

- Network
- Domains
- Document Analysis
 - Files (1/47)
 - .docx (1)
 - Metadata Summary
 - Malware Summary (DIARIO)

Search engines

- ☒ Google
- ☒ Bing
- ☒ DuckDuckGo

Extensions

All		None	
☐ doc	☒ docx	☐ sxw	☐ odp
☐ ppt	☒ pptx	☐ odt	☐ pdf
☐ pps	☐ ppsx	☐ ods	☐ wpd
☐ xls	☐ xlsx	☐ odg	☐ rtf

Custom search

Search All

Id	Type	URL	Download	Download Date	Size	Meta
0	docx	https://www.sans.org/event-downloads/35800/agenda...	☒	-	19,49 KB	☒
1		https://www.sans.org/blog/office-2007-metadata/	☒	-	150,48 KB	☒
2	docx	https://www.sans.org/media/35800/35800-checklist-for-platforms	☒	03/26/2022 16:38:00	138,04 KB	☒
3		https://www.sans.org/blog/	☒	-	160,97 KB	☒
4		http://www.sans.org/secure/	☒	-	-	☒
5		https://www.sans.org/blog/	☒	-	157,4 KB	☒
6	docx	https://www.sans.org/media/35800/35800-checklist-for-platforms	☒	-	26,92 KB	☒
7		https://www.sans.org/blog/	☒	-	161,25 KB	☒
8	docx	https://www.sans.org/media/35800/35800-checklist-for-platforms	☒	-	24,17 KB	☒
9	docx	https://www.sans.org/media/35800/35800-checklist-for-platforms	☒	-	23,91 KB	☒
10	docx	https://www.sans.org/media/35800/35800-checklist-for-platforms	☒	-	26,79 KB	☒

Download

Extract Metadata

Analyze Malware

Delete

Download All

Extract All Metadata

Analyze All Metadata

Analyze All Malware

Delete All

Add file

Add folder

Add URLs from file

Link(s)

Time	Source	Severity	Message
16:31:39	MetadataSearch	error	An error has occurred on DuckDuckGoWeb: Errore del serv
16:31:46	MetadataSearch	medium	BingWeb search finished successfully!! Total found result c
16:31:50	MetadataSearch	medium	GoogleWeb search finished successfully!! Total found resu

Settings Deactivate AutoScroll Clear

All documents have been downloaded

Save log to File

Utilizzo di Crawler

Ricerca di Metadati – FOCA – Esempio

SANS - FOCA Open Source 3.4.7.1

Project Plugins Options TaskList About

Metadati estratti dal file selezionato

Foca
OPEN SOURCE

Select project
Project name
Domain website
Alternative domains
Folder where to save documents
Project date
Project notes

Update Import Cancel

Time	Source	Severity	Message
16:31:39	MetadataSearch	error	An error has ocured on DuckDuckGoWeb: Errore del server remoto: (403) Non consentito..
16:31:46	MetadataSearch	medium	BingWeb search finished successfully!! Total found result count: 96
16:31:50	MetadataSearch	medium	GoogleWeb search finished successfully!! Total found result count: 17

Settings Deactivate AutoScroll Clear Save log to File

All documents were analyzed

Utilizzo di Crawler

Ricerca di Metadati – FOCA – Esempio

The screenshot shows the SANS - FOCA Open Source 3.4.7.1 application window. The left sidebar contains a tree view with the following items: SANS, Network, Domains, Document Analysis, Files (1/47), .docx (1), Metadata Summary, **Users (2)** (highlighted with a red box and arrow), Folders (23), Printers (0), Software (1), Emails (0), Operating Systems (0), Passwords (0), Servers (0), and Malware Summary (DIARIO). The main panel displays a table with the following data:

Attribute	Value
All users found (2) - Times found	
Name	Javier
Name	Neal

Below the main panel, there is a log table with the following data:

Time	Source	Severity	Message
16:31:39	MetadataSearch	error	An error has occurred on DuckDuckGoWeb: Errore del server remoto: (403) Non consentito..
16:31:46	MetadataSearch	medium	BingWeb search finished successfully!! Total found result count: 96
16:31:50	MetadataSearch	medium	GoogleWeb search finished successfully!! Total found result count: 17

At the bottom of the window, there are buttons for Settings, Deactivate AutoScroll, Clear, and Save log to File. The status bar at the very bottom indicates "All documents were analyzed".

Utilizzo di Crawler

Ricerca di Metadati – FOCA – Esempio

SANS - FOCA Open Source 3.4.7.1

Project Plugins Options TaskList About

SANS

- Network
- Domains
- Document Analysis
 - Files (1/47)
 - .docx (1)
 - Metadata Summary
 - Folders (23)**
 - Printers (0)
 - Software (1)
 - Emails (0)
 - Operating Systems (0)
 - Passwords (0)
 - Servers (0)
 - Malware Summary (DIARIO)

Folders

Attribute	Value
All folders found (23) - Times found	
Path	http://ioscentral.co.uk/
Path	http://csrc.nist.gov/publications/nistpubs/800-121/
Path	https://www.grc.com/
Path	http://askthegEEK.us/pwd_meter/
Path	http://www.theregister.co.uk/2011/03/10/apple_update_omits_iphone3g/
Path	http://krebsonsecurity.com/2011/05/weylend-yutani-crime-kit-targets-macs-for-b...
Path	http://en.wikipedia.org/wiki/
Path	http://www.reghardware.com/2011/06/15/pin_spy_app_pulled/
Path	http://www.zdziarski.com/blog/
Path	http://www.cellebrite.com/
Path	http://mocana.com/blog/2011/02/18/iphone-hackers-can-gain-access-to-your-...
Path	http://www.iosresearch.org/
Path	http://katanaforensics.com/
Path	http://ixam-forensics.com/
Path	http://blog.crackpassword.com/2011/05/elcomsoft-breaks-iphone-encryption-of-...
Path	http://iphone-forensics.com/
Path	http://www.apple.com/uk/ipad/built-in-apps/
Path	http://www.f-secure.com/en_UK/security/security-lab/tools-and-services/online...

Time	Source	Severity	Message
16:31:39	MetadataSearch	error	An error has occurred on DuckDuckGoWeb: Errore del server remoto: (403) Non consentito..
16:31:46	MetadataSearch	medium	BingWeb search finished successfully!! Total found result count: 96
16:31:50	MetadataSearch	medium	GoogleWeb search finished successfully!! Total found result count: 17

Settings Deactivate AutoScroll Clear Save log to File

All documents were analyzed

Utilizzo di Crawler

Ricerca di Metadati – FOCA – Esempio

SANS - FOCA Open Source 3.4.7.1

Project Plugins Options TaskList About

SANS

- Network
- Domains
- Document Analysis
 - Files (1/47)
 - .docx (1)
 - Metadata Summary
 - Users (2)
 - Folders (23)
 - Printers (0)
 - Software (1)**
 - Emails (0)
 - Operating Systems (0)
 - Passwords (0)
 - Servers (0)
 - Malware Summary (DIARIO)

Attribute Value

All software found (1) - Times found

Software	Microsoft Office
----------	------------------

Software

Time	Source	Severity	Message
16:31:39	MetadataSearch	error	An error has ocurred on DuckDuckGoWeb: Errore del server remoto: (403) Non consentito..
16:31:46	MetadataSearch	medium	BingWeb search finished successfully!! Total found result count: 96
16:31:50	MetadataSearch	medium	GoogleWeb search finished successfully!! Total found result count: 17

Settings Deactivate Auto Scroll Clear Save log to File

All documents were analyzed

Utilizzo di Crawler

Ricerca di Metadati – FOCA – Esempio

SANS - FOCA Open Source 3.4.7.1

Project Plugins Options TaskList About

SANS

- Network
- Domains
- Document Analysis
 - Files (46/47)
 - doc (12)
 - docx (26)
 - pdf (2)
 - Unknown (6)
 - Metadata Summary
 - Users (2)
 - Folders (23)
 - Printers (0)
 - Software (1)
 - Emails (0)
 - Operating Systems (0)
 - Passwords (0)
 - Servers (0)
 - Malware Summary (DIARIO)

Foca OPEN SOURCE

Search engines: ☒ Google ☒ Bing ☒ DuckDuckGo

Extensions: ☐ doc ☒ docx ☐ sww ☐ odp ☐ ppt ☒ pptx ☐ odt ☐ pdf ☐ pps ☐ ppsx ☐ ods ☐ wpd ☐ xls ☐ xlsx ☐ odg ☐ rtf

Custom search Search All

Id	Type	URL	Download	Download Date	Size	Meta
0	docx	https://www.sans.org/event-downloads/35800/agenda...	•	03/26/2022 17:05:56	19,49 KB	×
1		https://www.san...	•	03/26/2022 17:05:57	150,61 KB	×
2	docx	https://www.san...	•	03/26/2022 16:38:00	138,04 KB	•
3		https://www.san...	•	03/26/2022 17:05:57	161,11 KB	×
4		http://www.sans...	•	03/26/2022 17:05:57	0 B	×
5		https://www.san...	•	03/26/2022 17:05:57	157,53 KB	×
6	docx	https://www.san...	•	03/26/2022 17:05:57	26,92 KB	×
7		https://www.san...	•	03/26/2022 17:05:57	161,38 KB	×
8	docx	https://www.san...	•	03/26/2022 17:05:57	24,17 KB	×
9	docx	https://www.san...	•	03/26/2022 17:05:57	23,91 KB	×
10	docx	https://www.san...	•	03/26/2022 17:05:57	26,79 KB	×

Download Extract Metadata Analyze Malware Delete Stop All Downloads Download All

Extract All Metadata Analyze All Metadata Analyze All Malware Delete All Add file Add folder Add URLs from file Link(s)

Time	Source	Severity	Message
16:31:39	MetadataSearch	error	An error has occurred on DuckDuckGoWeb: Er...
16:31:46	MetadataSearch	medium	BingWeb search finished successfully!! Total fo...
16:31:50	MetadataSearch	medium	GoogleWeb search finished successfully!! Total

Settings Deactivate AutoScroll Clear

All documents have been downloaded

Save log to File

Information Gathering

Utilizzo di Crawler

Ricerca di Metadati – FOCA – Esempio

SANS - FOCA Open Source 3.4.7.1

Project Plugins Options TaskList About

Foca
OPEN SOURCE

Search engines: ☒ Google ☒ Bing ☒ DuckDuckGo

Extensions: ☒ doc ☒ docx ☒ pptx ☒ pps ☒ xls ☐ doc ☐ docx ☐ ppt ☐ pps ☐ xls ☐ sxw ☐ odt ☐ ods ☐ odg ☐ odp ☐ pdf ☐ wpd ☐ rtf

Custom search Search All

Id	Type	URL	Download	Download Date	Size	Meta
0	docx	https://www.sans.org/event-downloads/35800/agenda....	•	03/26/2022 17:05:56	19,49 KB	×
1		https://www.sans.org/blog/office-2007-metadata/	•	03/26/2022 17:05:57	150,61 KB	×
2	docx	https://www.sans.org/event-downloads/35800/agenda....	•	03/26/2022 16:38:00	138,04 KB	•
3		https://www.sans.org/event-downloads/35800/agenda....	•	03/26/2022 17:05:57	161,11 KB	×
4		https://www.sans.org/event-downloads/35800/agenda....	•	03/26/2022 17:05:57	0 B	×
5		https://www.sans.org/event-downloads/35800/agenda....	•	03/26/2022 17:05:57	157,53 KB	×
6	docx	https://www.sans.org/event-downloads/35800/agenda....	•	03/26/2022 17:05:57	26,92 KB	×
7		https://www.sans.org/event-downloads/35800/agenda....	•	03/26/2022 17:05:57	161,38 KB	×
8	docx	https://www.sans.org/event-downloads/35800/agenda....	•	03/26/2022 17:05:57	24,17 KB	×
9	docx	https://www.sans.org/event-downloads/35800/agenda....	•	03/26/2022 17:05:57	23,91 KB	×
10	docx	https://www.sans.org/event-downloads/35800/agenda....	•	03/26/2022 17:05:57	26,79 KB	×

Download Extract Metadata Analyze Malware Delete Download All

Extract All Metadata

Analyze All Metadata Analyze All Malware Delete All Add file Add folder Add URLs from file Link(s)

Time	Source	Severity	Message
16:31:39	MetadataSearch	error	An error has occurred on DuckDuckGo
16:31:46	MetadataSearch	medium	BingWeb search finished successfully
16:31:50	MetadataSearch	medium	GoogleWeb search finished successfully

Settings Deactivate AutoScroll Clear

Metadata analysis completed

Save log to File

Utilizzo di Crawler

Ricerca di Metadati – FOCA – Esempio

The screenshot displays the SANS - FOCA Open Source 3.4.7.1 application window. The interface includes a menu bar (Project, Plugins, Options, TaskList, About) and a toolbar. On the left, a tree view shows the project structure, with 'Users (23)' highlighted under the 'Metadata Summary' folder. A red arrow points to this item. The central pane shows a table of user names. The word 'Users' is written in red text over the table. The bottom pane shows a log of search results.

Attribute	Value
All users found (23) - Times found	
Name	Matthew Anderson
Name	kmarshall
Name	admuser
Name	Javier
Name	Neal
Name	Van Kirk, Tanya
Name	Schleisman, Sara
Name	Bank of Newport
Name	administrator
Name	Angela Loomis
Name	Schleisman, Sara
Name	C C
Name	Alan Paller
Name	Stephen Northcutt
Name	Pamela Levy
Name	knaidu
Name	vthiagarajan
Name	Dean Farrington

Time	Source	Severity	Message
16:31:39	MetadataSearch	error	An error has occurred on DuckDuckGoWeb: Errore del server remoto: (403) Non consentito..
16:31:46	MetadataSearch	medium	BingWeb search finished successfully!! Total found result count: 96
16:31:50	MetadataSearch	medium	GoogleWeb search finished successfully!! Total found result count: 17

Settings Deactivate AutoScroll Clear Save log to File

All documents were analyzed

Utilizzo di Crawler

Ricerca di Metadati – FOCA – Esempio

SANS - FOCA Open Source 3.4.7.1

Project Plugins Options TaskList About

Domains

- iphone-forensics.com
- mocana.com
- reghardware.com
- security.symantec.com
- www.iosresearch.org
- www.reghardware.com

Document Analysis

- Files (47/47)
 - .doc (12)
 - .docx (26)
 - .pdf (2)
 - Unknown (7)
- Metadata Summary
- Users (23)
- Printers (0)
- Software (8)
- Emails (1)
- Operating Systems (3)
- Passwords (0)
- Servers (0)
- Malware Summary (DIARIO)

Folders (59)

Folders

Attribute	Value
Path	http://www.color.org/
Path	C:\Documents%20and%20Settings\knaidu\Application%20Data\Microsoft\Word\
Path	D:\Data\sans\
Path	http://www.rediffmail.com/cgi-bin/
Path	http://www.wastelands.gen.nz/
Path	http://www.cirt.net/code/
Path	http://packetstormsecurity.nl/Crackers/
Path	http://www.netcraft.com/
Path	http://www.softbytelabs.com/
Path	http://www.mavensecurity.com/
Path	http://mypage.bluewin.ch/vogje01/e/nmapwin/
Path	http://www.insecure.org/
Path	http://www.gfi.com/lanntscan/
Path	http://www.networksolutions.com/cgi-bin/whois/
Path	http://network-tools.com/nslook/
Path	http://info.sen.ca.gov/pub/01-02/bill/sen/sb_1351-1400/
Path	C:\Documents%20and%20Settings\Administrator\Application%20Data\Microsoft...
Path	H:\SCORE\
Path	https://www.linkedin.com/in/

Time	Source	Severity	Message
16:31:39	MetadataSearch	error	An error has ocured on DuckDuckGoWeb: Errore del server remoto: (403) Non consentito..
16:31:46	MetadataSearch	medium	BingWeb search finished successfully!! Total found result count: 96
16:31:50	MetadataSearch	medium	GoogleWeb search finished successfully!! Total found result count: 17

Settings Deactivate Auto Scroll Clear Save log to File

All documents were analyzed

Utilizzo di Crawler

Ricerca di Metadati – FOCA – Esempio

SANS - FOCA Open Source 3.4.7.1

Project Plugins Options TaskList About

Domains

Document Analysis

Files (47/47)

.doc (12)

.docx (26)

.pdf (2)

Unknown (7)

Metadata Summary

Users (23)

Folders (59)

Printers (0)

Software (8)

Emails (1)

Operating Systems (3)

Passwords (0)

Servers (0)

Malware Summary (DIARIO)

Attribute	Value
All software found (8) - Times found	
Software	Adobe InDesign 16.0 (Macintosh)
Software	Adobe PDF Library 15.0
Software	Microsoft Office
Software	Adobe InDesign 15.0 (Macintosh)
Software	Microsoft Office 2000
Software	Microsoft Office XP
Software	OpenOffice
Software	Microsoft Office for Mac

Software

Time	Source	Severity	Message
16:31:39	MetadataSearch	error	An error has ocured on DuckDuckGoWeb: Errore del server remoto: (403) Non consentito..
16:31:46	MetadataSearch	medium	BingWeb search finished successfully!! Total found result count: 96
16:31:50	MetadataSearch	medium	GoogleWeb search finished successfully!! Total found result count: 17

Settings Deactivate AutoScroll Clear Save log to File

All documents were analyzed

Utilizzo di Crawler

Ricerca di Metadati – FOCA – Esempio

SANS - FOCA Open Source 3.4.7.1

Emails

Attribute	Value
All emails found (1) - Times found	
Email	emea@sans.org

Time	Source	Severity	Message
16:31:39	MetadataSearch	error	An error has occurred on DuckDuckGoWeb: Errore del server remoto: (403) Non consentito..
16:31:46	MetadataSearch	medium	BingWeb search finished successfully!! Total found result count: 96
16:31:50	MetadataSearch	medium	GoogleWeb search finished successfully!! Total found result count: 17

Settings Deactivate Auto Scroll Clear Save log to File

All documents were analyzed

Utilizzo di Crawler

Ricerca di Metadati – FOCA – Esempio

SANS - FOCA Open Source 3.4.7.1

Project Plugins Options TaskList About

Domains

Document Analysis

Files (47/47)

.doc (12)

.docx (26)

.pdf (2)

Unknown (7)

Metadata Summary

Users (23)

Folders (59)

Printers (0)

Software (8)

Emails (1)

Operating Systems (3)

Passwords (0)

Servers (0)

Malware Summary (DIARIO)

Attribute	Value
All operating systems found (0) - Times found	
OS	Windows Server 2000
OS	Windows XP
OS	Mac OS

Operating Systems

Time	Source	Severity	Message
16:31:39	MetadataSearch	error	An error has ocured on DuckDuckGoWeb: Errore del server remoto: (403) Non consentito..
16:31:46	MetadataSearch	medium	BingWeb search finished successfully!! Total found result count: 96
16:31:50	MetadataSearch	medium	GoogleWeb search finished successfully!! Total found result count: 17

Settings Deactivate AutoScroll Clear Save log to File

All documents were analyzed

Utilizzo di Crawler

Ricerca di Metadati – Metagoofil

- Strumento basato su Google Dork che permette di scaricare determinati tipi di file dall'asset
 - Originariamente permetteva anche di ottenere i relativi metadati

- Supporta vari tipi di file
 - *Word document (.docx, .doc)*
 - *Spreadsheet document (.xlsx, .xls, .ods)*
 - *Presentation file (.pptx, .ppt, .odp)*
 - *PDF file (.pdf)*



Utilizzo di Crawler

Ricerca di Metadati – Metagoofil

- Metagoofil effettua le seguenti azioni
 1. Cerca nel dominio analizzato alcuni o tutti i tipi di file mostrati in precedenza
 - Utilizzando query Google
 2. Scarica sul disco locale tutti i file trovati

- I file recuperati tramite Metagoofil possono essere successivamente valutati tramite altri strumenti per l'analisi dei metadati
 - Ad es., <https://www.metadata2go.com/>



Utilizzo di Crawler

Ricerca di Metadati – Metagoofil

- Le informazioni ricavate mediante Metagoofil possono essere di aiuto per le fasi successive del processo di penetration testing
 - In particolare per il Social Engineering ed il Post-Exploitation
- Metagoofil non è installato di default in Kali Linux
 - `apt-get install metagoofil`



Utilizzo di Crawler

Ricerca di Metadati – Metagoofil

- Per maggiori informazioni sul comando **metagoofil** è sufficiente digitarlo

➤ **metagoofil -h**

```
usage: metagoofil.py [-h] [-d DOMAIN] [-e DELAY] [-f] [-i URL_TIMEOUT]
                    [-l SEARCH_MAX] [-n DOWNLOAD_FILE_LIMIT] [-o SAVE_DIRECTORY]
                    [-r NUMBER_OF_THREADS] -t FILE_TYPES [-u [USER_AGENT]] [-w]

Metagoofil - Search and download specific filetypes

optional arguments:
  -h, --help                show this help message and exit
  -d DOMAIN                  Domain to search.
  -e DELAY                   Delay (in seconds) between searches. If it's too small
                             Google may block your IP, too big and your search may take
                             a while. Default: 30.0
  -f                         Save the html links to html_links_<TIMESTAMP>.txt file.
  -i URL_TIMEOUT             Number of seconds to wait before timeout for
                             unreachable/stale pages. Default: 15
  -l SEARCH_MAX              Maximum results to search. Default: 100
  -n DOWNLOAD_FILE_LIMIT     Maximum number of files to download per filetype. Default:
                             100
  -o SAVE_DIRECTORY          Directory to save downloaded files. Default is current
                             working directory, "."
  -r NUMBER_OF_THREADS       Number of downloader threads. Default: 8
  -t FILE_TYPES              file_types to download
                             (pdf,doc,xls,ppt,odp,ods,docx,xlsx,pptx). To search all
                             17,576 three-letter file extensions, type "ALL"
  -u [USER_AGENT]            User-Agent for file retrieval against -d domain.
```



Utilizzo di Crawler

Ricerca di Metadati – Metagoofil – Esempio

- `metagoofil -d nist.gov -t pdf,doc -l 30 -n 30 -o metagoofil_nist -w`
 - `-d nist.gov` Dominio su cui effettuare la ricerca
 - `-t doc,pdf` Tipi di file da scaricare (*doc* e *pdf*)
 - `-l 30` Limite sul numero di risultati da cercare. Di default è *100*
 - `-n 30` Limite sul numero di file da scaricare. Di default è *100*
 - `-o metagoofil_nist` Directory dove memorizzare i file scaricati
 - `-w` I file trovati vengono scaricati nella directory che segue il parametro `-o`



Utilizzo di Crawler

Ricerca di Metadati – Metagoofil – Esempio

Output parziale

➤ `metagoofil -d nist.gov -t pdf,doc -l 30 -n 30 -o metagoofil_nist -w`

Ricerca e download di file PDF

```
[*] Downloaded files will be saved here: metagoofil_nist
[*] Searching for 30 .pdf files and waiting 30.0 seconds between searches
[+] Downloading file - [378775 bytes] https://www.nist.gov/document/eppendorf5415rcentrifugemanualpdf
[+] Downloading file - [66449 bytes] https://math.nist.gov/~BALpert/evolution.pdf
[+] Downloading file - [1460582 bytes] https://www.nist.gov/document/measuredfsecuritywithcoverpagesjuly22finalpdf
[+] Downloading file - [180600 bytes] https://emtoolbox.nist.gov/Publications/NISTTechnicalNote1297s.pdf
[+] Downloading file - [1207384 bytes] https://www.nist.gov/document/smartgridinteroperabilitypdf
[+] Downloading file - [452543 bytes] https://www.nist.gov/document/cybersecurityframework6thworkshopintelcorppdf
[+] Downloading file - [233702 bytes] https://www.nist.gov/document-10863
[+] Downloading file - [270673 bytes] https://math.nist.gov/~BALpert/nrbc2.pdf
[+] Downloading file - [213999 bytes] https://www.nist.gov/document/r0001214pdf
[+] Downloading file - [1430251 bytes] https://www.nist.gov/document/4se7sealysengveefinalpdf
```



Utilizzo di Crawler

Ricerca di Metadati – Metagoofil – Esempio

Output parziale

➤ `metagoofil -d nist.gov -t pdf,doc -l 30 -n 30 -o metagoofil_nist -w`

Ricerca e download di file doc

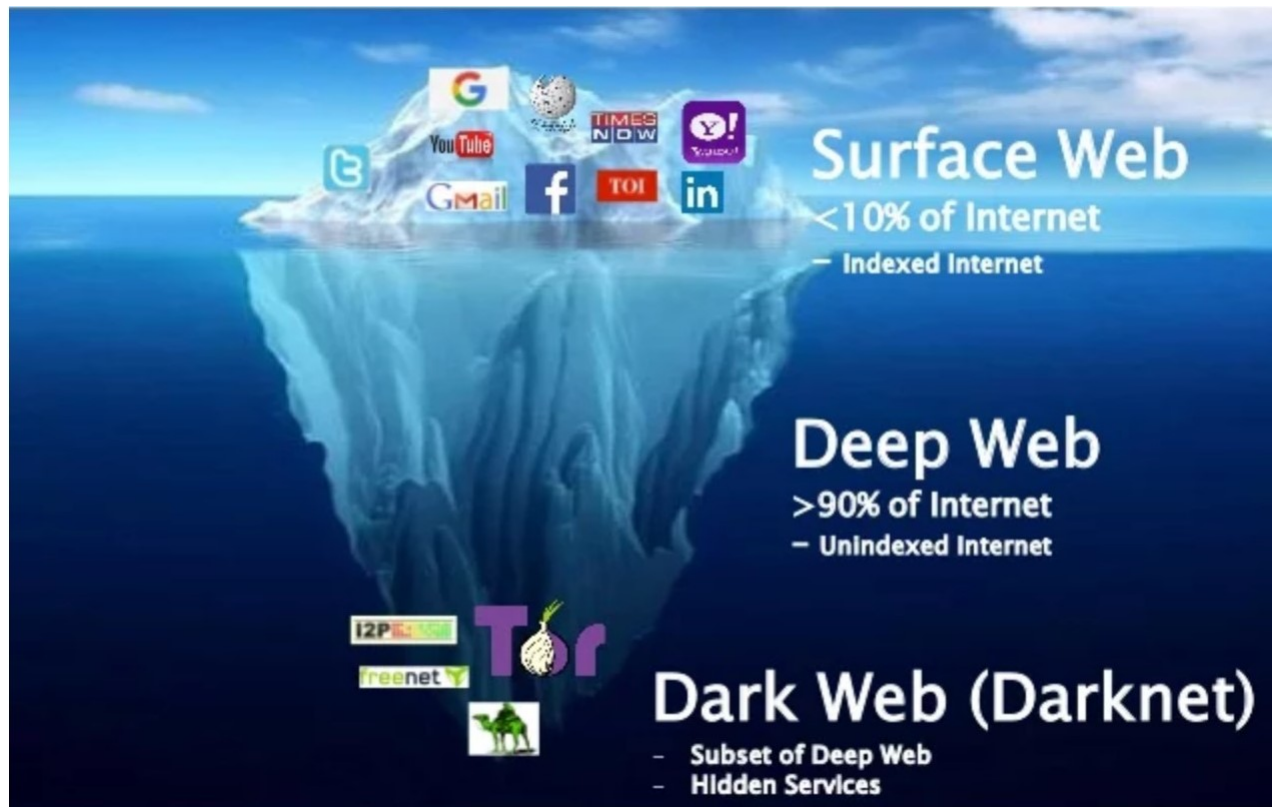
```
[*] Searching for 30 .doc files and waiting 30.0 seconds between searches
[+] Downloading file - [373248 bytes] https://www.nist.gov/document/appenadoc
[+] Downloading file - [38400 bytes] https://www.nist.gov/document/ststelecon
20060531doc
[+] Downloading file - [67584 bytes] https://www.nist.gov/document/ineap2011j
ulyfdoc
[+] Downloading file - [417280 bytes] https://www.nist.gov/document/oktextsit
esurveysdoc
[+] Downloading file - [98816 bytes] https://www.nist.gov/document/massachuse
ttsdoc
[+] Downloading file - [178688 bytes] https://www.nist.gov/document/procedure
13v401d03doc
[+] Downloading file - [4774400 bytes] https://www.nist.gov/document-15900
[+] Downloading file - [113664 bytes] https://www.nist.gov/document-11796
[+] Downloading file - [437760 bytes] https://www.nist.gov/document/xscript20
050118doc
[+] Downloading file - [98816 bytes] https://www.nist.gov/document-4050
[+] Downloading file - [51200 bytes] https://www.nist.gov/document/20050929pr
elimhfdoc
[+] Downloading file - [48640 bytes] https://www.nist.gov/document-14120
[+] Downloading file - [4760064 bytes] https://www.nist.gov/document-16197
[+] Downloading file - [68096 bytes] https://www.nist.gov/document/citationsp
art8doc
```



Outline

- Concetti Preliminari
- Raccolta di Informazioni da Risorse Web-Based
- Raccolta delle Informazioni di Registrazione
- Raccolta delle Informazioni di Routing
- Raccolta di Informazioni dall'Analisi dei Record DNS
- Raccolta di Informazioni mediante Crawler
- **Raccolta di informazioni dal Dark Web**
- Altri Strumenti e Servizi per Raccogliere Informazioni

Dark Web



Dark Web

➤ Il Dark Web

- Permette di accedere ad informazioni potenzialmente riservate
- Rappresenta un'area della rete Internet che non è indicizzata da motori di ricerca quali Google, Bing, Yandex, Baidu, etc

➤ In quest'area gli hacker si scambiano (o acquistano) informazioni su

- Vulnerabilità
 - Exploit
 - Malware
 - Credenziali
 - Etc
- Tipicamente 0-day**



Dark Web

- Il Dark Web ha una struttura diversa dal World Wide Web
 - I siti del Dark Web non sono indicizzati dai motori di ricerca e possono essere acceduti solo in modo diretto, digitando il loro URL
 - L'URL associato ad un sito nel Dark Web è costituito da una stringa (apparentemente) casuale seguita dal dominio di primo livello **.onion**



Dark Web

- **[Pro]** Il Dark Web potrebbe fornire informazioni che normalmente non si otterrebbero attraverso normali ricerche
- **[Cons]** Tuttavia
 - Il Dark Web potrebbe essere spesso popolato anche da cyber-criminali
 - E da criminali «non cyber»
 - Bisogna quindi fare molta attenzione
 - Possibilità di incorrere in truffe, raggiri, etc



Dark Web

TOR Browser

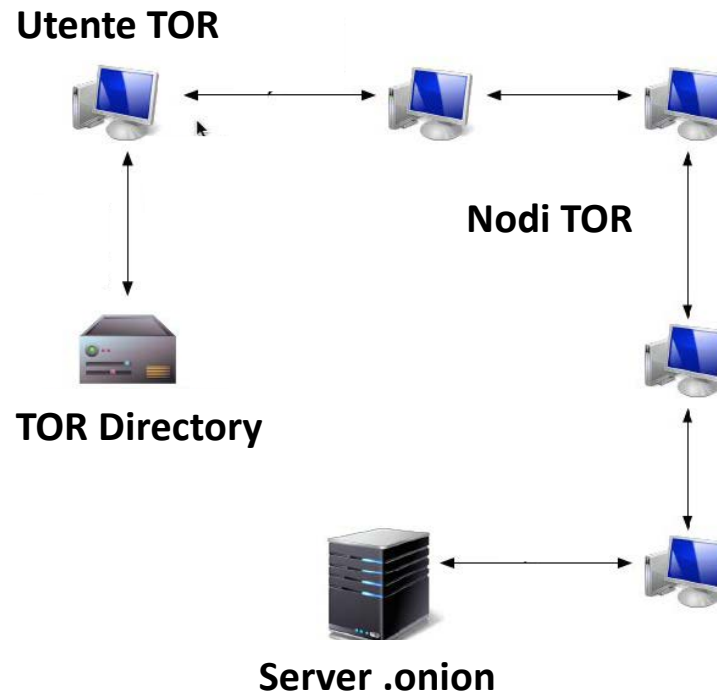
- Per accedere ai siti del Dark Web è necessario conoscere il loro URL
 - Non viene usato il DNS per la risoluzione dei nomi
- Il Dark Web può essere acceduto utilizzando il *The Onion Router (TOR) Browser*
- Il TOR Browser
 - È basato su Mozilla Firefox
 - Permette di accedere sia a siti con dominio **.onion** che a siti Web indicizzati in maniera convenzionale



Dark Web

TOR Browser

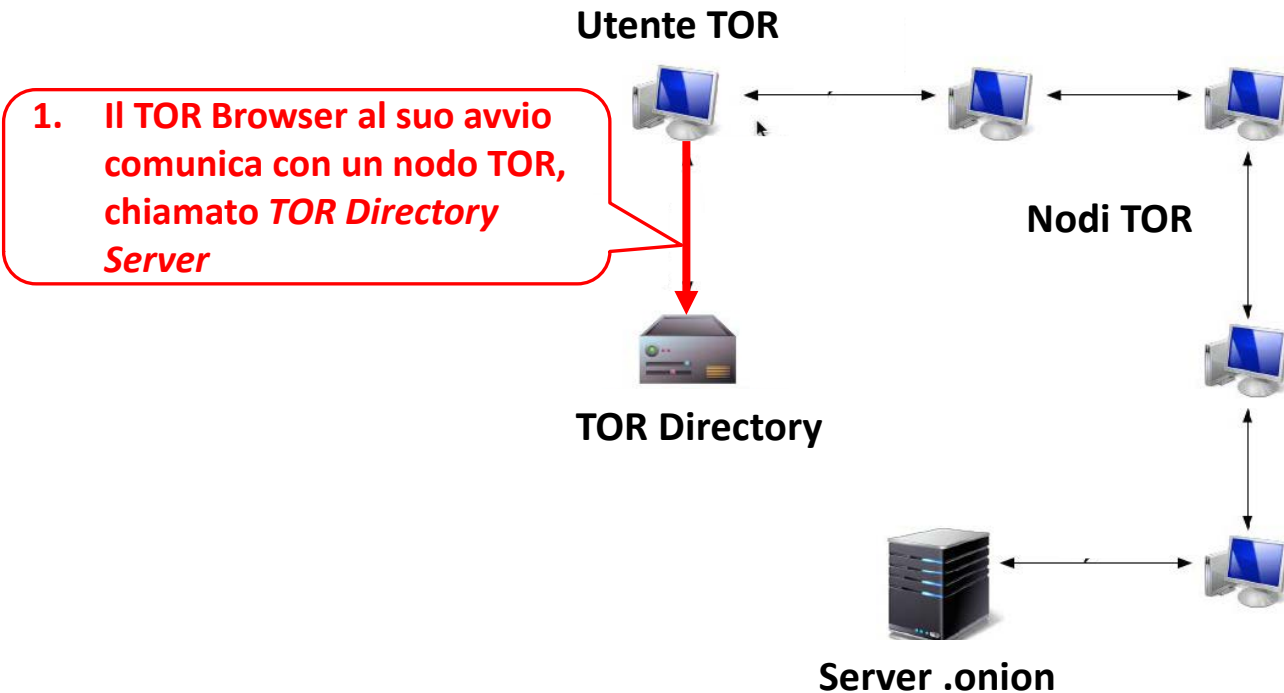
- Il TOR Browser accede ad un server contenente un sito con dominio **.onion** nel modo seguente



Dark Web

TOR Browser

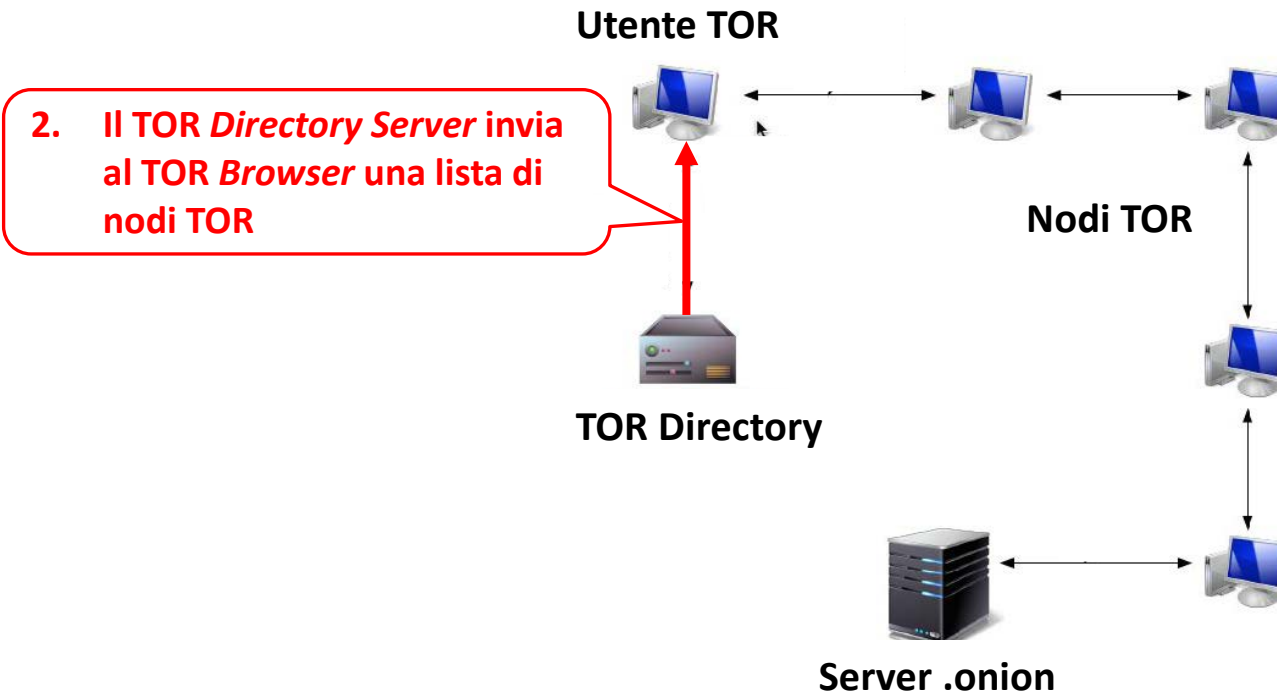
- Il TOR Browser accede ad un server contenente un sito con dominio **.onion** nel modo seguente



Dark Web

TOR Browser

- Il TOR Browser accede ad un server contenente un sito con dominio **.onion** nel modo seguente

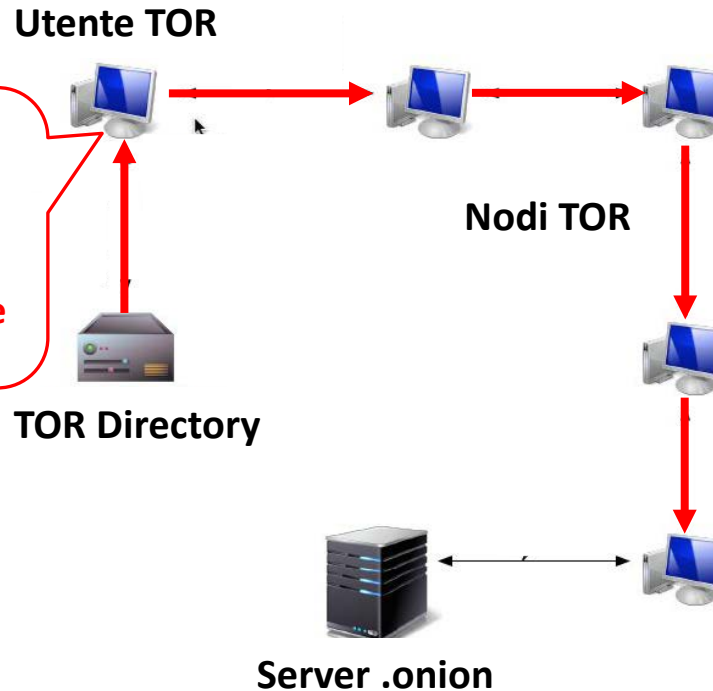


Dark Web

TOR Browser

- Il TOR Browser accede ad un server contenente un sito con dominio **.onion** nel modo seguente

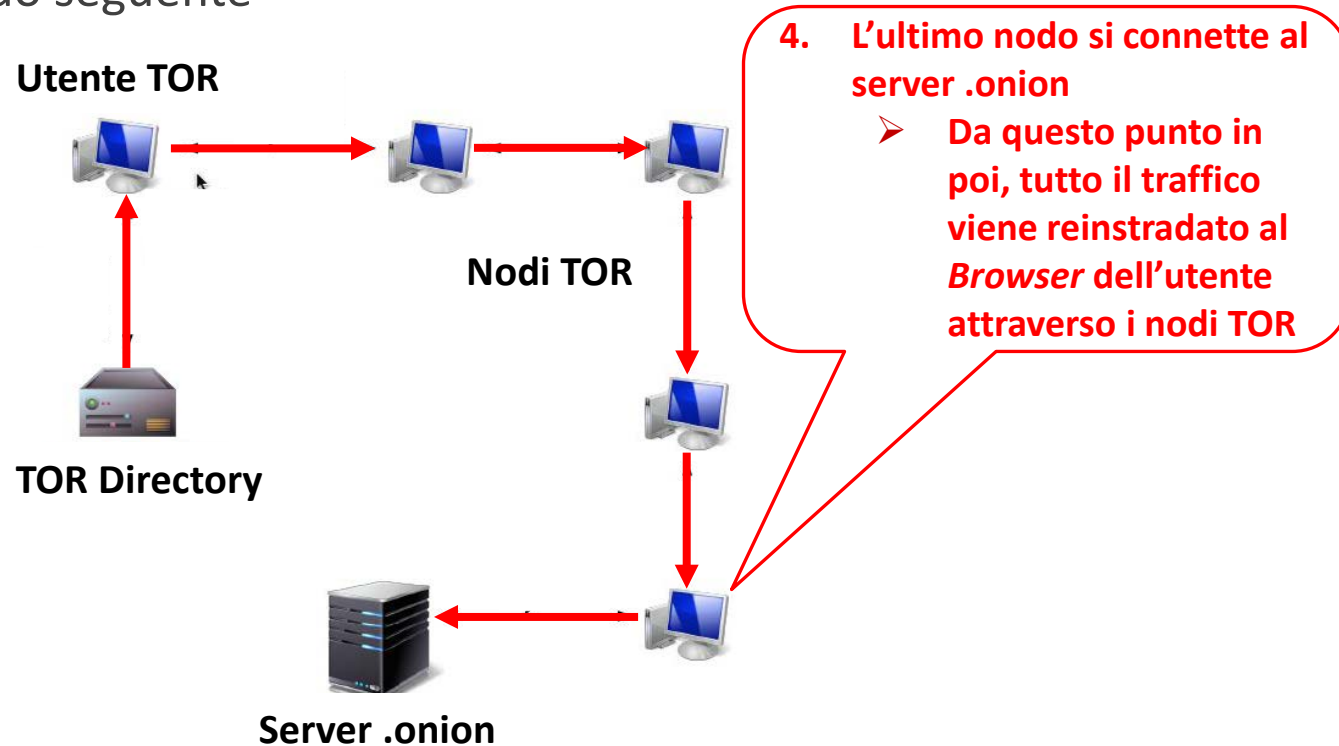
3. L'utente inserisce l'URL del sito .onion a cui intende accedere ed il TOR Browser effettua il forward del traffico attraverso una serie di nodi



Dark Web

TOR Browser

- Il TOR Browser accede ad un server contenente un sito con dominio **.onion** nel modo seguente



Dark Web

TOR Browser

➤ Grazie al TOR Browser

- La connessione tra tutti i nodi appartenenti al percorso di routing è protetta
 - Tranne quella tra l'ultimo nodo (*exit node* o *exit relay*) ed il Server **.onion**
- Il Server **.onion** sarà a conoscenza solo dell'indirizzo IP dell'ultimo nodo

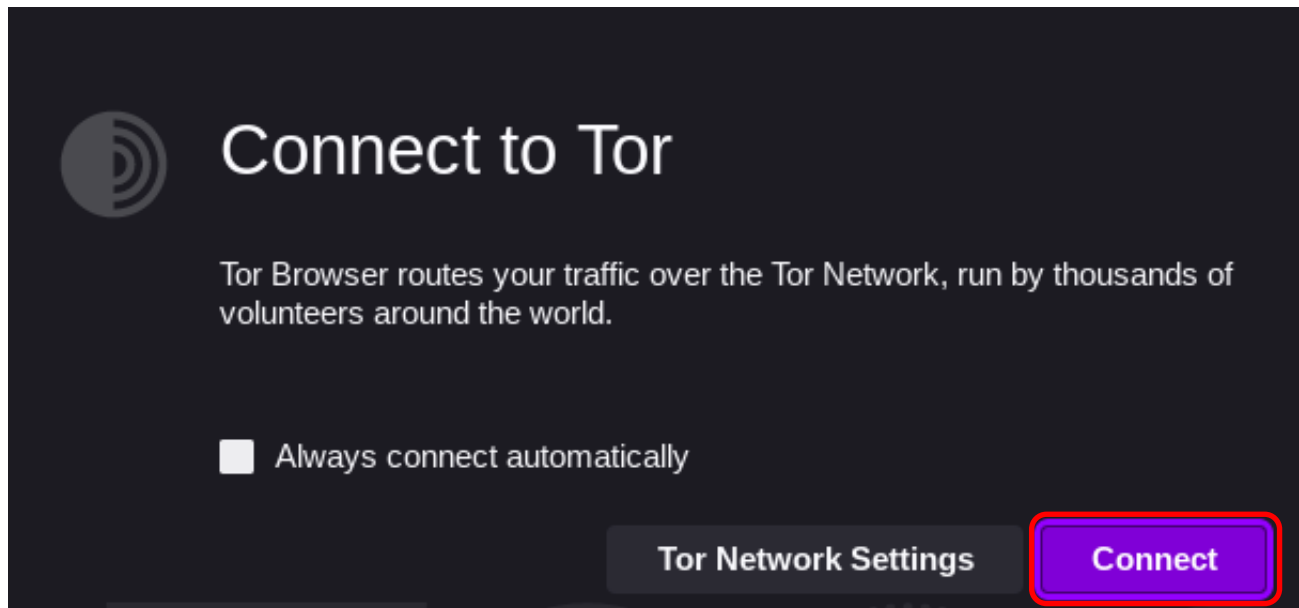
- Il TOR browser permette anche di anonimizzare il traffico verso i siti del World Wide Web (WWW)
 - Al browser verrà assegnato un indirizzo IP anonimo

Dark Web

TOR Browser – Installazione ed Utilizzo

➤ Per avviare il TOR Browser è sufficiente digitare il seguente comando (non presente di default in Kali Linux)

➤ **torbrowser-launcher**

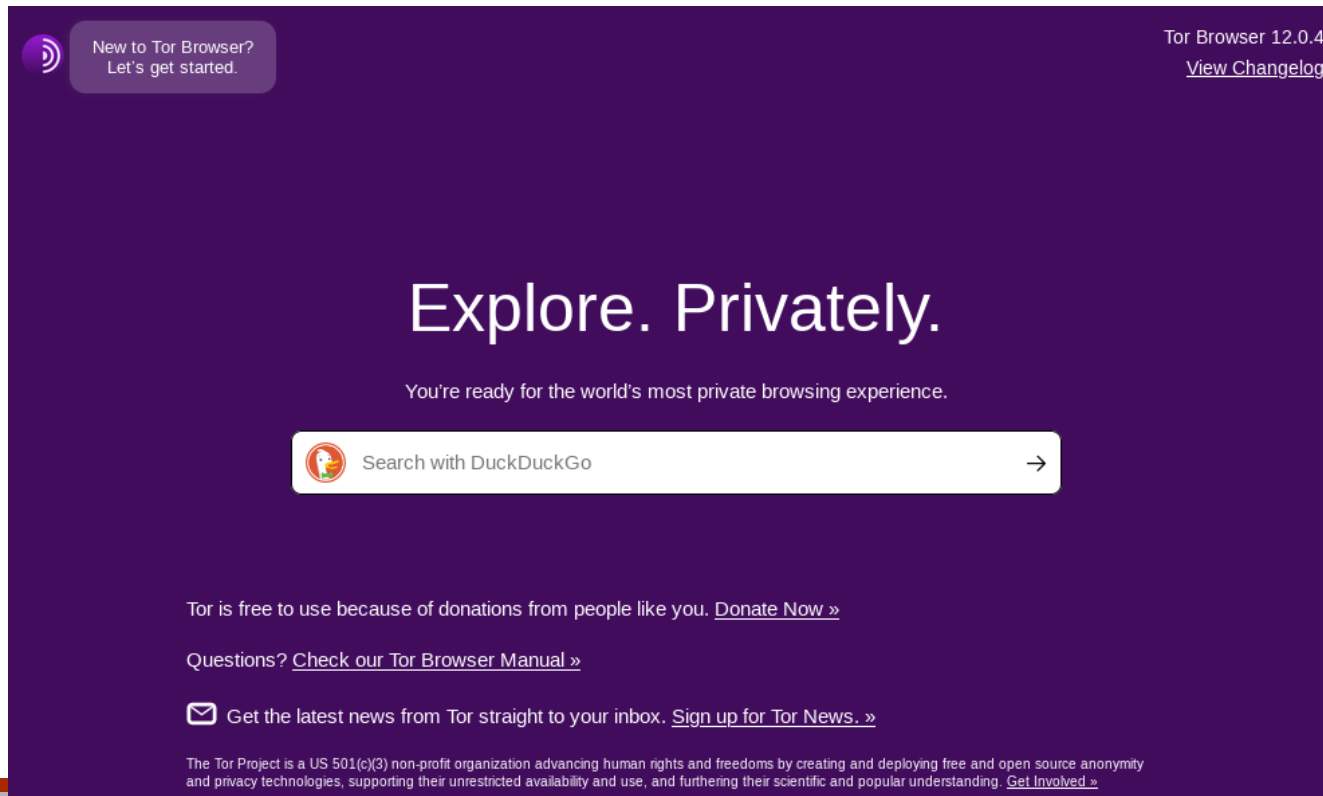


Dark Web

TOR Browser – Installazione ed Utilizzo

6. Avviare il TOR Browser

➤ **torbrowser-launcher**



Dark Web

Esempio 1 – Motori di Ricerca per il Dark Web

➤ DuckDuckGo (Accesso tramite Mozilla Firefox)

➤ `https://duckduckgogg42xjoc72x3sjasowoarfbgcmvfimaftt6twagswzczad.onion/`

Hmm. We're having trouble finding that site.

We can't connect to the server at `duckduckgogg42xjoc72x3sjasowoarfbgcmvfimaftt6twagswzczad.onion`.

If that address is correct, here are three other things you can try:

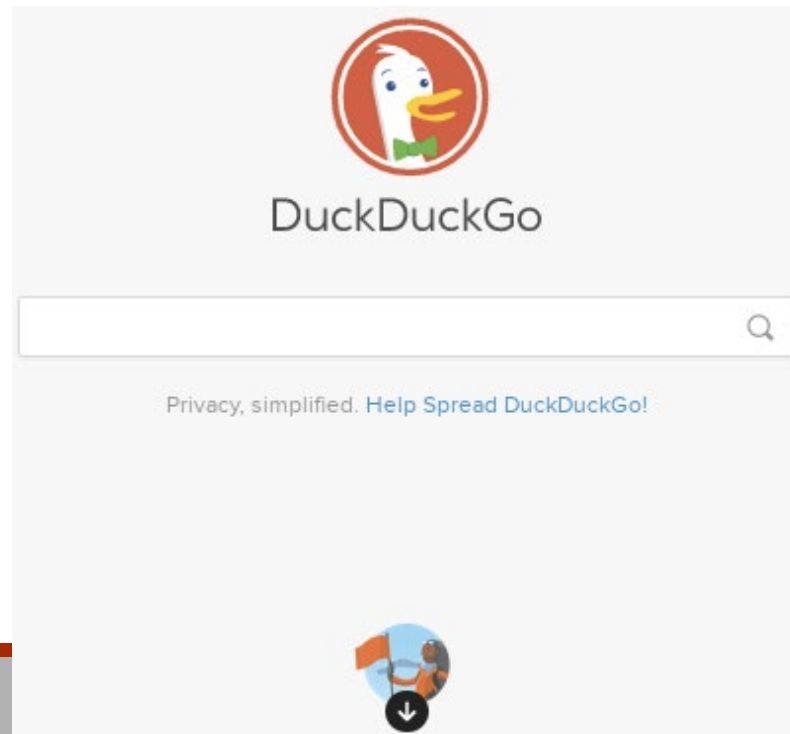
- Try again later.
- Check your network connection.
- If you are connected but behind a firewall, check that Firefox has permission to access the Web.

Try Again

Dark Web

Esempio 1 – Motori di Ricerca per il Dark Web

- DuckDuckGo (Accesso tramite TOR Browser)
 - `https://duckduckgogg42xjoc72x3sjasowoarfbgcmvfimaftt6twagswzczad.onion/`

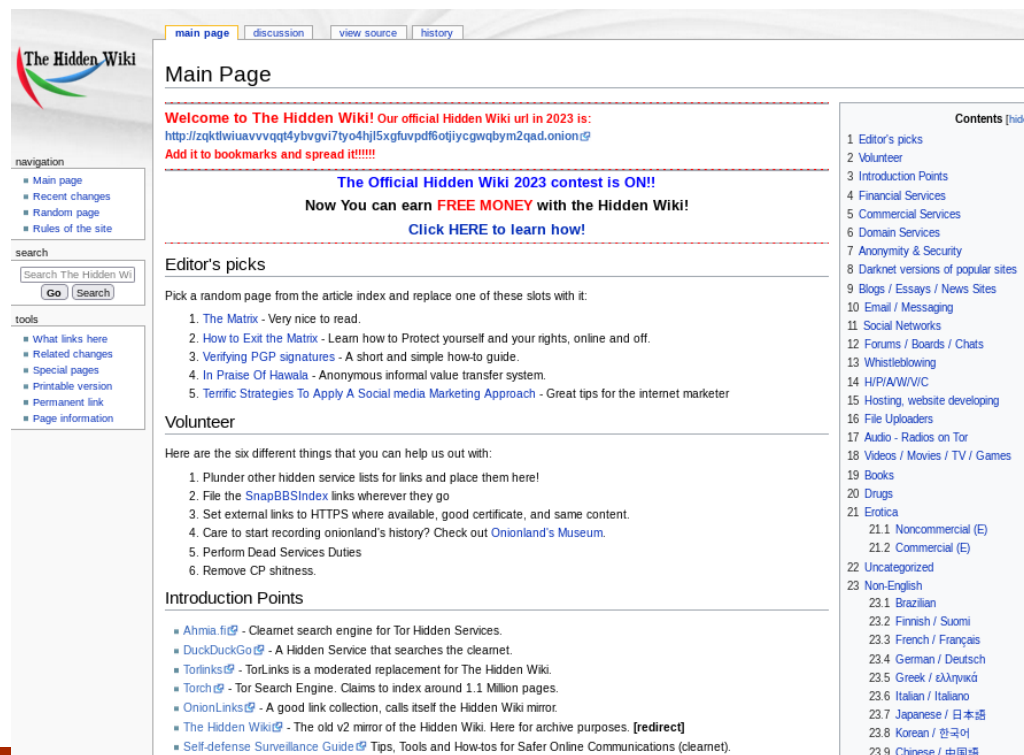


Dark Web

Esempio 2 – Motori di Ricerca per il Dark Web

➤ The Hidden Wiki

➤ <http://s4k4ceiapwwgcm3mkb6e4diqecpo7kvdnfr5gg7sph7jjppqkvw-wqtyd.onion/>



The screenshot shows the main page of 'The Hidden Wiki'. The page has a light blue header with navigation links: 'main page', 'discussion', 'view source', and 'history'. The main content area is titled 'Main Page' and contains a welcome message, a contest announcement for 2023, and a list of 'Editor's picks'. A 'Volunteer' section lists six tasks for users. An 'Introduction Points' section provides links to various services and languages. A sidebar on the left contains navigation links, a search box, and tools. A 'Contents' table of contents is on the right.

The Hidden Wiki

main page | discussion | view source | history

Main Page

Welcome to The Hidden Wiki! Our official Hidden Wiki url in 2023 is:
<http://zqkthwuaavvqq4ybvqvi7yo4hj5xgfuvpdl6otjycgwqbybm2qad.onion@>
Add it to bookmarks and spread it!!!!!!

The Official Hidden Wiki 2023 contest is ON!!
Now You can earn **FREE MONEY** with the Hidden Wiki!
[Click HERE to learn how!](#)

Editor's picks

Pick a random page from the article index and replace one of these slots with it:

1. [The Matrix](#) - Very nice to read.
2. [How to Exit the Matrix](#) - Learn how to Protect yourself and your rights, online and off.
3. [Verifying PGP signatures](#) - A short and simple how-to guide.
4. [In Praise Of Hawala](#) - Anonymous informal value transfer system.
5. [Terific Strategies To Apply A Social media Marketing Approach](#) - Great tips for the internet marketer

Volunteer

Here are the six different things that you can help us out with:

1. Plunder other hidden service lists for links and place them here!
2. File the [SnapBBSIndex](#) links wherever they go
3. Set external links to HTTPS where available, good certificate, and same content.
4. Care to start recording onionland's history? Check out [Onionland's Museum](#).
5. Perform Dead Services Duties
6. Remove CP shitness.

Introduction Points

- [Ahmia.fi](#) - Cleamnet search engine for Tor Hidden Services.
- [DuckDuckGo](#) - A Hidden Service that searches the cleamnet.
- [TorLinks](#) - TorLinks is a moderated replacement for The Hidden Wiki.
- [Torch](#) - Tor Search Engine. Claims to index around 1.1 Million pages.
- [OnionLinks](#) - A good link collection, calls itself the Hidden Wiki mirror.
- [The Hidden Wiki](#) - The old v2 mirror of the Hidden Wiki. Here for archive purposes. [\[redirect\]](#)
- [Self-defense Surveillance Guide](#) Tips, Tools and How-tos for Safer Online Communications (cleamnet).

Contents [hide]

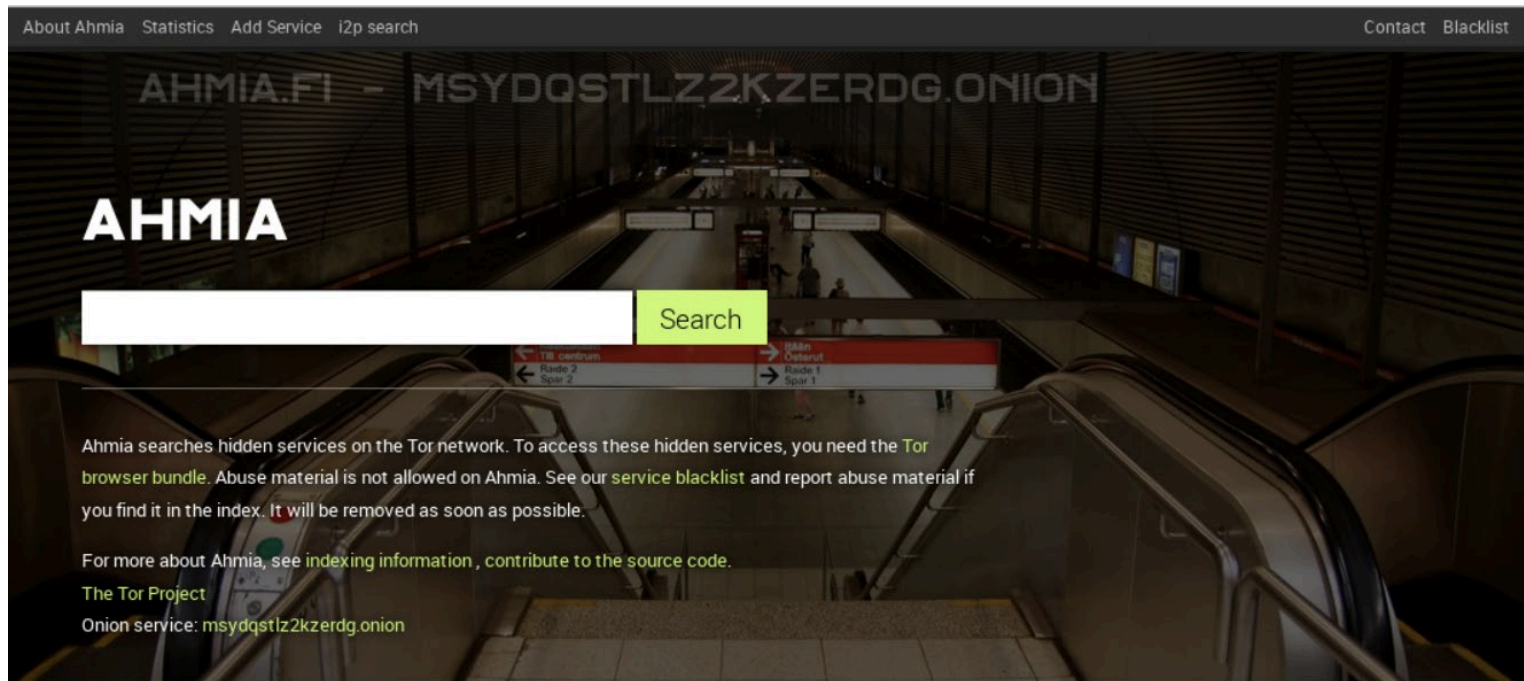
- 1 Editor's picks
- 2 Volunteer
- 3 Introduction Points
- 4 Financial Services
- 5 Commercial Services
- 6 Domain Services
- 7 Anonymity & Security
- 8 Darknet versions of popular sites
- 9 Blogs / Essays / News Sites
- 10 Email / Messaging
- 11 Social Networks
- 12 Forums / Boards / Chats
- 13 Whistleblowing
- 14 H/PIA/W/V/C
- 15 Hosting, website developing
- 16 File Uploaders
- 17 Audio - Radios on Tor
- 18 Videos / Movies / TV / Games
- 19 Books
- 20 Drugs
- 21 Erotica
 - 21.1 Noncommercial (E)
 - 21.2 Commercial (E)
- 22 Uncategorized
- 23 Non-English
 - 23.1 Brazilian
 - 23.2 Finnish / Suomi
 - 23.3 French / Français
 - 23.4 German / Deutsch
 - 23.5 Greek / ελληνικά
 - 23.6 Italian / Italiano
 - 23.7 Japanese / 日本語
 - 23.8 Korean / 한국어
 - 23.9 Chinese / 中国語

Dark Web

Esempio 3 – Motori di Ricerca per il Dark Web

➤ Ahmia

➤ `http://juhanurmihxlp77nkq76byazcldy2hlmovfu2epvl5ankdibst4csyd.onion/`

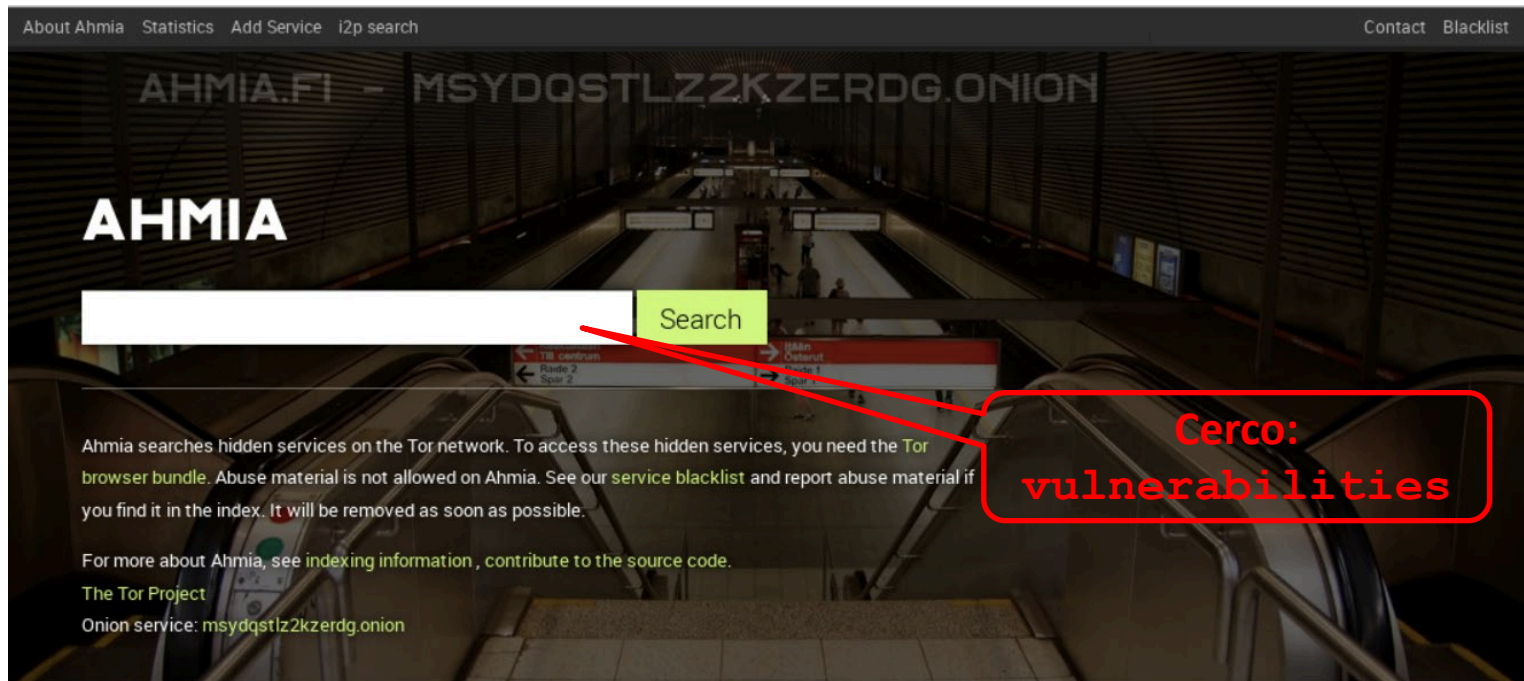


Dark Web

Esempio 3 – Motori di Ricerca per il Dark Web

➤ Ahmia

➤ `http://juhanurmihxlp77nkq76byazcldy2h1movfu2epv15ankdibsot4csyd.onion/`

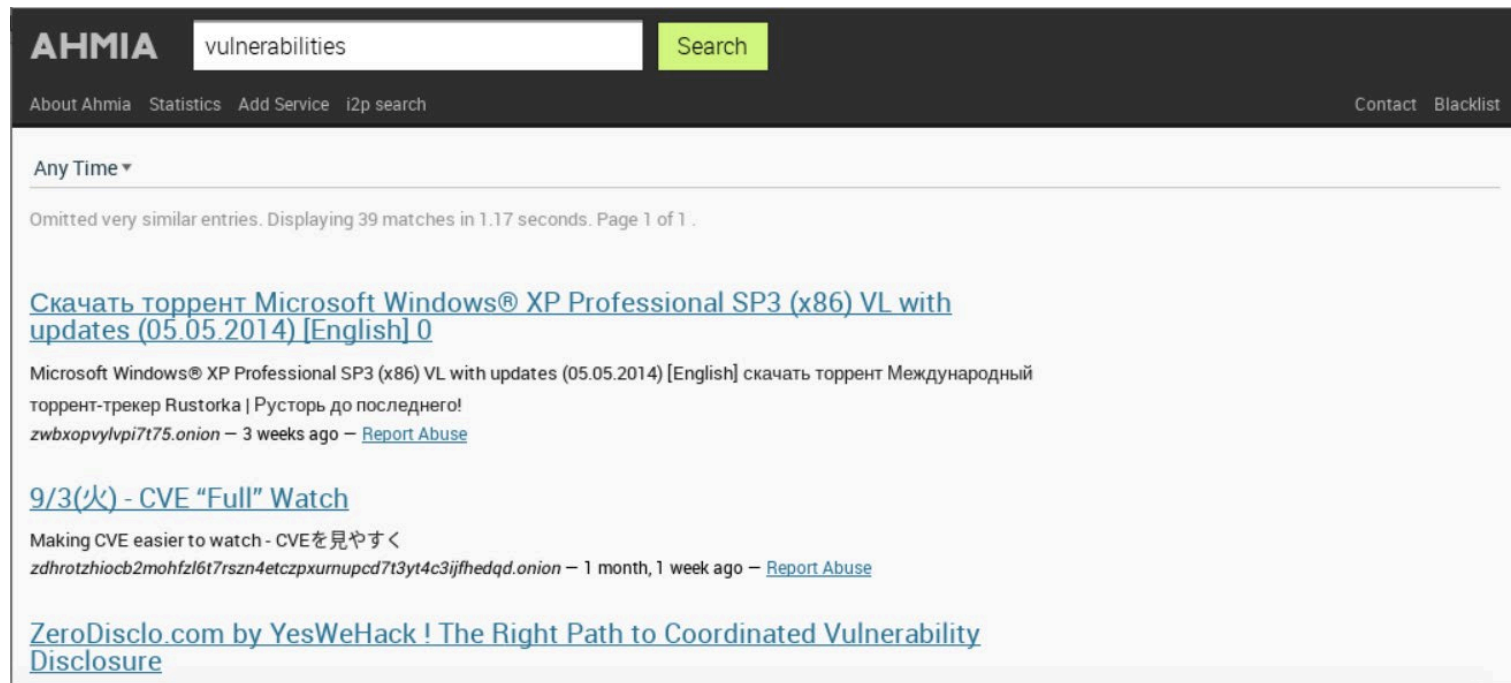


Dark Web

Esempio 3 – Motori di Ricerca per il Dark Web

➤ Ahmia

➤ <http://juhanurmihxlp77nkq76byazcldy2hlmovfu2epv15ankdibsot4csyd.onion/>



The screenshot displays the Ahmia search engine interface. At the top, there is a dark header with the Ahmia logo on the left, a search input field containing the text 'vulnerabilities', and a green 'Search' button on the right. Below the header, a navigation bar includes links for 'About Ahmia', 'Statistics', 'Add Service', and 'i2p search' on the left, and 'Contact' and 'Blacklist' on the right. The main content area shows a dropdown menu set to 'Any Time'. Below this, a message states: 'Omitted very similar entries. Displaying 39 matches in 1.17 seconds. Page 1 of 1.' The search results list three items:

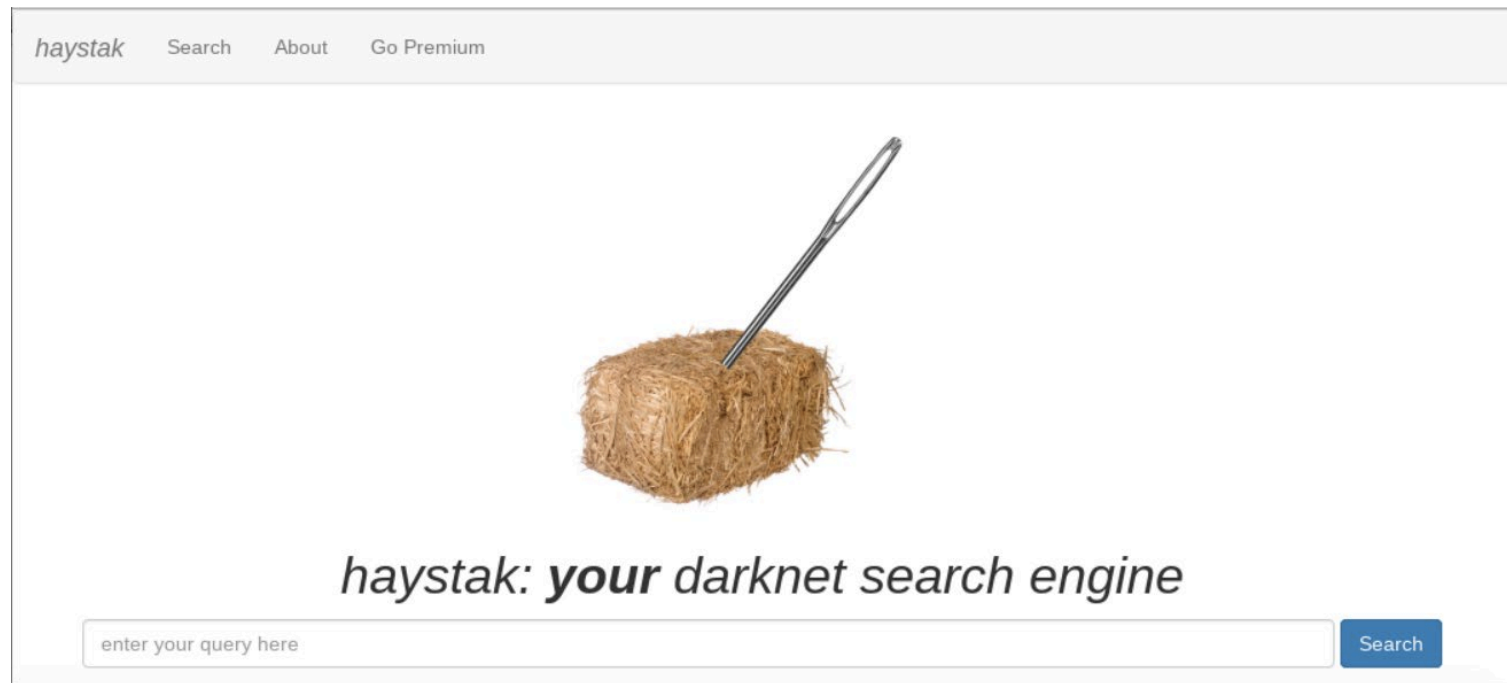
- [Скачать торрент Microsoft Windows® XP Professional SP3 \(x86\) VL with updates \(05.05.2014\) \[English\] 0](#)
Microsoft Windows® XP Professional SP3 (x86) VL with updates (05.05.2014) [English] скачать торрент Международный торрент-трекер Rustorka | Русторь до последнего!
[zwbxopvylpi7t75.onion](#) — 3 weeks ago — [Report Abuse](#)
- [9/3\(火\) - CVE "Full" Watch](#)
Making CVE easier to watch - CVEを見やすく
[zdhrotzhiocb2mohfzl6t7rszn4etcpxumupcd7t3yt4c3ijfhedqd.onion](#) — 1 month, 1 week ago — [Report Abuse](#)
- [ZeroDisclo.com by YesWeHack ! The Right Path to Coordinated Vulnerability Disclosure](#)

Dark Web

Esempio 4 – Motori di Ricerca per il Dark Web

➤ Haystak

➤ <http://haystak5njsmn2hqkewecpaxetahtwhsbsa64jom2k22z5afxhnpxfid.onion/>

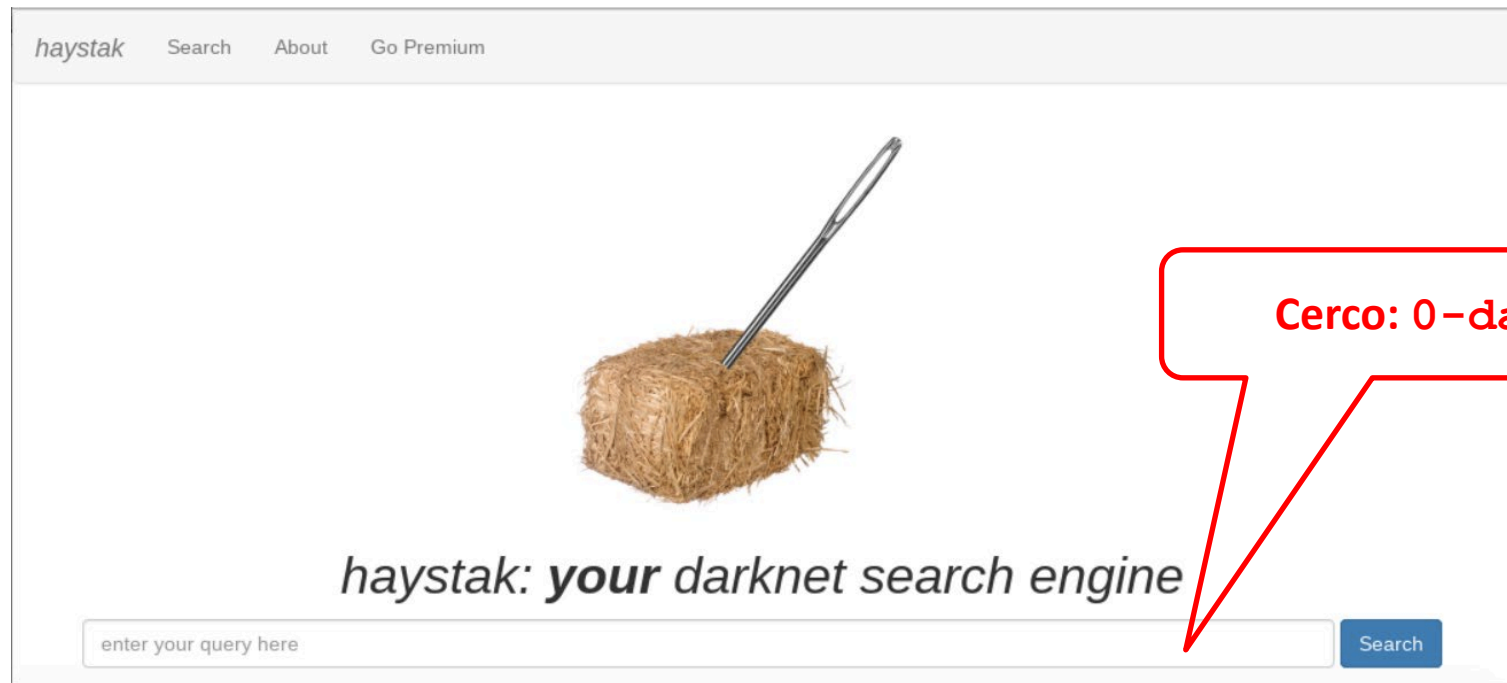


Dark Web

Esempio 4 – Motori di Ricerca per il Dark Web

➤ Haystak

➤ <http://haystak5njsmn2hqkewecpaxetahtwhsbsa64jom2k22z5afxhnpxfid.onion/>



Dark Web

Esempio 4 – Motori di Ricerca per il Dark Web

➤ Haystack

➤ `http://haystack5njsmn2hqkewecpaxetahtwhsbsa64jom2k22z5afxhnpxfid.onion/`

[haystack](#) [Search](#) [About](#) [Go Premium](#)

Search

Found Array results.

[Advertise here](#) - contact us via the form

exploit 0 day
http://damagelabraahzcu.onion/index.php?topic=27345.msg149910%3Btopicseen
exploit 0 day * ÐÐ*ÑÐ*Ð»Ð% [http://damagelabraahzcu.onion/index.php] * ÐÑÐ%Ð' [http://damagelabraahzcu.onion
board=56.0] » * exploit 0 day [http://damagelabraahzcu.onion/index.php?
action=printpage;topic=27345.0] Ð;ÑÑÐ*Ð%Ð,ÑÑ: [1] ÐÐ²ÑÐ%Ñ ÐcÐµÐ%Ð": exploit 0 day (ÐÑÐ%ÑÐ
action=profile;u=172692] * [http:] exploit 0 day [http://damagelabraahzcu.onion/index.php?
board=56.0] » * exploit 0 day [http://damagelabraahzcu.onion/index.php?
exploit 0 day
[Cached version](#) - [Historical versions](#) - [Datapoints \(e.g. bitcoin addresses\)](#) - [Report for removal](#)

exploit 0 day
http://damagelabraahzcu.onion/index.php?topic=27345.msg149910
exploit 0 day * ÐÐ*ÑÐ*Ð»Ð% [http://damagelabraahzcu.onion/index.php] * ÐÑÐ%Ð' [http://damagelabraahzcu.onion
board=56.0] » * exploit 0 day [http://damagelabraahzcu.onion/index.php?
action=printpage;topic=27345.0] Ð;ÑÑÐ*Ð%Ð,ÑÑ: [1] ÐÐ²ÑÐ%Ñ ÐcÐµÐ%Ð": exploit 0 day (ÐÑÐ%ÑÐ
action=profile;u=172692] * [http:] exploit 0 day [http://damagelabraahzcu.onion/index.php?

Dark Web

Esempio 5 – Motori di Ricerca per il Dark Web

➤ Torch

➤ `http://xmh57jrknzkhv6y3ls3ubitzfqnrwxhopf5aygthi7d6rplyvk3noyd.onion/`

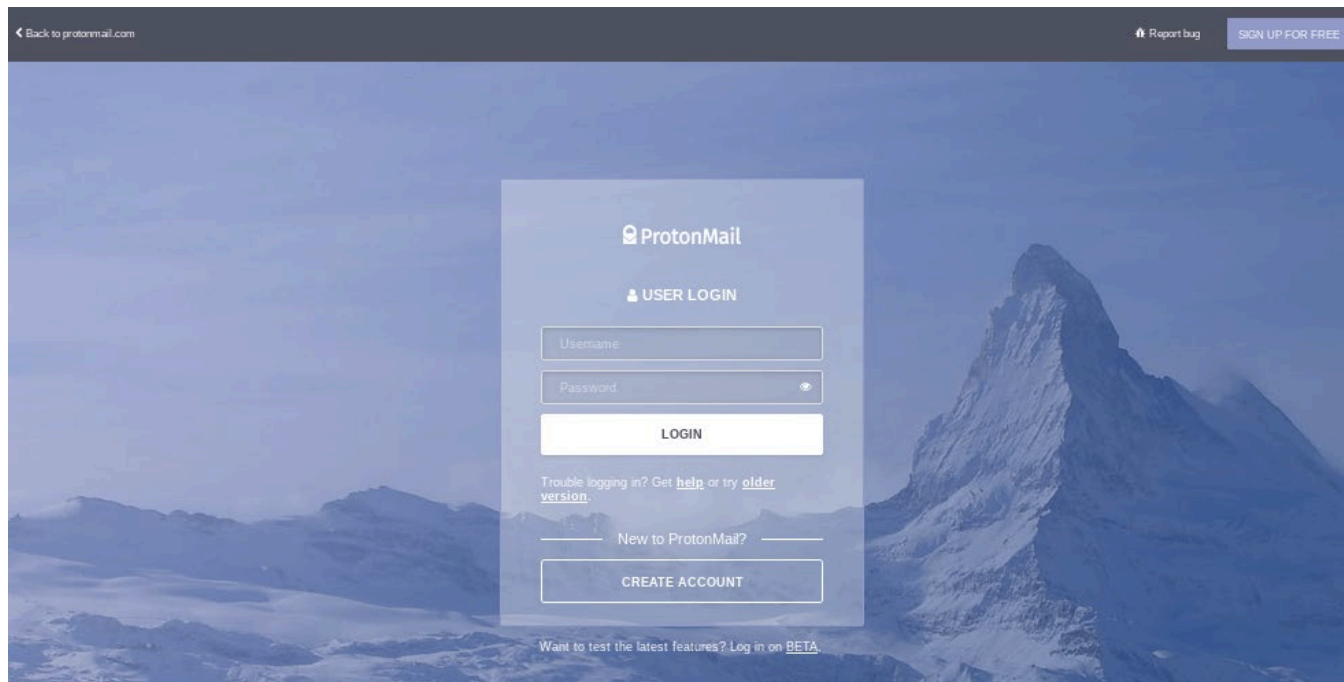


Dark Web

Esempio 6 – Servizi E-mail

➤ ProtonMail

➤ <https://protonmailrmez3lotccipshtkleegetolb73fuirgj7r4o4vfu7ozyd.onion/>



Outline

- Concetti Preliminari
- Raccolta di Informazioni da Risorse Web-Based
- Raccolta delle Informazioni di Registrazione
- Raccolta delle Informazioni di Routing
- Raccolta di Informazioni dai Record DNS
- Raccolta di Informazioni mediante Crawler
- Raccolta di informazioni dal Dark Web
- **Altri Strumenti e Servizi per Raccogliere Informazioni**

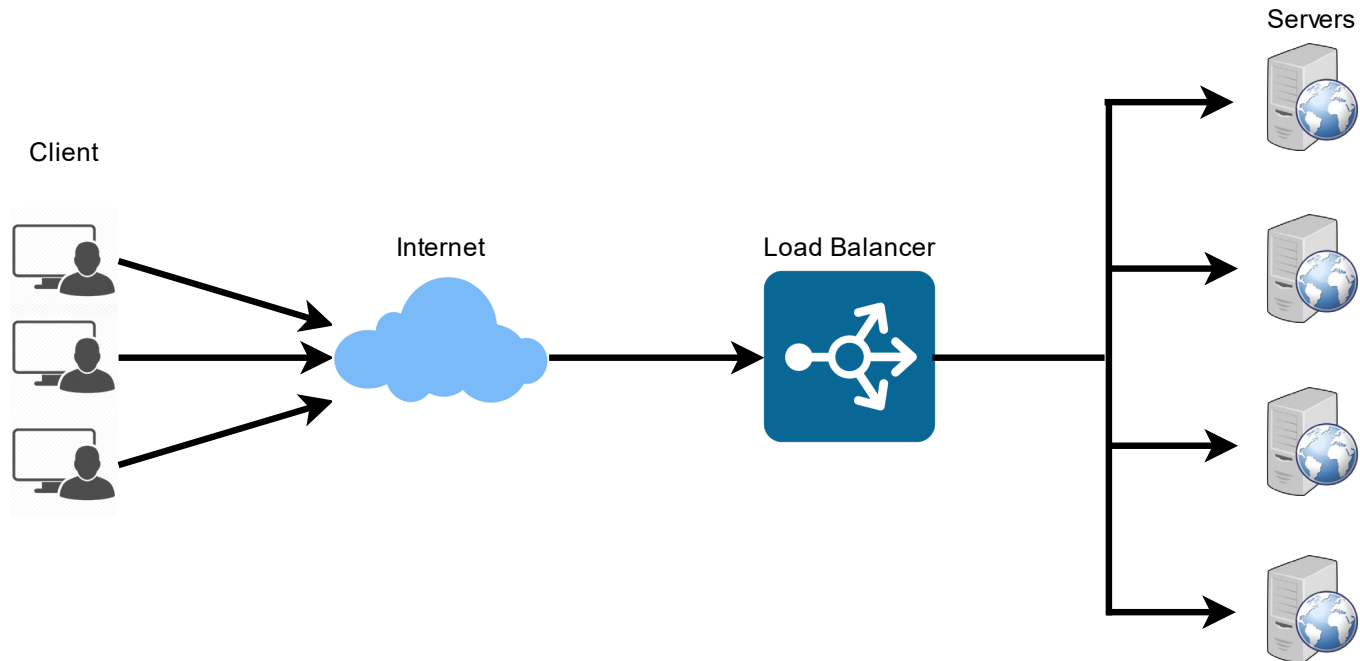
Altri Strumenti

Rilevamento Load Balancer

- Il *Load Balancing* è un metodo utilizzato per distribuire il carico applicativo su più Server
 - Permettendo alle applicazioni di funzionare in modo più efficiente ed affidabile
- I *Load Balancer* sono generalmente classificati in due categorie in base al protocollo utilizzato per effettuare il *Load Balancing*
 - *DNS Load Balancer*
 - *HTTP Load Balancer*

Altri Strumenti

Rilevamento Load Balancer



Rilevamento Load Balancer

Comando `ldb`

- **Idea:** Di solito se un singolo host è risolto in più indirizzi IP probabilmente sta utilizzando un servizio di *Load Balancing*
- Esiste uno strumento specifico per la rilevazione dei *Load Balancer*, chiamato **load balancer detector** (comando **ldb**)
 - **ldb** è in grado di rilevare *DNS load balancer* ed *HTTP load balancer*

Rilevamento Load Balancer

Comando lbd – Esempio 1

➤ lbd hackerone.com

```
root@kali:~# lbd hackerone.com

lbd - load balancing detector 0.4 - Checks if a given domain uses load-balancing.
                                   Written by Stefan Behte (http://ge.mine.nu)
                                   Proof-of-concept! Might give false positives.

Checking for DNS-Loadbalancing: FOUND
hackerone.com has address 104.16.99.52
hackerone.com has address 104.16.100.52

Checking for HTTP-Loadbalancing [Server]:
cloudflare
NOT FOUND

Checking for HTTP-Loadbalancing [Date]: 07:42:09, 07:42:09, 07:42:09, 07:42:09, 07:4
2:09, 07:42:10, 07:42:10, 07:42:10, 07:42:10, 07:42:10, 07:42:10, 07:42:11, 07:42:11, 07:42:11
, 07:42:11, 07:42:12, 07:42:12, 07:42:12, 07:42:12, 07:42:12, 07:42:12, 07:42:13, 07:42:13, 07
:42:13, 07:42:13, 07:42:14, 07:42:14, 07:42:14, 07:42:14, 07:42:14, 07:42:15, 07:42:
15, 07:42:15, 07:42:15, 07:42:16, 07:42:16, 07:42:16, 07:42:16, 07:42:17, 07:42:17,
07:42:17, 07:42:17, 07:42:18, 07:42:18, 07:42:18, 07:42:18, 07:42:19, 07:4
2:19, 07:42:19, 07:42:19, 07:42:19, NOT FOUND

Checking for HTTP-Loadbalancing [Diff]: FOUND
< CF-RAY: 5299d57f9b91e8f7-MXP
> CF-RAY: 5299d580f977be37-MXP

hackerone.com does Load-balancing. Found via Methods: DNS HTTP[Diff]
```

Rilevamento Load Balancer

Comando lbd – Esempio 1

➤ lbd hackerone.com

```
root@kali:~# lbd hackerone.com

lbd - load balancing detector 0.4 - Checks if a given domain uses load-balancing.
      Written by Stefan Behte (http://ge.mine.nu)
      Proof-of-concept! Might give false positives.

➔ Checking for DNS-Loadbalancing: FOUND
hackerone.com has address 104.16.99.52
hackerone.com has address 104.16.100.52

Checking for HTTP-Loadbalancing [Server]:
cloudflare
NOT FOUND

Checking for HTTP-Loadbalancing [Date]: 07:42:09, 07:42:09, 07:42:09, 07:42:09, 07:4
2:09, 07:42:10, 07:42:10, 07:42:10, 07:42:10, 07:42:10, 07:42:10, 07:42:11, 07:42:11, 07:42:11
, 07:42:11, 07:42:12, 07:42:12, 07:42:12, 07:42:12, 07:42:12, 07:42:12, 07:42:13, 07:42:13, 07
:42:13, 07:42:13, 07:42:14, 07:42:14, 07:42:14, 07:42:14, 07:42:14, 07:42:15, 07:42:
15, 07:42:15, 07:42:15, 07:42:16, 07:42:16, 07:42:16, 07:42:16, 07:42:17, 07:42:17,
07:42:17, 07:42:17, 07:42:18, 07:42:18, 07:42:18, 07:42:18, 07:42:19, 07:4
2:19, 07:42:19, 07:42:19, 07:42:19, NOT FOUND

Checking for HTTP-Loadbalancing [Diff]: FOUND
< CF-RAY: 5299d57f9b91e8f7-MXP
> CF-RAY: 5299d580f977be37-MXP

hackerone.com does Load-balancing. Found via Methods: DNS HTTP[Diff]
```

Rilevamento Load Balancer

Comando lbd – Esempio 2

➤ **lbd dazn.com**

```
root@kali:~# lbd dazn.com

lbd - load balancing detector 0.4 - Checks if a given domain uses load-balancing.
      Written by Stefan Behte (http://ge.mine.nu)
      Proof-of-concept! Might give false positives.

Checking for DNS-Loadbalancing: NOT FOUND
Checking for HTTP-Loadbalancing [Server]:
  AmazonS3
  NOT FOUND

Checking for HTTP-Loadbalancing [Date]: 22:58:32, 22:58:33, 22:58:33, 22:58:34, 22:58:35, 22:58:
36, 22:58:37, 22:58:37, 22:58:38, 22:58:39, 22:58:40, 22:58:41, 22:58:42, 22:58:43, 22:58:44, 22
:58:44, 22:58:45, 22:58:46, 22:58:47, 22:58:47, 22:58:48, 22:58:49, 22:58:50, 22:58:51, 22:58:52
, 22:58:53, 22:58:54, 22:58:55, 22:58:56, 22:58:56, 22:58:57, 22:58:58, 22:58:59, 22:59:05, 22:5
9:06, 22:59:06, 22:59:07, 22:59:08, 22:59:09, 22:59:10, 22:59:10, 22:59:11, 22:59:12, 22:59:13,
22:59:14, 22:59:15, 22:59:16, 22:59:17, 22:59:18, 22:59:18, NOT FOUND

Checking for HTTP-Loadbalancing [Diff]: FOUND
< x-amz-id-2: 0BdVFfZYKF0S8YaHSSk648dbywg2Z+S/yggkezC6tGRUc8iRcRmudKy+2gAShNwf3skr/Tl6ycQ=
< x-amz-request-id: CBC674B1AF90703E
> x-amz-id-2: PxxW5s0njuMQzpa3fS9ei39paKKlDtFu/tkGxphH8J6nvUzxRywmkjuRWckV0vRKAYTK4JqgZB0=
> x-amz-request-id: 117EB5353727DDCE

dazn.com does Load-balancing. Found via Methods: HTTP[Diff]
```

Rilevamento Load Balancer

Comando lbd – Esempio 2

➤ **lbd dazn.com**

```
root@kali:~# lbd dazn.com

lbd - load balancing detector 0.4 - Checks if a given domain uses load-balancing.
      Written by Stefan Behte (http://ge.mine.nu)
      Proof-of-concept! Might give false positives.

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36, 22:58:37, 22:58:37, 22:58:38, 22:58:39, 22:58:40, 22:58:41, 22:58:42, 22:58:43, 22:58:44, 22
:58:44, 22:58:45, 22:58:46, 22:58:47, 22:58:47, 22:58:48, 22:58:49, 22:58:50, 22:58:51, 22:58:52
, 22:58:53, 22:58:54, 22:58:55, 22:58:56, 22:58:56, 22:58:57, 22:58:58, 22:58:59, 22:59:05, 22:5
9:06, 22:59:06, 22:59:07, 22:59:08, 22:59:09, 22:59:10, 22:59:10, 22:59:11, 22:59:12, 22:59:13,
22:59:14, 22:59:15, 22:59:16, 22:59:17, 22:59:18, 22:59:18, NOT FOUND

➔ Checking for HTTP-Loadbalancing [Diff]: FOUND
< x-amz-id-2: 0BdVffZYKF0S8YaHSSk648dbywg2Z+S/yggkezC6tGRUc8iRcRmudKy+2gAShNwf3skr/Tl6ycQ=
< x-amz-request-id: CBC674B1AF90703E
> x-amz-id-2: PxxW5s0njuMQzpa3fS9ei39paKKlDtFu/tkGxphH8J6nvUzxRywmkjuRWckV0vRKAYTK4JqgZB0=
> x-amz-request-id: 117EB5353727DDCE

dazn.com does Load-balancing. Found via Methods: HTTP[Diff]
```

Altri Strumenti

Comando `sslscan`

- Permette di analizzare il supporto SSL/TLS lato server
 - Per maggiori informazioni su **sslscan**
 - **sslscan -h**

```
root@kali:~/Downloads/tor-browser_en-US/Browser# sslscan -h

sslscan

2.0.2-static
OpenSSL 1.1.1i-dev  xx XXX xxxx

Command:
sslscan [options] [host:port | host]
```

Altri Strumenti

Comando `sslscan` - Esempio

- Utilizzo del comando `sslscan` sul dominio `unisa.it`
- `sslscan unisa.it`

```
root@kali:~/Downloads/tor-browser_en-US/Browser# sslscan unisa.it
Version: 2.0.2-static
OpenSSL 1.1.1i-dev  xx XXX xxxx

Connected to 193.205.185.20

Testing SSL server unisa.it on port 443 using SNI name unisa.it

  SSL/TLS Protocols:
  SSLv2      disabled
  SSLv3      disabled
  TLSv1.0    enabled
  TLSv1.1    enabled
  TLSv1.2    enabled
  TLSv1.3    disabled

  TLS Fallback SCSV:
  Server supports TLS Fallback SCSV

  TLS renegotiation:
  Secure session renegotiation supported

  TLS Compression:
  Compression disabled
```

Altri Strumenti

Qualys – SSL Server Test

➤ <https://www.ssllabs.com/ssltest/>

 **Qualys.** SSL Labs

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SSL Server Test

This free online service performs a deep analysis of the configuration of any SSL web server on the public Internet. **Please note that the information you submit here is used only to provide you the service. We don't use the domain names or the test results, and we never will.**

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Submit

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Information Gathering

Altri Strumenti

Qualys – SSL Server Test

➤ <https://www.ssllabs.com/ssltest/>

 **Qualys.** SSL Labs

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SSL Server Test

This free online service performs a deep analysis of the configuration of any SSL web server on the public Internet. **Please note that the information you submit here is used only to provide you the service. We don't use the domain names or the test results, and we never will.**

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Submit

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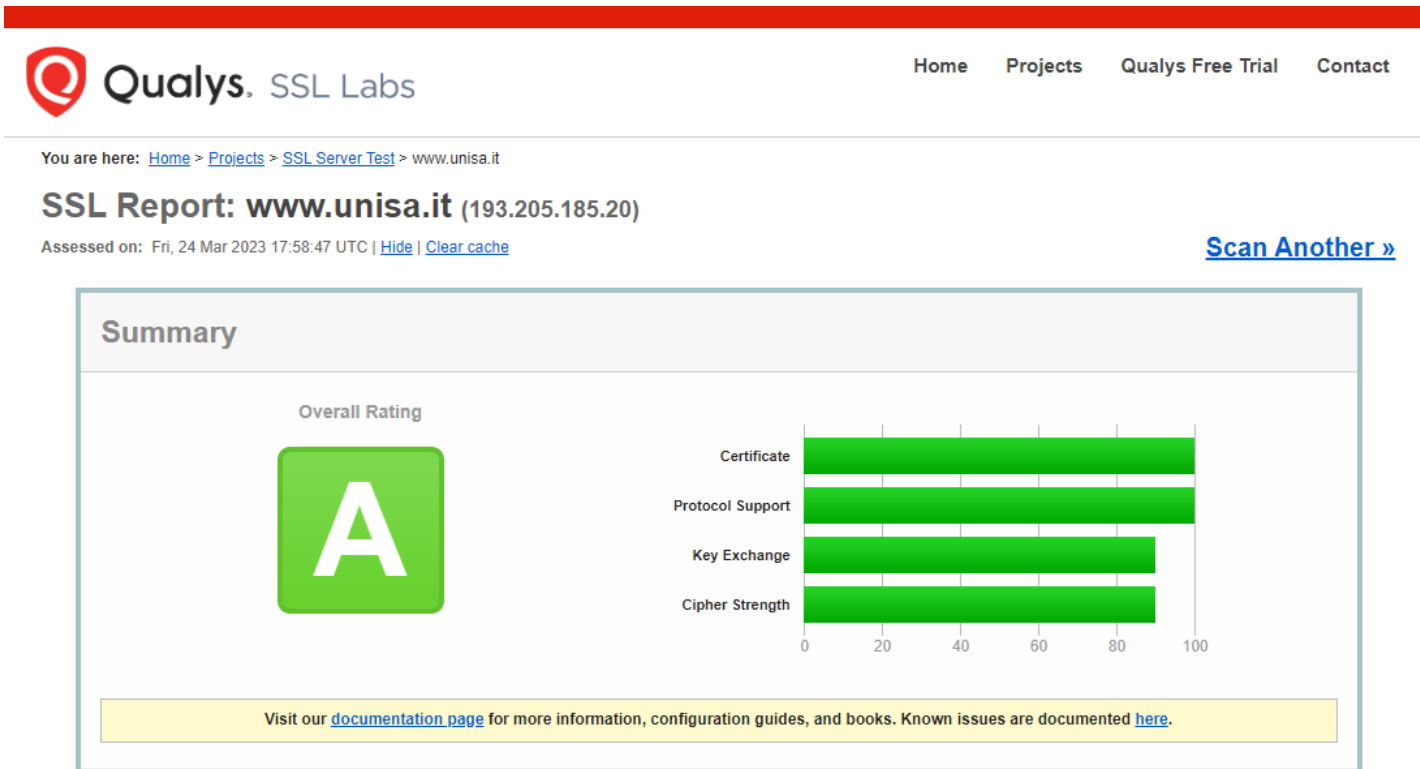
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www.unisa.it

Altri Strumenti

Qualys – SSL Server Test – Esempio

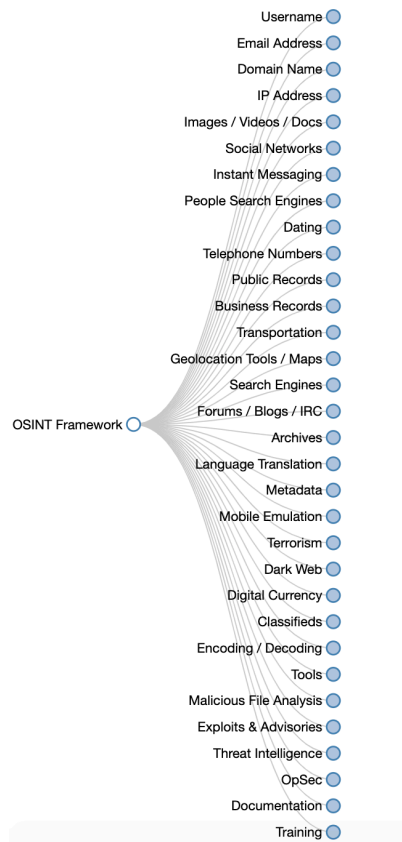
➤ <https://www.ssllabs.com/ssltest/>



Altri Strumenti

OSINT Framework

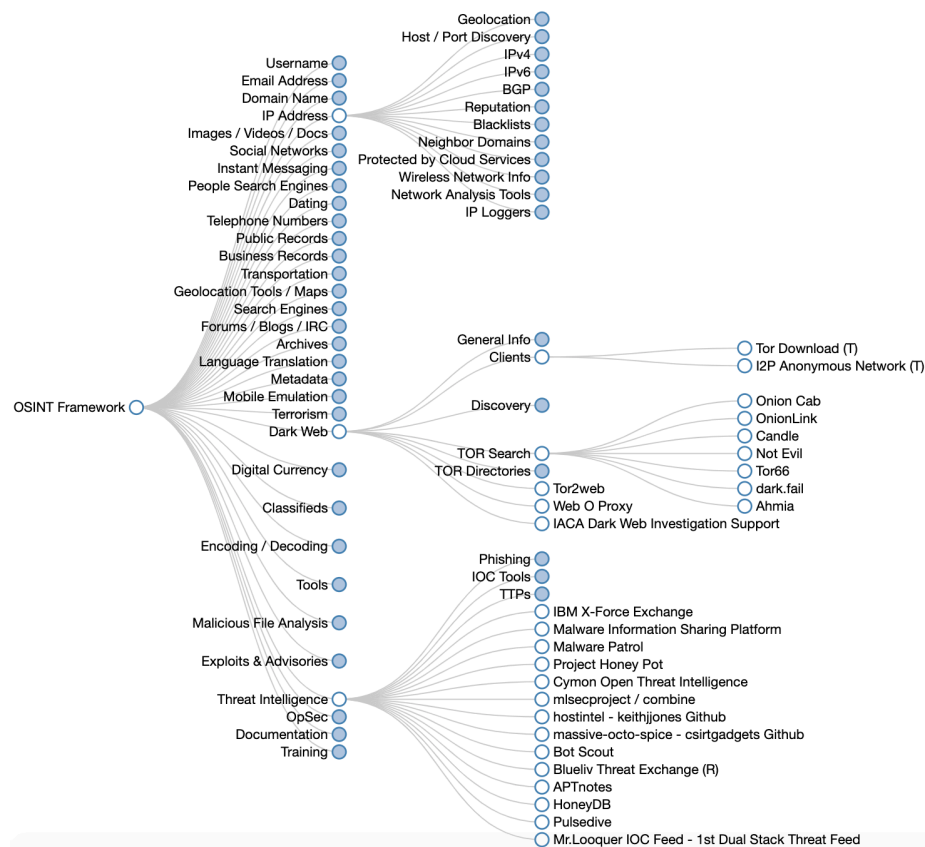
➤ <https://osintframework.com/>



Altri Strumenti

OSINT Framework

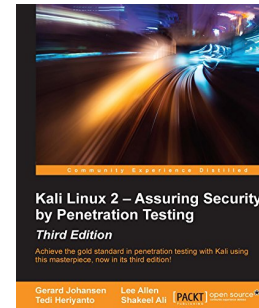
➤ <https://osintframework.com/>



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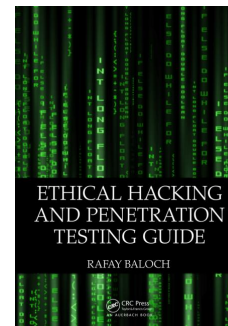
- **Kali Linux 2 - Assuring Security by Penetration Testing. Third Edition.** Gerard Johansen, Lee Allen, Tedi Heriyanto, Shakeel Ali. Packt Publishing. 2016

- Capitolo 4



- **Ethical Hacking and Penetration Testing Guide.** Rafay Baloch. CRC Press. 2014

- Capitolo 3



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- <https://ahrefs.com/blog/google-advanced-search-operators/>